

Admissible Record Construction in Physically Non-Selective Quantum Regimes

A Kernel-Role Determinacy Theorem for Record Fixation, Fixed-Domain Quotients, and Boundary-Trace Normal Form

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Overview. AASC is not an optional interpretive overlay on an already complete physical description. It states the jointly minimal necessity conditions under which any non-degenerate determinate object, state, carrier, record, transformation, comparison, or endpoint can exist as such. Once a physical measurement target is asserted as determinate, admissibility, standing, reference, and irreversibility are already active at the relevant scope. This manuscript specializes that kernel to measurement-record construction and then occupies the specialization with an explicit finite quantum measurement domain. On a two-branch carrier, squared-amplitude weights, density weights, branch-map traces, effect weights, pointer records, instrument records, environment records, channel records, and zero-product projector/repeatability surfaces agree on the same branch architecture. Fine records remain separated; every declared non-selective surface agrees only on the merged record; and the disjunction of those surfaces as candidate hidden fine-record selectors is eliminated. The Lean formalization is synchronized with v0.9.1, the public finite measurement-boundary release.

The appendices expose the quotient presentation, Lean/mathematical crosswalk, dependency ledger, and audit path; the finite formal development is integrated into the main theorem ladder.

Abstract

The measurement problem is usually framed as a problem about outcome production, state reduction, or probability. This manuscript treats it first as a problem of determinate measurement objecthood. The governing AASC dependency is not that an independently complete physics later receives an optional constructional interpretation. Rather,

$$\text{NDObj}(x) \Rightarrow K(x) = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}),$$

so any determinate physical state, outcome, carrier, transformation, record, or endpoint already presupposes admissibility, standing, reference, and irreversibility at the scope on which it is asserted as the same object. The measurement-record packet is the domain-specific occupation of that kernel when a record must remain the same standing-bearing physical and evidential object across retained use, prospective use, deliberation, comparison, and report.

Within that target, the manuscript proves fixed-domain assignment-space forcing, quotient uniqueness for standing-bearing content, same-scope equivalence collapse, Everettian non-selection for the bare unitary-entangling-decohering package, fixation commutation for evidential ordering, diachronic inheritance for downstream reuse, record-asymmetry exhaustion, and selector exhaustion. These results do not introduce collapse dynamics, hidden variables, branch selection, or a new record-producing mechanism. They show what any such mechanism or interpretation must satisfy once it claims non-degenerate determinate measurement-record objecthood.

The paper also develops an explicit finite quantum occupation of the kernel. The finite domain has two outcomes, two fine records and one merged record, denominator-two branch weights, an integer-amplitude balanced state, a density-table realization, selective and non-selective instrument maps, environment-record maps, projective branch maps, complementary effects, zero-product projectors, and repeatability operations. In that construction,

$$\tilde{T}(\mathcal{K}_o(\tilde{\rho})) = \tilde{e}_o(\tilde{\rho}) = \tilde{w}_{\tilde{\rho}}(o),$$

and, on the balanced amplitude/density witness,

$$\tilde{T}(\mathcal{K}_o(\tilde{\rho}_\psi)) = \tilde{e}_o(\tilde{\rho}_\psi) = \tilde{w}_{\tilde{\rho}_\psi}(o) = \tilde{w}_{\tilde{\psi}}(o) = 1.$$

The selective fine-record maps agree across outcome, pointer, instrument, channel, and effect surfaces; all declared non-selective surfaces agree only on the merged record; and the two fine records remain separated.

The finite selector-boundary theorem then eliminates the disjunction of candidate hidden selectors at the state, instrument, environment, channel, effect, and repeatability surfaces:

$$\neg \text{Sel}_{\text{finite}}.$$

Thus normalized weighting, coherence removal, branch correlation, instrument decomposition, channel closure, effect completeness, projector orthogonality, and repeatability can all agree while no declared non-selective surface supplies a fine-record selector. The missing work is not branch production or stabilization; it is admissible fine-record role fixation.

The formalization is finite, conditional on its declared carriers, and manuscript-facing. It constrains determinate physical measurement structure, but it does not reconstruct Hilbert-space quantum mechanics, general POVM theory, full PVM theory, full operator algebra, a general Kraus representation theorem, physical decoherence theory, the continuum Born rule, physical branch selection, interpretive truth, or a physical no-collapse theorem. The public Lean 4 release v0.9.1 is the controlling finite measurement-boundary formal artifact for the effect/projector, compatibility, closed-stack, boundary-certificate, and audit-manifest layers [21–23].

Main integrated claim. Determinate measurement objecthood activates the AASC kernel. The measurement-record specialization then forces admissible record-role construction on a fixed domain. The finite quantum occupation developed below shows that the familiar physical surfaces—amplitude weights, density tables, branch maps, instruments, environment records, effects, projectors, and repeatability maps—can commute on branch and weight content without selecting a standing fine record at any declared non-selective surface. The residual burden is therefore admissible record-role fixation, not further branch differentiation, weighting, decoherence, or repeatability alone.

Constraint-formalism orientation. AASC touches real physics because real physics asserts determinate objects, states, carriers, records, comparisons, and transformations. The kernel roles do not add a collapse process or hidden selector; they specify the necessity conditions under which such physical contents possess determinate standing at all. The finite development is therefore not external audit decoration. It is a kernel-governed finite physical specialization whose exact scope is declared.

Front-loaded dependency lock: record use is not raw record existence

Dependency lock. This manuscript has one nonclassical dependency: theory-level evidential use of a measurement record is governed record use. Once that is granted, the rest of the manuscript is target instantiation, quotient construction, non-selection diagnosis, fixation-commutation analysis, and selector exhaustion. The dependency is not hidden in the Everettian branch structure, in decoherence, in the assignment space, in the quotient, or in the boundary trace. Bare branch-local record content may exist as physical structure, local report, support, decision-theoretic input, instrumental update, or global branch-state descriptor. But when a record is used as the same standing evidential object across retained-record reuse, prospective-record reuse, deliberation, or evidential comparison, that use fixes target, record role, outcome role, evidential role, downstream reuse, reportability, act-time status, and lawful-redescription stability. That is governed record use. A same-mode denial should therefore attack the record-use-to-kernel theorem directly; otherwise the objection has changed the target practice.

Layer	Measurement-record statement	Status
Physical carrier	Branch-local records, decohered sectors, detector states, observer-successor memories.	Lawful physical content; not yet theory-level evidential-role fixation.
Local record report	“This branch contains record r_i .”	Branch-relative reportability.
Evidential anchor	“This is the record reused across retained, prospective, deliberative, and comparative practice.”	Governed record use; kernel-triggering.
Assignment space	$\rho : \mathcal{D} \rightarrow \mathcal{B} \times \mathcal{R}$.	Forced typing of standing record-role realization.
Measurement-record quotient	$\text{ARef}/\approx_{\text{MR}}$.	Exact standing-bearing content space.
Boundary trace	Singleton admissible residual fixation when internal and provenance/indexical routes fail.	Physical record-role anchoring; not a collapse dynamics.
Everettian dynamics	Unitary branching, entanglement, decoherence, and branch-local record stability.	Physical multiplicity and local stability; not singleton standing collapse.
Finite formalization	Machine-checked finite physical specialization over declared carriers, including v0.9.1 zero-effect and zero-product projector orthogonality.	Integrated finite theorem anchored to the public Lean release.

Imported closure spine and local use in this manuscript

Import discipline. The proof class is nonstandard by design: it is a kernel-forced record-use exhaustion theorem, not a hidden-variable model, collapse dynamics, Born-rule derivation, decoherence theorem, decision-theoretic derivation, or Everettian self-location argument. The measurement-record construction, assignment-space forcing, quotient theorem, Everettian non-selection diagnosis, and measurement-specific boundary-trace specialization are developed locally. The manuscript imports only a narrow AASC closure spine for the general fixed-domain consequences activated by governed record use: kernel necessity [16], claim-standing and report legitimacy [17], anchor/tensor/skin routing [19], same-scope operator closure by exhaustion [20], and singleton admissible-interior closure [18]. A standing objection must break the record-use-to-kernel link, defeat one of those named imported closure results, deny the construction-relative evidential target, or provide a faithful same-domain record-use alternative preserving the same target without the kernel roles.

Kernel theorem and kernel roles. When referring to the upstream result, this manuscript uses *the kernel theorem*. When describing what becomes operational in the measurement-record target, it uses *kernel roles*: admissibility, standing, reference, and irreversibility. The point is not to ask the reader to accept a monolithic framework; it is to show what these roles force once construction-relative evidential record use is instantiated as a non-degenerate target.

Support paper	DOI	Imported role in this manuscript
<i>Non-Degenerate Construction and the Kernel of Admissibility</i>	10.5281/ zenodo.19324199	Kernel necessity: target adequacy forces admissibility, standing, reference, and irreversibility.
<i>Claim Standing and Legitimacy</i>	10.5281/ zenodo.20312075	Claim-status and report discipline: record reports cannot exceed carrier, support, quotient, fixation, or trace status.
<i>Anchor, Tensor, and Skin</i>	10.5281/ zenodo.18678041	Fixed-domain description routing: anchors fix record/domain identity, tensor content carries standing-bearing record load, and skin is presentation surplus.
<i>Closure by Exhaustion for Same-Scope Operators under Admissibility</i>	10.5281/ zenodo.19183407	Same-scope closure: no independent same-domain classifier, hidden fifth route, or admissibility-relevant operator outside the admitted interface.
<i>Unus Solus Possibilis Est</i>	10.5281/ zenodo.18518484	Singleton admissible-interior closure: no second same-domain admissible record interior or rival evidential-use space.

Kernel-role activation in the measurement-record target. The imported closure spine is not used as an opaque authority. The target activates specific roles locally: reference fixes the same construction token, record-bearing target, and lawful redescription class; standing fixes evidential eligibility and licensed reuse; admissibility separates record construction from support, bookkeeping, selector import, and target shift; irreversibility prevents later success from repairing an unfixed same act. The subsequent quotient, Everettian non-selection, boundary-trace, and selector-exhaustion results are the measurement-specific consequences of those activated roles.

construction-relative evidential record use $\Rightarrow K_{MR}$ [kernel necessity]

$$K_{MR} \Rightarrow \begin{cases} \text{report preservation} & [\text{claim standing}] \\ \text{anchor/tensor/skin routing} & [\text{ATS}] \\ \text{same-scope closure and no fifth route} & [\text{same-scope operators}] \\ \text{singleton admissible record interior} & [\text{Unus Solus}] \end{cases}$$

$\Rightarrow$ measurement-record quotient, Everettian non-selection diagnosis,
and boundary-trace exhaustion.

Why this measurement-record target makes the kernel roles visible

The measurement-record target is used here because physically non-selective quantum regimes make the kernel roles operationally visible. Such regimes supply branch-local physical records, decoherent stability, and local reportability, but they do not by themselves supply a singleton standing evidential record for construction-relative reuse. The gap between physical record occurrence and standing evidential record use is exactly where admissibility, standing, reference, and irreversibility become load-bearing. The present manuscript therefore functions as a kernel-role consequence theorem: it asks what these roles force once the target is not merely branch-local record occurrence, but retained, prospective, deliberative, and comparative use of the same record-bearing construction token.

What the kernel roles do here. The table states the operational difference made by admissibility, standing, reference, and irreversibility in the measurement-record target.

Without the kernel roles	With the kernel roles active
Branch-local record stability can be mistaken for evidential-role fixation.	Physical realization and standing evidential use are separated.
Later success can be used ambiguously as if it repaired an earlier unfixed record-use act.	Irreversibility separates later support, new acts, and declared updates from same-act repair.
Global branch-state, branch-relative, or decision-theoretic structures can be treated as same-target solutions without preserving the same record object.	Reference and admissibility force target retyping to be declared.
Extra classifiers can hide as interpretation, book-keeping, branch labels, or record labels.	Same-scope closure forces terminal standing-role classification.
Boundary trace can be misread as a new physical posit.	It is identified as the forced residual admissible normal form after inadmissible routes are exhausted.
Reports can exceed their carrier, support, or fixation status.	Claim standing and report preservation block over-reporting.

Paired quantum determinacy arc

This manuscript is Paper I of the *AASC Quantum Determinacy Pair*. Its endpoint is record-role fixation: a physical record occurrence can function as a non-degenerate determinate evidential object only if its record-bearing role is fixed under the kernel roles. The companion Bell-scope manuscript is Paper II: it proves the no-signaling compatibility endpoint for Bell-witness determinacy, namely that a Bell witness set can function as a non-degenerate reusable object only when the entangled preparation is typed as a joint standing class carrying tensor-level compatibility and no same-domain signal, metric-transport, marginal-modulation, or distant-repair typing survives. Together, the manuscripts show that quantum record and witness determinacy are not free-standing physical occurrences; they are kernel-governed standing structures. The companion theorem is a paired result, not a premise of the present proof.[24]

Read this first: measurement-record proof spine

Reader orientation: practical success and target determinacy. This manuscript does not deny that branch-relative, instrumental, decision-theoretic, decoherence-based, or ordinary laboratory uses of quantum theory can be predictively and experimentally successful. The claim is narrower and *target-relative*. Such uses become non-degenerate realizations of the present target only if they preserve construction-relative measurement-record use: fixed record reference, standing-bearing eligibility, admissibility discipline, act-time finality, and lawful redescription across retained-record reuse, prospective-record reuse, deliberation, and evidential comparison. A view that refuses that non-degenerate record target adopts a different target practice; it is not thereby a same-mode counterexample. A view that accepts the target but weakens the kernel loses determinacy relative to the fixed target.

Why determinacy, not conventionality, is at issue. The manuscript is not motivated by the claim that this target yields better predictions. Prediction is a different burden. The present burden arises when a measurement record must function as the same evidential object across lawful redescription, retained-record reuse, prospective-record reuse, deliberation, and hypothesis comparison. At that point, the record must occupy a licensed evidential role whose sameness is governed rather than merely locally stable, branch-relative, or assignment-relative. The target is not an optional strengthening; it is what remains when the record is instantiated as a non-degenerate determinate evidential object.

Governed sameness and role occupancy. Scientific use of measurement records repeatedly depends on treating a later record as the same evidential object under admissible reuse. Memory-like retention, replication, comparison, deliberation, and future inference all presuppose that the record's evidential role remains fixed through lawful redescription and downstream use.

The central problem is therefore not merely whether records occur, or whether they are locally stable, but whether evidential-role occupancy is unique, lawful, and resistant to hidden selectors, assignment-relative ambiguity, and post-hoc repair. These governance conditions can be satisfied implicitly through rigorous experimental and theoretical practice even when the explicit structural vocabulary used in this manuscript is not adopted.

Technical use of “degenerate”. The term “degenerate” is used here in its fixed-domain technical sense. A regime is degenerate relative to the present target iff it fails to preserve the conditions required for construction-relative measurement-record use: fixed reference, standing-bearing eligibility, admissibility discipline, act-time finality, and lawful same-domain redescription. Degeneracy does not mean empirical failure, predictive uselessness, scientific irrationality, or incoherence for every purpose. It means failure to instantiate this target as non-degenerate construction. Thus a branch-relative, instrumental, decision-theoretic, global-branch, or prediction-only practice may be successful for its own target while still being degenerate relative to construction-relative evidential record reuse. The concession is not that degenerate same-target practice is acceptable. The concession is that other targets may be legitimate. Once the present target is fixed, degeneracy remains bivalent and fail-closed.

The live proof spine is intentionally short:

$$\text{RecordUse}_{\text{MR}}(d) \Rightarrow \text{NonDegRecordConstr}_{\text{MR}}(d) \Rightarrow K_{\text{MR}}(d),$$

$$K_{\text{MR}}(d) \Rightarrow \text{AMetric no-selector discipline}$$

$$\Rightarrow \text{no hidden branch-weight, environment, detector, centered-index, or decision selector.}$$

$$\text{PhysNonSel}_{\text{Everett}} \Rightarrow \text{multiple stable branch-local records} \\ \text{without singleton role-fixation.}$$

$$\begin{array}{l} \text{no singleton admissible fixation} \\ \text{and no standing-equivalence collapse} \end{array} \Rightarrow \text{ARef}/\approx_{\text{MR}} \text{ remains non-collapsed,}$$

$$\text{ARef}/\approx_{\text{MR}} \text{ non-collapsed} \Rightarrow \text{record use is assignment-relative rather than construction-relative.}$$

Therefore, in a physically non-selective quantum regime, stable branch-local records are not sufficient for theory-level evidential use. Measurement records require admissible record-fixation on a fixed domain.

Target-refusal boundary. A branch-relative or construction-deflationary Everettian may refuse this target practice. That is a coherent exit from the conditional, not a counterexample to it. The manuscript concerns theories assigned the determinate target of supporting theory-level retained-record reuse, prospective-record reuse, deliberation, and evidential comparison through measurement records. A different target may be legitimate; it is not a non-degenerate realization of the same target.

Implicit satisfaction. The manuscript does not require working physicists, experimental protocols, or users of measurement records to speak AASC. Successful construction-relative practice may satisfy reference, standing, admissibility, irreversibility, selector control, and repair discipline implicitly. The formalism makes explicit the governance conditions already required when record use succeeds as non-degenerate construction-relative evidence. It audits scientific practice; it does not legislate scientific vocabulary.

Vocabulary shortcut. For readability, “fixed-domain ordering of constructed hypotheses under admissible record reuse” is often called *fixed-domain evidential ordering*. The full phrase remains available when a theorem statement needs the complete constructional role.

Object	What it gives	What it does not give
Branch-local record	Local stability and within-branch reportability.	Singleton standing for the pre-measurement construction token.
Evidential anchor	Admissible record-role fixation for downstream reuse.	Physical outcome production or collapse.
Theory-level evidential output	Construction-relative comparison between hypotheses.	Assignment-relative branch bookkeeping.

Example 0.1 (Minimal witness). Take two decohered branches, b_U and b_D , with stable branch-local records r_U and r_D . Let a single pre-measurement construction token d be used as “the record I will use as evidence.” One admissible assignment sends d to (b_U, r_U) and another sends d to (b_D, r_D) . If the first record favors H_U over H_D and the second favors H_D over H_U , the same physical branching episode supports opposed evidential orderings unless a singleton admissible fixation asymmetry, or an explicit restriction collapsing the admissible assignments to one standing-equivalence class, is supplied. Without that fixation or collapse, the record-use output is assignment-relative rather than construction-relative.

Branch-set evidence boundary. A global branch-set, measure-valued, or weight-indexed evidential object may be introduced as a different target construction. It is not a same-domain solution to singular record-role fixation unless it preserves the retained-record, prospective-record, deliberative, and evidential role of the original construction token. If the target is retyped as branch-set evidence, the manuscript classifies that move as a change of target practice rather than as a refutation of the fixed-domain record-fixation theorem. Retyping may be coherent; it is not non-degenerate realization of the original record-role target.

Claim-class, proof-class, and formalization lock

Claim-class lock. This manuscript is a kernel-forced theorem of determinate measurement objecthood with an explicit finite quantum occupation. It is not a collapse theory, hidden-variable theory, decision-theoretic derivation, Born-rule derivation, empirical alteration of Hilbert-space quantum mechanics, or general metaphysical ranking of interpretations. It proves that once a measurement record is required to function as a standing physical/evidential object for downstream reuse, its record-bearing role must be fixed by admissible same-domain structure.

Proof-class lock. The proof class combines general AASC necessity, fixed-domain quotient/exhaustion analysis, and finite formalized physical specialization. A right-mode objection must target a named theorem link: objecthood-to-kernel necessity, measurement-record specialization, assignment-space forcing, quotient uniqueness, Everettian non-selection, boundary-trace exhaustion, finite weight/record/operation compatibility, finite selector-exclusion, weakening resistance, diachronic inheritance, or no-fifth-case selector exhaustion.

Formalization lock. The mechanized finite development is integrated into the manuscript argument. Its mathematical role is to occupy the kernel-governed measurement-record target with an explicit two-branch quantum domain and to verify, over declared finite carriers, the agreement of record, amplitude, density, instrument, environment, channel, effect, and repeatability surfaces. The public v0.9.1 formalization is the controlling artifact for this finite development. It includes the zero-effect finite effect carrier, zero-product projector orthogonality, branch-map annihilation, compatibility, closed-stack, measurement-boundary, and audit-manifest endpoints. The manuscript package does not vendor the full Lean source tree; reproducibility is routed through the public repository and release archive.

Integrated theorem node	Role in this manuscript
Universal AASC kernel	Necessity conditions of non-degenerate determinate objecthood.
Measurement-record specialization	Fixed-domain admissible construction of a standing record object.
Finite quantum occupation	Explicit two-branch carrier with amplitudes, density tables, instruments, environment records, branch maps, effects, projectors, and repeatability operations.
Cross-surface compatibility	Equality of fine records, coarse records, and branch weights across the declared physical surfaces.
Finite selector boundary	Exhaustion of state, instrument, environment, channel, effect, and repeatability surfaces as hidden fine-record selectors.

Integrated endpoint. The central result is not a general argument merely accompanied by a Lean audit. It is a kernel-forced theorem of determinate measurement objecthood with an explicit machine-checked finite quantum occupation and selector-boundary closure. The finite theorem remains finite; the physical consequence is that branch differentiation, weighting, decoherence-style coherence removal, effect completeness, zero-product orthogonality, and repeatability do not by themselves supply admissible fine-record role fixation.

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1 Introduction

The measurement problem is usually introduced as a tension among unitary dynamics, the superposition principle, and the apparent definiteness of observed outcomes. On that framing, the central question is how a linear, non-collapsing state description yields a world in which observers report definite results. Collapse theories answer by changing dynamics, hidden-variable theories by adding ontological structure, Everettian theories by retaining unitary dynamics while reinterpreting branching, and instrumentalist or QBist views by weakening the descriptive burden placed on the theory. The debate is then conducted primarily as a dispute over outcome production.¹

This manuscript does not treat representation, record-fixation, or record-use processing as primitive explanatory mechanisms. It analyzes admissible construction on a fixed domain. Quantum-foundational material enters because determinate quantum states, outcomes, carriers, records, and transformations are already kernel-governed objects when they are asserted as non-degenerate physical objects. The manuscript therefore does not offer a philosophical reinterpretation of measurement language. It proves a kernel-role determinacy theorem: once physical record occurrence is claimed to stand as a non-degenerate determinate object, admissibility, standing, reference, and irreversibility govern the record's determinacy.

That question is prior to any dispute over outcome production. Outcome reports enter scientific use through measurement records. Those records are not treated as linguistic items first and physical items second. They are fixed-domain constructed objects reused in stabilization, comparison, retained-record reuse, prospective-record reuse, and deliberative guidance. If a theory underwrites that reuse as a non-degenerate determinate physical/evidential target, it must fix the records' standing-bearing role. Before asking how one outcome is produced, one must ask how an admissible record-bearing continuation is fixed for later same-domain reuse.

Core distinction. Physical realization means that multiple branch-local records exist and remain stable under the physical dynamics. Evidential-role fixation means that one record is fixed as the standing evidential anchor for same-domain admissible downstream reuse. The first does not imply the second: a physically realized record can be locally determinate while its evidential role remains unfixed on the target construction.

Constraint formalism, not constructive dynamics. The present analysis belongs to a constraint formalism whose object is ontic non-degenerate determinacy. The kernel roles do not generate or physically explain the records, correlations, or branch-local contents to which they apply. They constrain when such physical contents may count as standing-bearing objects under identity-preserving reuse. What survives after inadmissible routes are closed is therefore a forced admissible determinacy normal form, not a newly introduced collapse process, hidden selector, or branch-selection mechanism.

Why this target matters. Prediction alone does not require the full target analyzed here. Branch-local reporting alone does not require it. Decision-theoretic expectation over future branch-relative

¹For standard landmarks see Einstein et al. [1], Bell [2], Zurek [4], Schlosshauer [5], and Wallace [10].

records does not require it by itself. The present target arises when the same measurement record is required to function as a standing evidential anchor across retained-record reuse, prospective-record reuse, deliberative guidance, and evidential comparison of constructed hypotheses. At that point, the record is not merely a local report, a calculational input, or a branch-relative inscription. It is being used as a construction-relative object whose identity and admissible reuse must be preserved. The target is not chosen because the formalism requires a heavy object; it is the determinate object fixed by the work assigned to measurement records when they are treated as reusable evidential anchors on a fixed domain. In this sense, the measurement-record target functions as a kernel-role consequence theorem: it shows what admissibility, standing, reference, and irreversibility force when a physical record occurrence is claimed as non-degenerate record objecthood rather than merely branch-local content.

Finite formal occupation. The main text now includes the finite quantum construction rather than relegating it to an external companion note. The two-branch carrier, denominator-two amplitude weights, density table, instrument, environment record map, branch maps, effects, zero-product projectors, and repeatability maps are used to prove that the declared physical surfaces agree on branch and weight content while no non-selective surface selects a fine record. This finite theorem is the concrete physical occupation of the same kernel-role structure, not a detachable illustration.

Prediction is easier than evidential reuse. A predictive theory can answer which statistics, frequencies, or branch-relative reports are expected without thereby fixing which record is the same evidential object across reuse. A prediction may succeed even when the later record's construction-relative role is not fixed; evidential reuse requires the additional fact that the record remains the same standing evidential object across admissible downstream use. Construction-relative science carries the additional burden of record identity: what exactly is this record, what role does it occupy, and why may it be reused as the same evidential anchor under lawful redescription? The manuscript concerns that second burden.

Governed sameness and role occupancy. Scientific use of measurement records repeatedly depends on treating a later record as the same evidential object under admissible reuse. Memory-like retention, replication, comparison, deliberation, and future inference all presuppose that the record's evidential role remains fixed through lawful redescription and downstream use. In physically non-selective regimes, multiplicity of stable local records makes this sameness nontrivial. The central problem is therefore not merely whether records occur, or whether they are locally stable, but whether evidential-role occupancy is unique, lawful, and resistant to hidden selectors, assignment-relative ambiguity, and post-hoc repair. These governance conditions can be satisfied implicitly through rigorous experimental and theoretical practice even when the explicit structural vocabulary used in this manuscript is not adopted.

The paired Bell-scope result is not a later appendix to this argument. The present manuscript proves the record-determinacy side: non-degenerate evidential record objecthood requires admissible record-role fixation. The companion manuscript *Bell-Nonlocal Quantum Entanglement as Boundary-Level Compatibility: A Kernel-Role Determinacy Theorem for Bell-Witness Fixation, No-Signaling, and Tensor-Level Compatibility* proves the witness-compatibility side: fixed Bell witness reuse

requires declared setting/outcome and detector-channel interfaces, entangled preparations are typed as joint standing classes carrying tensor-level compatibility, and no same-domain signal, metric-transport, marginal-modulation, or distant-repair typing survives [24]. Together, the two manuscripts locate record determinacy and Bell-witness no-signaling compatibility as kernel-governed standing structures.

The issue becomes acute in physically non-selective theories. In Everettian quantum mechanics, for example, unitary dynamics and decoherence can stabilize multiple post-measurement branch-histories, each containing a determinate branch-local observer-state and a determinate branch-local record, without providing a physical selector that privileges one such history as the fixed downstream continuation for pre-measurement construction. Branch-local determinacy is therefore not yet fixed-domain diachronic determinacy. The issue is not whether post-branch observers have determinate local states. The issue is whether the theory fixes which future record-bearing construction stands as the continuation relevant to pre-measurement constructions of the form “I will observe o ”, “I remember observing o ”, or “this record favors H_1 over H_2 ”. Those familiar expressions are retained only as external bridge language. The load-bearing work of the manuscript concerns admissible record-fixation, not representation.

The necessity claim is conditional only in the precise sense in which any necessity theorem for a target is conditional: the target must be instantiated. In the present case,

$$\text{NDRecTarget}_{\text{MR}}(d) \Rightarrow K_{\text{MR}}(d),$$

and contrapositively,

$$\neg K_{\text{MR}}(d) \Rightarrow \text{loss, refusal, or retyping of NDRecTarget}_{\text{MR}}(d).$$

This is not conditional on adopting AASC as an optional framework; it is conditional only on the target being instantiated as a non-degenerate determinate record object at all. Once that target is present, the kernel is upstream of the physical description rather than an interpretation added after it.

The argument is conditional but exact. The analysis works throughout after local activation of the imported fixed-domain closure package: once construction-relative evidential record use triggers the measurement-record kernel, the cited closure-spine consequences govern the evaluated fragment. Under that activation order, the manuscript does not stipulate an assignment space and then prove theorems about it. It shows that same-domain reuse of measurement records forces admissible record-fixation assignments as the unique constructional typing of that burden. The central mathematical construction is then a *measurement-record quotient theorem*: any admissibility-preserving presentation of the standing-bearing constructional content of measurement-record practice factors uniquely through a quotient of admissible assignments by standing-equivalence. Differences that do not survive that quotient are representational skin. Differences that do survive it are standing-relevant. The next hardening step is an *exact standing-content equivalence-collapse theorem*: any same-scope quotient relation that claims to carry exactly and only standing-bearing content collapses to that same quotient. In particular, divergence that changes comparative ordering role for constructed hypotheses is not a harmless repackaging of one content; it is quotient-relevant divergence.

Once that quotient structure is in place, the rest of the theorem ladder tightens around one

bridge. First, quotient uniqueness and exact standing-content equivalence collapse show that same-scope standing-bearing content is fixed by the measurement-record quotient rather than by a discretionary presentation. Second, fixation commutation is identified as the bridge invariant: if lawful identity-preserving transformation does not preserve the relevant quotient class, then the theory fails to preserve its own admissible evidential target. Third, the bare Everettian package is shown to be physically non-selective and therefore not to supply that bridge by dynamics alone. Fourth, fixed-domain evidential ordering is undefined whenever fixation fails. Fifth, retained-record reuse, prospective-record reuse, deliberation, and comparison inherit the same dependence because admissible diachronic construction factors through the same quotient-level lineage. Finally, interpretation packages are exhausted by the structural locus at which singular record-fixation is done, if it is done at all; no fifth same-scope locus remains.

Two restrictions on scope matter throughout. First, the manuscript is not a general representation of quantum language. It concerns a narrower target practice: admissible diachronic construction in which measurement records function as evidential anchors for downstream reuse. Second, the manuscript does not re-derive the downstream closure package on which it works. That package is imported explicitly and cited where needed; the present manuscript uses it to sharpen the measurement argument rather than to reproduce the imported closure spine.

The resulting diagnosis is different from the standard presentation of the measurement problem. The first failure is not yet that the theory does not produce a unique outcome. The first failure is that, in a physically non-selective setting, the theory may leave underfixed the admissible construction of the very records through which outcomes are reported and used as evidence. On that fixed domain, the measurement problem is therefore a problem of admissible record construction before it is a problem of outcome production.

Central thesis. Let T be a physically non-selective theory used to underwrite admissible diachronic construction about measurement and downstream reuse. Then admissible record-fixation assignments for measurement records must either collapse into a single standing-equivalence class on that fixed domain or else make downstream use assignment-relative rather than construction-relative. Bare Everettian dynamics, entanglement, decoherence, and branch-local record stability do not by themselves collapse the admissible class. Accordingly, the measurement problem is a problem of fixed-domain record-fixation, not merely a problem of physical outcome production.

The proof order is as follows. Section 2 identifies the imported fixed-domain closure package, proves its local activation for measurement-record use, and fixes the exact standing conditional. Section 3 defines admissible record-fixation assignments in gate-governed fixed-domain form, proves the measurement-record quotient theorem, proves the exact standing-content equivalence-collapse theorem that blocks the charge that the quotient was stipulated rather than forced, and then promotes fixation commutation to the bridge invariant between quotient structure and evidential use. Section 5 proves the Everettian non-selection results as the physical trigger for the bridge problem. Section 7 proves the fixed-domain evidential ordering criterion and the resulting stabilization consequences. Section 8 imports the boundary-trace fixation primitive, proves measurement-record asymmetry exhaustion, and closes the bare Everettian asymmetry classes. Section 10 exhibits a worked reversal witness. Section 11 proves the diachronic closure and inheritance theorems

for downstream reuse across retained-record reuse, prospective-record reuse, deliberation, and comparison. Section 13 derives selector exhaustion from structural constraints before stating the comparative interpretation taxonomy. Section 14 opens with a strict denial table so that standard objections are tied to exact labeled failures. Section 15 marks the manuscript's precise limits. Appendix A supplies a compact formal appendix for the quotient presentation, Appendix B states the exact companion closure package on which the main body is conditional, and Appendix C records the dependency and objection maps.

2 Fixed-Domain Measurement-Record Use and the Local Kernel

The present manuscript is downstream of the general kernel theorem, but the kernel is not imported as optional vocabulary. The measurement-record target itself forces the local roles once it is instantiated as a non-degenerate determinate record object. If records are required to stand, be reused, preserve identity, and guide retained-record reuse, prospective-record reuse, deliberation, and evidential comparison, the practice is already a non-degenerate fixed-domain construction practice. AASC names and audits the roles already required by that practice.

Theorem 2.1 (Non-degenerate constructional objecthood activation). *Let x be asserted as a non-degenerate determinate object within a fixed domain of reference, preservation, comparison, transformation, or endpoint use. Then*

$$\text{NDObj}(x) \Rightarrow \text{Adm}(x) \wedge \text{Stand}_{\text{obj}}(x) \wedge \text{Ref}(x) \wedge \text{Irr}(x).$$

Reference is forced because a determinate object must be referable as the same target rather than as an arbitrary surrogate. Admissibility is forced because there must be a boundary between transformations that preserve the object and transformations that abandon, replace, or retype it. Local object-standing, $\text{Stand}_{\text{obj}}$, is forced because the object must be available for theorem-bearing, comparison-bearing, or reuse-bearing employment at the scope on which it is asserted as the same object rather than remaining a mere inscription. Irreversibility is forced because determinate endpoint status cannot be retroactively repaired as the same act after the act has failed to fix its object.

Proof. If any one of the four roles is absent, the asserted object loses the feature by which it is being treated as non-degenerate and determinate. Without reference there is no same target; without admissibility there is no preservation boundary; without local object-standing there is no object eligible for stable comparison, transformation, or reuse at the asserted scope; without irreversibility there is no fixed endpoint rather than revisable same-act repair. Hence non-degenerate determinate constructional objecthood entails the kernel roles at that scope. $\square$

The theorem does not claim that every informal mention of an object has already discharged every downstream measurement-record burden. It claims that once an entity is asserted as a non-degenerate determinate object within a fixed domain of reference, preservation, comparison, transformation, or endpoint use, the kernel roles are already active at that scope. The stronger measurement-record packet is a specialization for record-bearing physical/evidential reuse:

$$\text{EvidentialRecordUse}(x) \Rightarrow \text{Stand}_{\text{MR}}(x),$$

where Stand_{MR} is measurement-record standing across retained use, prospective use, deliberation, and evidential comparison. Thus branch-local physical determinacy is not kernel-free; it simply carries $\text{Stand}_{\text{obj}}$ at its local physical scope rather than automatically discharging the stronger Stand_{MR} burden. This activation lemma is a denial-cost result rather than a stipulative relabeling: removing any kernel role destroys the asserted sameness, preservation boundary, theorem-bearing availability, or endpoint finality required for non-degenerate constructional use.

Proof-class and claim-class lock. This manuscript is a fixed-domain constructional theorem about admissible measurement-record use. It is not a collapse theory, hidden-variable theory, Born-rule derivation, empirical modification of Hilbert-space quantum mechanics, or ranking of all interpretations by total metaphysical virtue. Its claim is narrower: if a physically non-selective quantum theory is used to underwrite measurement records as standing evidential anchors, then those records must be fixed by admissible same-domain role-fixation. Stable branch-local records are not thereby theory-level evidential anchors.

Constraint-formalism scope lock. The kernel and its closure results function here as constraints on admissible construction, not as the starting point of a dynamical measurement theory. The manuscript does not choose nature’s mechanism for producing branch-local contents. It supplies the determinacy constraint any such mechanism must satisfy if those contents are claimed as standing-bearing record objects. The task is therefore ontic and eliminative: given a non-degenerate measurement-record target and the activated kernel roles, what record-construction forms remain admissible on the fixed domain, and which familiar moves are ruled out?

Lean 4 finite specialization boundary. The mechanized finite development is part of the manuscript’s mathematical argument: it gives the two-branch quantum occupation of the measurement-record kernel and verifies record, weight, instrument, environment, channel, effect, and repeatability compatibility over declared carriers. The public v0.9.1 release is the controlling formal artifact for the zero-effect and zero-product projector layer. The mechanized development is finite and scope-declared; it does not reconstruct Hilbert-space quantum mechanics, general POVM/PVM theory, physical decoherence, the continuum Born rule, physical selection, interpretive truth, or physical no-collapse [21–23].

2.1 Bare record content and governed record use

The first distinction is between kernel-governed branch-local physical determinacy and the stronger kernel-governed role of a record as the same standing evidential object across retained reuse, prospective reuse, deliberation, and comparison. Physical branch-local record content is not kernel-free: if it is asserted as determinate physical content, it already instantiates admissibility, standing, reference, and irreversibility at its local physical scope. What it does not yet instantiate is the stronger cross-use measurement-record packet required for theory-level evidential record objecthood.

Definition 2.2 (Bare branch-local record content). For a branch or post-measurement history b_i with record r_i , *bare branch-local record content*, written schematically as $\text{BranchLocalRecord}(b_i, r_i)$, means that r_i is physically realized, locally stable, or reportable within b_i . It does not assert theory-level evidential standing, cross-branch singularity, retained reuse, prospective reuse, deliberative force, or fixed-domain evidential ordering.

Definition 2.3 (Ordinary branch-relative record report). An *ordinary branch-relative record report* is a report such as “observer-successor O_i records o_i ” or “branch b_i contains r_i .” Such a report may be physically true, decoherence-stable, and scientifically useful within its role while still not being the standing record-role realization of a pre-measurement construction token.

Definition 2.4 (Theory-level evidential record use). For a construction token $d \in \mathcal{D}$, *theory-level evidential record use*, written $\text{EvidentialRecordUse}_{\text{MR}}(d)$, is the use of a physically realized record as the same evidential object across retained-record reuse, prospective-record reuse, deliberation, or evidential comparison. This use fixes target, record role, outcome role, evidential role, downstream reuse, reportability, act-time status, and lawful-redescription profile.

Definition 2.5 (Local record-counterexample use). A candidate branch-local record has *local record-counterexample use* for d only when it is opened inside the governed analysis as the live candidate standing realization of d . Physical occurrence alone is not local record-counterexample use.

Theorem 2.6 (Bare branch-local record content is not governed record use). *For any branch-record pair (b_i, r_i) and construction token d ,*

$$\text{BranchLocalRecord}(b_i, r_i) \not\Rightarrow \text{RecordUse}_{\text{MR}}(d).$$

Branch-local record existence, stability, and reportability are therefore preserved as lawful physical or branch-relative content without being promoted into theory-level evidential standing.

Proof. Bare branch-local record content states that a record exists or is reportable inside a particular branch. It does not fix the record as the same evidential object across retained reuse, prospective reuse, deliberation, or comparison. Nor does it fix act-time finality or lawful same-domain redescription for the record-bearing role. Thus the record may be physically realized and locally determinate while the governed record-use role for d remains unfixed. $\square$

Theorem 2.7 (Theory-level evidential record use is governed record use). *For every construction token d ,*

$$\text{EvidentialRecordUse}_{\text{MR}}(d) \Rightarrow \text{RecordUse}_{\text{MR}}(d).$$

Proof. Theory-level evidential record use is not the mere presence of a mark. It treats a record as the same standing evidential object across retained reuse, prospective reuse, deliberation, or comparison. That use fixes the relevant record target, role, reportability, downstream continuation, act-time status, and lawful-redescription profile. Those are precisely the features of record use under audit. $\square$

Theorem 2.8 (Governed record use satisfies target adequacy). *If $\text{RecordUse}_{\text{MR}}(d)$ is theory-level governed record use in the target practice, then it satisfies the kernel-neutral adequacy packet of Definition 2.22:*

$$\text{RecordUse}_{\text{MR}}(d) \Rightarrow \text{TargetAdeq}_{\text{MR}}(d).$$

Proof. Governed record use fixes the same record role and evidential target under lawful redescription, which is target determinacy. It treats the act as eligible or ineligible for retained reuse, prospective reuse, deliberation, and comparison, which is step evaluability. It does not let later decoherence, calibration, posterior success, or future data repair an unfixed record-role as the same act, which is act-time finality. It preserves record identity, outcome role, evidential role, downstream reuse, and admissibility classification across lawful same-scope redescription, which is same-regime fidelity. $\square$

Corollary 2.9 (No autonomous evidential-record role). *If a branch-local record is not governed record use for d , then it cannot carry the same construction-relative evidential role for d :*

$$\neg \text{RecordUse}_{\text{MR}}(d) \Rightarrow \neg \text{EvidentialRecordUse}_{\text{MR}}(d).$$

It may remain branch-local content, support, prediction input, instrumental update, or a different target practice; it is not a non-degenerate realization of the original evidential-record role.

Proof. By Theorem 2.7, theory-level evidential record use implies governed record use. The displayed statement is its contrapositive, with the lawful lower-status roles made explicit. $\square$

2.2 Physical occurrence and ontic record determinacy

Definition 2.10 (Physical record occurrence). *A physical record occurrence*, written schematically as $\text{PhysRecordOcc}_{\text{MR}}(r)$, is a branch-local, detector-local, environment-local, memory-local, or apparatus-local physical state that may be stable, decohered, correlated with a prior interaction, and locally reportable. Physical occurrence is lawful physical content. As determinate physical content it is kernel-governed at its local physical scope, but it is not yet standing-bearing record objecthood for a construction token.

Definition 2.11 (Ontic non-degenerate record determinacy). For a construction token d and physical record occurrence r , write $\text{NDRecDet}_{\text{MR}}(d, r)$ when r functions as the same record-bearing object for d under stable target identity, standing-bearing evidential use, admissible reuse, lawful redescription, no same-act repair, and quotient-level record-role stability. This is ontic non-degenerate record determinacy: not mere report language, and not raw physical occurrence alone, but standing-bearing record objecthood on the fixed domain.

Proposition 2.12 (Physical occurrence is not yet ontic record determinacy). *For any physical record occurrence r and construction token d ,*

$$\text{PhysRecordOcc}_{\text{MR}}(r) \not\Rightarrow \text{NDRecDet}_{\text{MR}}(d, r).$$

A record may occur physically and remain locally stable while its record-bearing role for d remains unfixed.

Proof. Physical occurrence supplies branch-local or apparatus-local content. It does not by itself fix the construction token for which the record stands, the admissible record role, the record's evidential standing, the quotient class of its role-bearing assignment, or the lawful redescription class preserving that role. Those are determinacy conditions, not mere occurrence conditions. Therefore physical occurrence alone does not imply ontic non-degenerate record determinacy. $\square$

Theorem 2.13 (Ontic record determinacy forces kernel-governed fixation). *If a physical record occurrence r is used as a non-degenerate determinate evidential object for d , then its determinacy is governed by the local kernel and by record-role fixation:*

$$\text{NDRecDet}_{\text{MR}}(d, r) \Rightarrow K_{\text{MR}}(d) \Rightarrow (\text{Fix}_{\text{MR}}(d, r) \vee \text{FailClosed}_{\text{MR}}(d, r)).$$

If no admissible fixation is supplied, r may remain physical content, branch-local report, support, or global-state content, but it does not stand as the same evidential record object for d .

Proof. By Definition 2.11, ontic record determinacy requires stable reference to the same record-bearing object, standing-bearing evidential eligibility, admissible reuse, lawful redescription, and no same-act repair. These are precisely the local kernel roles named by K_{MR} . Once the kernel is active, a record-bearing role cannot remain unfixed while still carrying the same standing evidential use. The role must be fixed up to the measurement-record quotient, or the act fails closed relative to the non-degenerate target. Hence $\text{NDRecDet}_{\text{MR}}(d, r)$ forces kernel-governed fixation or fail-closed status. $\square$

Physical selection as a stronger fixation route. Physical selection is not degenerate. It is an admissible and stronger realization of record-role fixation when supplied by a theory. Collapse dynamics, ontic variables, detector-channel dynamics, or boundary-condition physics may occupy the dynamic-selection or ontic-privileging route if they fix the record-bearing role without hidden selector import, same-act repair, or target shift. What fails in physically non-selective regimes is not physicality. What fails is the attempt to treat plural branch-local physical records, without any physical or constructional fixation of record role, as a single standing evidential record for the same construction token.

$$\text{PhysSel}_{\text{MR}}(d, r) \Rightarrow \text{Fix}_{\text{MR}}(d, r),$$

while a physically non-selective theory supplies no such physical selector unless it is retyped into a selective or enriched theory.

Level	Endpoint	Status
Physical occurrence	Branch-local detector, memory, apparatus, or environment record exists.	Lawful physical content; not yet standing evidential object.
Ontic non-degenerate determinacy	Physical occurrence stands as the same record-bearing object.	Kernel roles active.
Record-role fixation	Record-bearing role fixed up to $\approx_{\text{MR}}$.	Required by non-degenerate determinacy.
Physical selection	A physical mechanism selects or fixes the record.	Stronger admissible realization when supplied.
Boundary-trace fixation	Declared stable trace fixes role without physical branch selection.	Residual admissible determinacy normal form.
Target refusal/retyping	Branch-relative, global-state, self-locating, or instrumental target replaces record objecthood.	Coherent different target, not a same-target solution.

2.3 Non-degenerate target instantiation

Definition 2.14 (Non-degenerate measurement-record target). For a construction token d , write $\text{NDRecTarget}_{\text{MR}}(d)$ when the measurement-record target is instantiated as a determinate same-record evidential object. This means that the target preserves:

1. the same record-bearing target under lawful redescription;
2. evaluability as carrying or failing evidential standing;
3. act-time distinction between admitted continuation and later support, update, or repair;
4. non-assignment-relative record-role identity across incompatible candidate realizations.

A practice that lacks one of these conditions may remain coherent for prediction, branch-relative reporting, global-state evidence, self-location, decision policy, or instrumental update. It is not, without further bridge work, a non-degenerate realization of the same construction-relative record target.

Theorem 2.15 (Non-degenerate target instantiation forces the local kernel). *For every construction token d ,*

$$\text{NDRecTarget}_{\text{MR}}(d) \Rightarrow K_{\text{MR}}(d).$$

Equivalently, denying the local kernel while claiming to preserve the same record target entails loss, refusal, or retyping of the non-degenerate target:

$$\neg K_{\text{MR}}(d) \Rightarrow \neg \text{NDRecTarget}_{\text{MR}}(d).$$

Proof. If $\text{NDRecTarget}_{\text{MR}}(d)$ holds, the record target must preserve fixed record identity, evidential standing, admissible entry, lawful redescription, and act-time finality. These are exactly the local roles named reference, standing, admissibility, and irreversibility. If any of those roles is removed,

the practice may continue as a different target, but the original construction-relative same-record evidential object is no longer instantiated as non-degenerate. $\square$

Identity-condition note. The adequacy requirements below are not strengthened evidential ideals. They are the identity conditions under which the object under analysis remains a non-degenerate measurement-record target. A practice may decline them by adopting branch-relative evidence, global-state evidence, decision-theoretic policy, instrumental update, or another target. Such a practice is not thereby incoherent; it no longer preserves the same construction-relative evidential record object.

Kernel role	Measurement-record activation
Reference	Fixes the same construction token, record-bearing target, candidate branch-record role, and lawful redescription class.
Standing	Fixes whether a record may function as evidential anchor for retained reuse, prospective reuse, deliberation, and comparison.
Admissibility	Separates record-role fixation from support-only dynamics, bookkeeping, hidden selector import, carrier shift, and target retyping.
Irreversibility	Blocks later decoherence, memory success, calibration, or report success from retroactively repairing the same unfixed act.

2.4 Target explanatory practice and fixed scope

Definition 2.16 (Target explanatory practice). The *target explanatory practice* of this manuscript is the practice in which measurement records function as evidential anchors for theory-level construction tokens about what was observed, what follows for retained-record reuse and prospective-record reuse, how deliberation is guided, and how those records bear on rival hypotheses.

Definition 2.17 (Fixed scope of the manuscript). The fixed scope of the manuscript is the same-scope, construction-committing use of a physically non-selective measurement theory within the target explanatory practice of Definition 2.16. Moves that import new ontology, new dynamics, new admissibility-relevant structure, new boundary selectors, new target gates, or new record-use roles count as scope changes rather than same-scope repairs.

Target-refusal boundary. A branch-relative Everettian, QBist, instrumentalist, or other deflationary strategy may refuse the target explanatory practice. Such a refusal is not a same-mode counterexample to the theorem ladder; it changes the burden from theory-level record reuse to a branch-relative or deflationary practice. The arguments below apply only once the construction-committing target of Definition 2.16 is instantiated as a non-degenerate determinate record target.

Burden	Typical question	Governance needed	Status in this manuscript
Predictive / laboratory coordination	What outcomes, frequencies, or statistics should be expected?	Stable calculations, empirical outputs, and operational record handling.	Granted as legitimate; not the full target.
Branch-relative record use	What record exists locally within this branch, sector, or relative state?	Local stability, decoherence, and branch-internal reportability.	Granted as physically important; not yet theory-level evidential-role fixation.
Construction-relative evidential reuse	What is the same evidential object across retained reuse, prospective reuse, deliberation, and comparison?	Fixed reference, standing, admissibility, irreversibility, and lawful redescription.	Target of the theorem; failure appears as assignment-relativity, hidden selector import, post-hoc repair, or loss of evidential role occupancy.

Cost of refusing construction-relative records. A branch-relative, global-branch, instrumental, QBist, or decision-theoretic practice may refuse the construction-relative target, but that refusal has a precise cost. It must retype or give up the same-target role of pre-measurement claims such as “I will observe o ,” retained-record claims such as “I remember observing o ,” use of one record as the same evidential object in hypothesis comparison, deliberative reliance on future record-bearing continuations, cross-context reuse of the same measurement record, and ordinary report stability under lawful redescription. Those may be coherent costs for a different burden, but they are not same-mode solutions to the singular construction-relative record-use target. Kernel necessity, report legitimacy, role decomposition, same-scope closure, and singleton-interior closure are the imported closure disciplines activated when that target is retained [16–20].

The manuscript’s use of degeneracy applies to the third burden. It does not imply that the first two burdens are scientifically worthless. It says that practical prediction and branch-local reportability do not by themselves certify non-degenerate construction-relative evidential reuse.

Proposition 2.18 (Different target, different work). *A framework that treats measurement records only as branch-relative reports, instrumental update devices, decision-theoretic inputs, or global branch-state descriptors may be coherent and predictively successful. But unless it also fixes the same-domain record-bearing role required for retained-record reuse, prospective-record reuse, deliberation, and fixed-domain evidential ordering, it does not perform the same explanatory job as the target*

practice of this manuscript.

Proof. The target practice requires construction-relative reuse of measurement records. A branch-relative or instrumental practice may supply prediction, local reportability, or agent-relative update without fixing the record-bearing role as a standing evidential anchor on the same domain. Such a practice is not refuted; it is classified as a different target. If it nevertheless claims to preserve the present target, then failure to fix the record-bearing role is degeneracy relative to that target. $\square$

Dependency inversion exclusion. The dependency order is not

AASC as optional overlay $\Rightarrow$ measurement-record constraints.

The dependency order is

measurement records as standing evidential anchors
 $\Rightarrow$ non-degenerate fixed-domain record construction
 $\Rightarrow K_{MR}$ already active.

AASC supplies the audit vocabulary for roles forced by the target practice. It does not impose those roles from outside quantum theory.

Closure-package activation order. The fixed-domain closure package is not the first premise of the measurement-record argument. The first premise is governed record use. Once a record is used as the same standing evidential object under retained reuse, prospective reuse, deliberation, and comparison, Theorems 2.7–2.25 activate the local kernel. The package below is the closure package that becomes locally active after that kernel-triggering use has occurred.

2.5 Imported fixed-domain closure package

Definition 2.19 (Fixed-domain closure package activated below). The *fixed-domain closure package* is the imported AASC closure spine whose activation conditions are verified below for the measurement-record target. It is not an additional local premise and not a definition chosen to settle the measurement problem in advance. Once activated, it supplies the following fixed-scope consequences.

1. Admissibility is a forced governance boundary for the target class of non-degenerate, constructed-object-bearing systems.
2. Standing is bivalent on the evaluated fragment and fail-closed when evaluation is not instantiated.
3. Same-scope admissibility classification is unique: there is no second admissibility predicate, no admissibility-relevant boundary selector, and no admissibility-relevant parameterization on the fixed scope.
4. The admissible interior is unique up to lawful equivalence.

5. Standing is conserved: there is no same-scope repair, no generator, no primitive carrier, and no same-scope transfer without fresh target gating.
6. Any second same-scope invariant either collapses to bookkeeping or constitutes a scope change.

Remark 2.20 (Provenance of the closure package). Definition 2.19 records the local measurement-record form of the five-source closure spine identified in the introduction: kernel necessity, claim-standing and report legitimacy, anchor/tensor/skin routing, same-scope operator closure, and singleton admissible-interior uniqueness [16–20]. The present section proves only the local activation claim: construction-relative evidential record use satisfies the conditions under which that imported package applies to the measurement-record target.

2.6 Vocabulary-neutral adequacy after non-degenerate target instantiation

Definition 2.21 (Measurement-record use under audit). For a measurement construction token d , *record use under audit*, written schematically as $\text{RecordUse}_{MR}(d)$, means theory-level use of a physically realized record as a standing candidate for retained-record reuse, prospective-record reuse, deliberation, or evidential comparison within the fixed scope of Definition 2.17. The expression is target-fixing; it does not assert that the relevant record role is already fixed.

Definition 2.22 (Vocabulary-neutral adequacy for non-degenerate measurement-record targets). A raw measurement-record practice is adequate for the target explanatory practice only if the following four requirements hold before a record is used as theory-level evidential content. The neutrality claimed here is not neutrality among every possible epistemic target. It is neutrality with respect to AASC vocabulary after the measurement-record target has been instantiated as a non-degenerate determinate object of evidential use.

(A1) Target determinacy.

The same measurement record role, outcome role, evidential target, candidate record set, hypothesis-comparison role, and downstream reuse practice are fixed across lawful same-scope redescription.

(A2) Step evaluability.

A record-use act is evaluable as eligible or ineligible for retained-record reuse, prospective-record reuse, deliberation, and evidential comparison, rather than merely occurring as a branch-local mark, report, or representation.

(A3) Act-time finality.

Later decoherence success, calibration, environmental stabilization, branch-relative report, posterior success, or future data cannot repair an unfixed record-role as the same admissible act.

(A4) Same-regime fidelity.

Lawful redescription preserves record identity, outcome role, evidential role, downstream reuse profile, act identity, and admissibility classification. Failure of such preservation is domain shift, not same-record continuation.

Remark 2.23 (Later support is not same-act repair). Act-time finality does not deny later branch-relative success, decoherence, memory retention, calibration, environmental stabilization, or theory

update. Such later material may supply support, admissible continuation, or a new act. The prohibited move is narrower: later material may not retroactively confer standing on the same earlier construction-relative record-use act while preserving that act as numerically the same act. If the later material changes the standing basis, the result is a new act, a declared update scope, a support relation, or target retyping.

Definition 2.24 (Local measurement-record kernel). The *local measurement-record kernel* for a construction token d is

$$K_{MR}(d) = \{\text{Adm}_d, \text{St}_d, \text{Ref}_d, \text{Irr}_d\},$$

where Adm_d is admissibility of record-use entry, St_d is standing for licensed downstream reuse, Ref_d is reference to the same record/outcome/evidential target, and Irr_d is irreversibility of failed or unfixed record-role status at act-time.

Theorem 2.25 (Measurement-record adequacy forces the local kernel). *For every construction token d , if $\text{RecordUse}_{MR}(d)$ is adequate for the target explanatory practice in the sense of Definition 2.22, then $K_{MR}(d)$ is already in force.*

Proof. Target determinacy requires a role preserving the same record, outcome, evidential target, and reuse profile across lawful redescription. That is Ref_d . Step evaluability requires a role by which record use can be eligible or ineligible for licensed downstream continuation. That is St_d . Act-time finality requires a role preventing later branch-relative success, calibration, environmental stabilization, or posterior report from repairing an earlier unfixed record-role as the same act. That is Irr_d . Same-regime fidelity requires the boundary between lawful same-record continuation and selector import, domain shift, report overreach, or repair. That is Adm_d . Therefore the measurement-record adequacy requirements force the local kernel before AASC vocabulary is used downstream. $\square$

Corollary 2.26 (No measurement-record kernel opt-out). *A critic cannot preserve measurement records as standing evidential anchors for retained reuse, prospective reuse, deliberation, and evidential comparison while rejecting the local kernel roles. To opt out of K_{MR} , the critic must weaken at least one of target determinacy, step evaluability, act-time finality, or same-regime fidelity.*

Same-mode denial fork. A same-mode denial must preserve the same target and do one of the following: deny that retained reuse, prospective reuse, deliberation, and comparison require same-record role identity; deny that same-record role identity requires reference; deny that evidential use requires standing; deny that record-use eligibility requires admissibility; deny that later repair changes the act; supply a same-domain record-use practice preserving all four kernel roles while omitting one of them; or show that branch-local, global-branch, or instrumental evidence preserves the same singular construction token rather than retyping the target. Otherwise the response is vocabulary refusal, target refusal, support-only practice, or a different evidential object.

Theorem 2.27 (No cost-free target retyping). *If a proposed solution replaces singular construction-relative record use for d by a branch-relative, global branch-set, self-locating, decision-theoretic, observer-stage-relative, or instrumental evidence object E' , then one of two terminal roles is forced:*

either E' preserves non-degenerate same-record evidential objecthood, in which case K_{MR} and the quotient framework apply; or E' does not preserve that objecthood, in which case it changes the target practice and is not a same-mode solution to the original record-role problem.

Proof. If E' preserves the original construction token, retained reuse, prospective reuse, deliberative role, and evidential comparison role, then it preserves the identity conditions of $\text{NDRecTarget}_{MR}(d)$ and must identify the same standing record-bearing realization required by the target practice. Then the record-use-to-kernel and quotient theorems apply. If E' does not preserve those roles, it may be a coherent branch-relative, global, instrumental, or decision-theoretic practice, but it is a different target. There is no third route on which a view rejects the kernel while retaining the same non-degenerate record target. $\square$

Theorem 2.28 (Target refusal is not kernel failure). *If a theory refuses construction-relative evidential record use and instead adopts branch-relative, global-state, self-locating, decision-theoretic, or instrumental evidence, then it has not produced a kernel-free realization of the same target. It has declined or retyped the target. Such refusal may be coherent for a different target, but it does not falsify*

$$\text{NDRecTarget}_{MR}(d) \Rightarrow K_{MR}(d).$$

Proof. The displayed implication is a necessity claim for the non-degenerate target. A refusal of that target removes the antecedent; it does not exhibit a non-degenerate same-target realization without the kernel. To refute the theorem, a rival must preserve the same record-bearing construction token, standing-bearing reuse, act-time finality, and lawful redescription while omitting the kernel roles. A branch-relative, global-state, self-locating, decision-theoretic, or instrumental replacement that does not preserve those conditions is a target retyping, not a kernel-free counterexample. $\square$

2.7 Local measurement-record kernel packet

The following packet is the local measurement-record form of the fixed-domain kernel. It does not ban physical multiplicity, branch-local determinacy, environmental interaction, detector stability, or ordinary probabilistic practice. It only blocks their promotion into same-domain record-role authority unless they preserve the kernel conditions.

Table 1: Local measurement-record kernel packet.

Node	General necessity	Measurement-record form
KMR1	Fixed-domain kernel	Evidential record use requires fixed record target, standing, admissibility boundary, and act-time irreversibility.
KMR2	Reference fixation	A record-use act cannot silently change outcome, branch, observer-stage, evidential target, or hypothesis-comparison role.

Node	General necessity	Measurement-record form
KMR3	Physical-record / evidential-anchor separation	A branch-local record may be physically stable without being the standing evidential anchor.
KMR4	Bivalence and fail-closed status	A record-use act is admitted or not admitted for standing-bearing downstream use; unfixed status is not a third evidential status.
KMR5	Unique admissible interior	There cannot be plural same-scope admissible interiors for the same record-fixation problem.
KMR6	Standing-admissibility identity	A record cannot carry evidential standing while inadmissibly fixed.
KMR7	No generators	Branch weights, labels, reports, memories, or decision policies cannot generate record standing for themselves.
KMR8	No carriers	Decoherence, environment, detector states, memories, amplitudes, or branch-local reports cannot carry evidential role-fixation without a lawful bridge.
KMR9	No post-selection repair	Later calibration, posterior success, future data, or branch-relative report cannot repair an unfixed record-role as the same act.
KMR10	AMetric no-selector boundary	No branch weight, environmental salience, centered index, detector preference, probability ranking, or caring measure decides boundary standing.
KMR11	Report preservation	A record report may not exceed its target, carrier, fixation, and quotient support.
KMR12	ATS skin non-authority	Labels, centered descriptions, branch names, diagrams, and representational packages do not become standing-bearing by being written.
KMR13	Record-role exhaustion	Once the record target, standing packet, reuse role, and admissibility boundary are fixed, record-use statuses are exhausted by admitted fixation, support/bookkeeping, scope change, or forbidden second classifier.

Node	Local activation in this manuscript	Closure source activated
KMR1–KMR3	Governed evidential record use fixes target, reference, and the physical-record/evidential-anchor distinction.	Local record-use-to-kernel bridge plus kernel necessity [16].
KMR4–KMR6	The target reaches bivalent/fail-closed standing and standing-admissibility identity once the local kernel is active.	Kernel plus claim-standing and singleton-interior closure [17, 18].
KMR7–KMR10	Generators, surrogate carriers, post-selection repair, and AMetric selectors are excluded only after the local kernel activates the fixed-domain closure spine.	Kernel non-derivability and same-scope closure [16, 20].
KMR11–KMR12	Report preservation and skin non-authority route record claims and presentations without letting reports or labels generate standing.	Claim-standing/report legitimacy and ATS [17, 19].
KMR13	Record-role exhaustion is the measurement-record specialization of same-scope no-fifth-route closure.	Same-scope operator closure plus singleton-interior closure [18, 20].

Derivation ledger. The expanded KMR packet is not asserted to follow from A1–A4 alone. A1–A4 activate the local kernel; the cited closure spine supplies the fixed-domain consequences listed in KMR4–KMR13. This is the import boundary for the table.

Theorem 2.29 (Local measurement-record packet). *For every adequate measurement-record use under audit, KMR1–KMR13 in Table 1 are in force on the fixed record-use domain.*

Proof. By Theorem 2.25, adequate record use instantiates reference, standing, admissibility, and irreversibility. That local activation supplies KMR1–KMR3 directly. The remaining packet nodes are fixed-domain closure consequences activated at the measurement-record target: KMR4–KMR6 use fail-closed standing, standing-admissibility identity, and singleton-interior closure; KMR7–KMR10 use non-derivability, no surrogate carrier, no same-act repair, and AMetric no-selector discipline; KMR11–KMR12 use claim-standing/report preservation and ATS skin discipline; and KMR13 uses same-scope no-fifth-route exhaustion. Therefore the packet is local in application but not presented as a fresh derivation of the whole closure spine from A1–A4 alone. $\square$

2.8 Cost of denying the measurement-record kernel

The following table is not a criticism of every use of quantum theory that lacks explicit AASC vocabulary. It is a same-target denial table. It records what is lost if one keeps the construction-relative record-use target while denying the structural role listed in the left column. The table should be read against the governed-sameness burden: each denial below either breaks

sameness of the record, removes its standing role, imports hidden role-selection authority, or permits repair after the fact. Practical success does not by itself establish non-degeneracy for this target. Non-degeneracy requires fixed reference, standing, admissibility, irreversibility, and lawful same-domain redescription for record reuse.

Table 2: Cost of denying the local measurement-record kernel.

Denied role	Cost in measurement-record terms
Reference	The record no longer fixes the same outcome, record-bearing construction, evidential target, or hypothesis-comparison role across lawful redescription.
Standing	The record may be physically present but is not eligible for retained-record reuse, prospective-record reuse, deliberation, or evidential comparison.
Admissibility	There is no governed distinction between lawful record-fixation and selector import, branch salience, environment appeal, decision-weight substitution, or scope reset.
Irreversibility	Later decoherence, calibration, observer report, posterior evidential success, or future data may retroactively repair an unfixed record-role as the same admissible act.
Bivalence / fail-closed status	Unfixed, deferred, branch-relative, weighted, environmentally suggested, or pragmatically selected status becomes a third standing-bearing record status.
Standing-admissibility identity	A record may carry evidential standing while inadmissibly fixed, or admissibility may perform record-use work without standing.
No second classifier	A branch weight, centered-world index, environmental marker, detector salience, or decision-theoretic policy may govern the same record-fixation problem beside the declared admissibility gate.

Refusal route	May preserve	Does not preserve as the same target unless further bridge is supplied
Branch-relative evidence	Local branch record stability and branch-internal reportability.	Non-assignment-relative same-record evidential objecthood for the original construction token.
Global branch-state evidence	Full unitary-state or measure-valued evidential information.	Singular record-bearing realization of the construction token.
Decision-theoretic evidence	Rational action guidance, credence policy, or caring-measure use.	Record fixation as a standing evidential object rather than action-guidance input.
Self-locating uncertainty	Centered successor credence practice.	Non-centered same-record target identity.
Instrumental update	Prediction and update success.	Theorem-bearing record standing for construction-relative reuse.
Decoherence-only account	Effective classicality and stable local records.	Singleton standing fixation for cross-context evidential reuse.

These routes are not declared incoherent. They are not same-target counterexamples unless they preserve the non-degenerate record object whose kernel conditions are under analysis.

Theorem 2.30 (Kernel-denial cost for measurement-record use). *Any denial of K_{MR} while preserving the target explanatory practice abandons at least one of record reference, record standing, admissible same-domain discipline, act-time finality, or lawful redescription. Therefore kernel denial is not a harmless refusal of terminology; it exits the target practice or imports forbidden selector authority.*

Proof. The target practice requires fixed records to remain the same evidential anchors through downstream reuse. By Theorem 2.25, that requirement forces the local kernel. If a denial leaves all four adequacy requirements intact, then the kernel roles remain in force under renaming. If it rejects reference, the record target drifts. If it rejects standing, the record is not eligible for reuse. If it rejects admissibility, selector import and scope reset are no longer separated from lawful fixation. If it rejects irreversibility, later material can repair an unfixed record-role as the same act. If it rejects lawful redescription, the practice no longer has same-domain record use. These are exactly the costs listed in Table 2. $\square$

Remark 2.31 (Tacit satisfaction is enough). The kernel roles need not be explicitly represented by working physicists, laboratory protocols, observers, or users of measurement records. A practice may rely on stable target identity, record eligibility, report discipline, and no-retroactive-repair without naming reference, standing, admissibility, or irreversibility. The theorem is therefore structural rather than sociological: practitioners need not name the kernel roles for their practice to depend on them. What matters is whether the practice preserves fixed reference, standing-bearing eligibility, admissibility discipline, act-time finality, and lawful redescription when records are used as construction-relative evidential anchors. A proof-assistant user need not explicitly theorize

admissibility for proof-checking to rely on it. Likewise, a laboratory practice need not name the record-use kernel for its construction-relative evidential use to depend on it.

Remark 2.32 (Scientific practice already attempts this). Careful experimental and theoretical practice already attempts to secure stable reference, reproducibility, controlled comparison, calibration discipline, invariance under admissible changes of presentation, and limits on post-hoc repair. The present framework does not replace those norms. It identifies the minimal governance required when those norms are treated as construction-relative record use rather than as merely predictive or branch-local procedure.

2.9 Weakening-resistance of the local record kernel

Definition 2.33 (Core measurement-record adequacy packet). The core measurement-record adequacy packet for a construction token d is

$$C_0^{MR}(d) = \{\text{Ref}_d, \text{Stand}_d, \text{Adm}_d, \text{Irr}_d\},$$

together with fixed record target, record-use evaluability, retained-record reuse, prospective-record reuse, deliberative guidance, evidential comparison, act-time finality, and same-regime fidelity.

Definition 2.34 (Record-governance-permissive weakening). A *record-governance-permissive weakening* is a proposed same-domain regime W that claims to preserve the target explanatory practice while weakening at least one local consequence of the measurement-record kernel. More precisely, W preserves the same fixed measurement-record domain, candidate record set, record-use target, retained-record reuse, prospective-record reuse, deliberative guidance, evidential comparison, lawful redescription class, minimal report evaluability, and act-time finality. It then permits at least one of the following to carry theory-level evidential use:

1. an unfixed record-role;
2. multiple admissible record-fixation classes producing one theory-level evidential output;
3. branch-weighted, pointer-stabilized, detector-salient, or environment-suggested status without declared singleton fixation;
4. later calibration, posterior success, or future data as repair of a prior unfixed record-role.

The definition records a *purported preservation claim*. The theorem below shows that no such claim can preserve the non-degenerate same-domain target while remaining strict and governance-effective.

Theorem 2.35 (No strict same-domain weakening permits evidentially usable unfixed record roles). *No record-governance-permissive weakening W can be both strict and governance-effective on the same fixed measurement-record domain. Every such W falls into one of the following cases:*

1. ***governance-equivalent preservation***: *it performs the same kernel work under another description;*
2. ***support or bookkeeping***: *it describes branch-local, diagnostic, probabilistic, or evidential-support material without fixing record standing;*

3. **domain or target shift:** *it changes the explanatory practice, ontology, dynamics, candidate record set, or record target;*
4. **second same-domain classifier:** *it introduces a forbidden selector, gate, or admissibility-relevant parameter beside the declared record-fixation gate;*
5. **same-act repair:** *it retroactively validates an unfixed record-role as the same admissible act.*

Proof. Let W preserve the target practice while weakening some local kernel consequence. If W changes no standing-bearing record-use verdict, it is support or bookkeeping. If it preserves all standing-bearing record-use verdicts, lawful continuations, and reports, it is governance-equivalent for the target. If it changes the record target, candidate set, ontology, dynamics, or explanatory practice, it is a domain or target shift. If it changes record-use status on the same fixed domain without collapsing to the kernel interface, then the change must be carried by a branch weight, environment appeal, centered index, detector salience, decision policy, calibration channel, or comparable classifier. KMR10 and KMR13 exclude that as second same-domain record-status authority unless it is declared as a lawful singleton role-fixing trace. If the alleged weakening lets later material validate an earlier unfixed role, KMR9 and irreversibility classify it as same-act repair. These cases exhaust the ways a strict same-domain weakening could affect evidential record use. Hence no such weakening is both strict and governance-effective. $\square$

Good enough is not a third status. “Good enough for prediction,” “stable enough for branch-local report,” or “usable enough for decision” are not third admissibility statuses for the present target. They may identify legitimate support or different target practices. But once the target is construction-relative record reuse, the record-role is either admitted for standing-bearing use or not admitted. Anything else is support, bookkeeping, target shift, or forbidden second classifier.

Corollary 2.36 (Kernel weakening consequence). *If the measurement-record kernel is weakened, one of five costs follows: physical records lose evidential standing, record use becomes assignment-relative, hidden selector authority is imported, same-act repair is permitted, or the target practice is deflated. There is no harmless weakening that preserves retained-record reuse, prospective-record reuse, deliberation, and evidential comparison while denying admissibility, standing, reference, irreversibility, AMetric no-selector discipline, and standing conservation.*

2.10 Local activation of the imported fixed-domain closure package

This subsection is an import-activation result, not a reproof of the whole AASC closure spine. The kernel and closure package are constraint-theoretic: they classify admissible record construction after non-degenerate target instantiation; they do not generate a new physical mechanism for records, correlations, or branch selection. The present manuscript proves that construction-relative evidential record use instantiates the local measurement-record kernel K_{MR} . The consequences concerning report preservation, anchor/tensor/skin routing, no independent same-domain classifier, standing conservation, same-scope closure, and singleton admissible-interior uniqueness are then activated from the cited AASC closure spine [16–20].

Definition 2.37 (Minimal coherence triggers). A same-scope construction-practice satisfies the *minimal coherence triggers* relevant to the present manuscript iff it preserves fixed record-fixation under lawful identity-preserving transformation, treats admissible continuation as compositionally available candidate material, carries non-trivial constructional content, and executes at least one irreversible commitment on the fixed scope.

Lemma 2.38 (Stable record-fixation is necessary for the target explanatory practice). *If measurement records are used as theory-level evidential anchors under lawful same-scope redescription, then lawful identity-preserving transformation must preserve what those records are about. Otherwise the same evidential construction token changes target under admissible use and no standing-bearing evidential role is fixed.*

Proof. A theory-level evidential construction is supposed to remain the same construction across lawful same-scope identity-preserving transformation. If lawful transformation changes what the construction is fixed to, then the evidential role of the record is presentation-dependent rather than construction-relative. On the fail-closed closure package recorded in Definition 2.19, that is not a weaker kind of ordering of constructed hypotheses under admissible record reuse; it is loss of determinate standing-bearing use. So stable record-fixation is necessary. $\square$

Lemma 2.39 (Identity quotient necessity). *If same-scope measurement-record construction is stable under lawful identity-preserving transformation, then admissible record-fixation assignments admit a standing-equivalence relation whose classes exhaust same-scope standing-bearing content for the target construction. Any same-scope standing-bearing presentation of that content therefore factors through a quotient of admissible assignments.*

Proof. Once lawful identity-preserving transformation preserves standing-bearing use, assignments that agree on every standing-bearing value in the target construction cannot be treated as distinct same-scope contents unless one introduces an additional same-scope invariant. But the fixed-domain closure package excludes any second same-scope invariant except as bookkeeping or scope change. Therefore standing-bearing content must collapse to equivalence classes of assignments, and any same-scope standing-bearing presentation must factor through the corresponding quotient. $\square$

Lemma 2.40 (Irreversibility forces bivalent gating). *If same-scope illicit evaluated acts cannot be repaired as the same acts, then evaluated acts must fall under a fail-closed eligible/ineligible gate on the evaluated fragment. No third standing-relevant status remains.*

Proof. Without a gate there is no stable distinction between eligible and ineligible evaluated acts, so irreversibility has nothing determinate to attach to. If a third standing-relevant status were admitted, then some acts would be neither fully eligible nor fully ineligible and illegitimacy could be softened or reclassified without a fresh gate. That is exactly the kind of same-act repair excluded by the irreversibility condition. Hence irreversibility forces bivalent fail-closed gating. $\square$

Lemma 2.41 (Bivalence plus the AMetric boundary excludes parameterization). *If the admissibility interface is bivalent on the evaluated fragment and the pre-admissible side is AMetric, then no same-scope admissibility-relevant parameter, ranking, or selector can refine the boundary.*

Proof. A same-scope admissibility-relevant parameter would introduce more than the two bivalent statuses of eligible and ineligible, or else it would provide a selector among allegedly admissible

alternatives. Either way the boundary would cease to be AMetric and non-parameterized. So bivalence together with the AMetric boundary excludes same-scope admissibility-relevant parameterization. $\square$

Theorem 2.42 (Measurement-record activation of the fixed-domain closure spine). *Let a same-scope construction-practice satisfy the minimal coherence triggers of Definition 2.37 and let it be used to underwrite the target explanatory practice of Definition 2.16. Then the local measurement-record kernel is active, and the imported fixed-domain closure spine applies to the measurement-record target: admissibility is forced, report preservation is in force, anchor/tensor/skin routing is fixed, same-scope admissibility classification is unique, the admissible interior is unique up to lawful equivalence, standing is conserved, and any second same-scope invariant collapses to bookkeeping or scope change.*

Proof. Lemma 2.38 forces stable record-fixation for the target explanatory practice. Lemma 2.39 then forces quotient-level standing-bearing content rather than representative-dependent content. Lemma 2.40 forces fail-closed bivalent gating on the evaluated fragment, and Lemma 2.41 excludes same-scope admissibility-relevant boundary parameterization or selection. These local lemmas verify that the measurement-record target satisfies the activation conditions for the imported closure spine. The remaining fixed-domain consequences are imported at this exact point: kernel necessity from Maley [16], claim-status and report legitimacy from Maley [17], anchor/tensor/skin routing from Maley [19], same-scope operator closure and no-fifth-route exhaustion from Maley [20], and singleton admissible-interior uniqueness from Maley [18]. Thus the closure package recorded in Definition 2.19 is not an optional overlay; it is the imported fixed-domain closure package locally activated by governed measurement-record use. $\square$

Proposition 2.43 (Immediate consequences of the closure package). *Within the activated fixed-domain closure package, any same-scope admissibility-bearing distinction about measurement-record construction must appear in one of only two places:*

1. *in the standing classification of evaluated acts, or*
2. *in the licensed continuation structure of the construction built on those acts.*

No third same-scope locus of admissibility-bearing work remains.

Proof. By Definition 2.19, same-scope admissibility classification is unique, standing is conserved, and no second same-scope invariant survives except as bookkeeping or scope change. Therefore any admissibility-bearing difference must either change which acts stand, or change what same-scope continuation is licensed for standing-bearing acts. If it changes neither, then by the same fixed-domain package it is representational or representational surplus rather than admissibility-bearing content. So no third locus remains. $\square$

Local theorem node	Local manuscript role	External closure invoked
Record-use-to-kernel theorem	Evidential record use fixes target, record role, standing-bearing reuse, admissibility boundary, reference, and irreversibility.	Kernel necessity [16].
Claim-status ledger	Record claims must not exceed carrier, support, fixation, quotient, or trace status.	Claim-standing/report legitimacy [17].
Record anchor/tensor/skin routing	Record identity, standing-bearing content, and presentation skin are separated on the same fixed domain.	ATS decomposition [19].
No fifth record-fixation route	Residual same-domain classifiers collapse into admitted fixation, support/bookkeeping, scope change, repair, or forbidden selector work.	Same-scope operator closure [20].
No plural record interior	Same-domain record use cannot split into plural admissible evidential interiors.	Singleton interior closure [18].

2.11 Physically non-selective theories and the record-fixation problem

Definition 2.44 (Physical non-selection). A theory is *physically non-selective* with respect to a measurement process if, after the measurement interaction, the theory’s physical description realizes multiple successor histories containing stable measurement records and physically continuous observer-successors, while supplying no physical law, state variable, or dynamically privileged trajectory that singles out one such successor history as physically distinguished from the others.

Definition 2.45 (Diachronic record-fixation problem). Given a physically non-selective theory, the *diachronic constructional selection problem* is the problem whether the theory, together with its admissible interpretive use on the activated fixed-domain closure package, determines which successor history is the standing realization of pre-measurement diachronic construction.

Definitions 2.44 and 2.45 separate two questions that are often run together. Physical non-selection states what the physical theory itself delivers: multiplicity without a physically distinguishing selector. The diachronic constructional selection problem is the further question whether that physical package, together with whatever admissible same-scope use it supports, determines which successor history is the standing realization of construction about what the pre-measurement subject will observe, remember, or use as evidence.

Everettian quantum mechanics is the paradigm case. Unitary dynamics and decoherence can stabilize multiple post-measurement branch-histories, each carrying a determinate branch-relative observer and a determinate branch-relative record, while leaving the physical theory itself non-selective among them. The issue is not whether the theory assigns numerical weights to those histories. The issue is whether the theory itself fixes which physically realized history is the standing realization of the measurement record construction of an internal subject.

2.12 Exact conditional and public vocabulary

The manuscript runs under one standing conditional.

Standing conditional. If a quantum theory is used to underwrite admissible diachronic construction about measurement, retained-record reuse, prospective-record reuse, deliberation, and evidential ordering, then admissible record-fixation assignments for measurement records must either

1. collapse into a single standing-bearing class on that construction, or
2. leave evidential ordering and diachronic practice assignment-relative rather than construction-relative.

This conditional is not a disguised collapse postulate. It says only that if a theory is assigned explanatory work involving records as evidence, then the standing realizations of those records cannot vary in same-scope standing-bearing ways while the theory still claims a single theory-level evidential output.

The public vocabulary below is retained only where it names familiar external targets. The governing ontology of the manuscript is fixed-domain admissible construction, and the listed terms are to be read constructionally rather than semantically.

- **Measurement record:** a physically realized outcome-bearing structure used as an evidential anchor.
- **Diachronic measurement construction:** construction linking a pre-measurement subject to post-measurement records by way of retained-record reuse, prospective-record reuse, deliberation, or evidential ordering.
- **Record-fixation assignment:** a map from diachronic measurement constructions to candidate branch-record standing realizations.
- **Admissible record-fixation assignment:** a same-scope record-fixation assignment eligible under the gate-governed fixed-domain constraints stated in Section 3.
- **Selector:** any principle that fixes singular record-fixation by privileging one continuation, history, or record as the standing realization relevant to the target construction.
- **Standing-bearing content:** content that survives into standing classification or licensed continuation on the fixed-domain closure package.
- **Skin:** representational difference that does not survive into standing-bearing content.

2.13 Scope of the present manuscript

Theorems in the present manuscript do not settle every metaphysical question in quantum foundations. Their function is narrower. On the fixed-domain closure package,

1. the manuscript identifies the measurement problem as a fixed-domain problem of record-fixation to measurement records,

2. proves a quotient uniqueness theorem for admissible measurement-record content,
3. shows that the bare Everettian package is physically non-selective with respect to that target construction,
4. proves that evidential-order divergence among admissible assignments is quotient-relevant and therefore not mere representational skin,
5. proves that fixed-domain evidential ordering requires fixation commutation under admissible redescription,
6. proves that retained-record reuse, prospective-record reuse, and deliberative guidance inherit the same lineage dependence, and
7. classifies interpretation packages by the structural locus at which singular record-fixation is fixed, if it is fixed at all.

The manuscript is therefore neither a general representation of quantum theory nor a re-derivation of the imported closure spine. It is a fixed-domain measurement-record manuscript built on, and locally reactivating, that spine.

3 Measurement-record quotient and minimal formal setting

This section fixes the minimum formal vocabulary required by the proof bridge. The key upgrade is a quotient theorem. The assignment space used below is not introduced as a discretionary representation of the problem. By Theorem 3.6, once one demands admissible same-domain reuse of measurement records on the fixed scope, admissible record-fixation assignments are forced as the unique constructional typing of that burden. Standing-bearing differences among those assignments are then separated from representational skin by an explicit quotient construction.

We work with a fixed measurement setting consisting of a pre-measurement subject-stage s_0 , a nonempty set $\mathcal{B}$ of physically realized post-measurement histories, a record map $\text{Rec} : \mathcal{B} \rightarrow \mathcal{R}$, a continuant map $\text{Cont} : \mathcal{B} \rightarrow \mathcal{C}$, a set $\mathcal{D}$ of diachronic measurement constructions, a set $\mathcal{H}$ of rival hypotheses, a set $\mathcal{E}$ of comparison-state for constructed hypotheses, and an evidential-order-role assignment

$$\text{CR} : \mathcal{E} \rightarrow (\mathcal{H} \times \mathcal{H} \rightarrow \{-1, 0, 1\}).$$

Here 1 means that the induced comparison-state for constructed hypotheses favors the first hypothesis over the second, -1 means that it favors the second over the first, and 0 means that it leaves their comparative ordering unchanged. This is only a minimal comparative scaffold; the manuscript does not presuppose a full evidential ordering theory.

Definition 3.1 (Measurement record). A *measurement record* is a physically realized structure that the target explanatory practice uses as an evidential anchor for constructed states about what outcome was observed.

A *diachronic measurement construction* is any constructed state whose admissibility conditions depend on a relation between the pre-measurement subject-stage s_0 and one or more post-measurement record-bearing histories. Typical examples are construction tokens of the form

“I will observe outcome o ,” “I remember observing outcome o ,” or “this record confirms H_i over H_j .”

For a construction token $d \in \mathcal{D}$, let $\mathbf{Cand}(d) \subseteq \mathcal{B} \times \mathcal{R}$ be the set of candidate branch-record pairs physically realized by the theory and eligible to serve as standing realizations for d .

Definition 3.2 (Boundary-trace admissibility relation). For a candidate record-role set $S_d \subseteq \mathbf{Cand}(d)$, a *boundary-trace admissibility relation* is a fixed-domain admissibility-preserving predicate

$$\mathrm{Tr}_{Ch}^d : S_d \rightarrow \{0, 1\}$$

whose valuation is declared before evidential use, stable under lawful channel-preserving identity transformations,

$$\sigma \in \mathrm{Aut}_{\mathrm{Ch}} \implies \mathrm{Tr}_{Ch}^d(\sigma x) = \mathrm{Tr}_{Ch}^d(x),$$

outcome-independent, and non-generative. It does not inspect the environment as an oracle and does not alter the candidate record set, report content, or induced outcome profile. It fixes a record-bearing role only when its positive set is a singleton.

Definition 3.3 (Record-fixation assignment). A *record-fixation assignment* is a rule

$$\rho : \mathcal{D} \rightarrow \mathcal{B} \times \mathcal{R}$$

that maps each diachronic measurement construction token $d \in \mathcal{D}$ to a candidate branch-record pair $\rho(d) \in \mathbf{Cand}(d)$.

Definition 3.4 (Admissible record-fixation assignment). A record-fixation assignment ρ is *admissible* on the fixed scope of Definition 2.17 iff it satisfies all of the following.

1. **Same-scope condition.** ρ constructs the tokens in $\mathcal{D}$ on the fixed scope and does not rely on scope reset, fresh target gating, or post hoc domain change.
2. **Ontology-compatibility.** For every $d \in \mathcal{D}$, $\rho(d)$ maps only to a physically realized candidate in $\mathbf{Cand}(d)$.
3. **Discourse-compatibility.** ρ is suitable to interpret the target explanatory practice, namely admissible downstream construction involving measurement, retained-record reuse, prospective-record reuse, deliberation, and evidential comparison.
4. **Non-question-begging use.** Admissibility of ρ is not defined by stipulating in advance the very evidential-order conclusion the later theorems are meant to establish.
5. **No second classifier / no boundary selector.** The admissibility of ρ does not depend on introducing a second same-scope admissibility predicate, a boundary selector, or an admissibility-relevant parameterization external to the fixed-domain closure package.
6. **Boundary-trace admissibility.** If the admissibility of ρ depends on detector, apparatus, channel, calibration, environment, branch-interface, or a predeclared record-location relation fixed by the measurement-domain architecture rather than by later outcome salience, that information must enter through a declared boundary-trace admissibility relation in the sense of

Definition 3.2. Otherwise it is an extra-domain selector, a scalarized parameter, delayed repair, or scope reset.

The set of all admissible record-fixation assignments for the fixed setting is written $\mathbf{ARef}$.

Clauses (5) and (6) are the crucial strengthening. They prevent the assignment space from being widened by a same-scope selector principle, narrowed by a second admissibility classifier, or silently stabilized by undeclared boundary information whose only role would be to manufacture uniqueness.

The assignment space is therefore not introduced as a modeling convenience. By the local kernel packet (Theorem 2.29), record-use must preserve reference, standing, admissibility, and irreversibility. Any same-domain realization of retained-record reuse, prospective-record reuse, deliberation, or evidential comparison must specify which candidate branch-record pair bears the standing role for each construction token, or else fail closed.

Definition 3.5 (Singular record-role target). For a fixed measurement setting,

$$\text{SingRecRoleTarget}_{\text{MR}}(\mathcal{D}, \mathcal{B}, \mathcal{R})$$

means that the construction tokens in $\mathcal{D}$ are assigned the burden of having standing branch-record realizations for downstream reuse on the fixed scope. The predicate does not assert which candidate is fixed. It states the target burden: each token must have a record-bearing role whose standing can be reused, anticipated, deliberately used, or compared as evidence without changing domain.

Theorem 3.6 (Assignment-space forcing theorem (constructional form)). *Work on the fixed scope of Definition 2.17 and under the standing conditional of Section 2. Let the target constructional burden be $\text{SingRecRoleTarget}_{\text{MR}}(\mathcal{D}, \mathcal{B}, \mathcal{R})$: supporting admissible diachronic reuse of measurement records for evidential comparison, retained-record reuse, prospective-record reuse, and deliberative guidance on that same fixed domain.*

Then any admissible same-domain realization of that burden induces, up to lawful same-scope equivalence, a canonical record-fixation assignment structure

$$\rho : \mathcal{D} \rightarrow \mathcal{B} \times \mathcal{R}$$

such that for every construction token $d \in \mathcal{D}$:

1. $\rho(d) \in \mathbf{Cand}(d)$ *is a physically realized branch-record pair eligible to serve as the standing realization of d on the fixed scope;*
2. *the standing-bearing value of d is determined only through such eligible branch-record realization;*
3. *any two admissible same-domain realizations of the target burden that agree on the induced assignment structure are lawfully equivalent with respect to standing-bearing content; and*
4. *any purported same-domain realization of the target burden that does not induce such an assignment structure either*
 - (a) *fails to fix a standing realization for some $d \in \mathcal{D}$,*
 - (b) *imports an additional same-scope selector, classifier, or admissibility-relevant parameterization,*

- (c) *changes the target constructional burden, or*
- (d) *changes domain.*

Equivalently: the admissible assignment space used in this section is not an optional modeling layer. It is the unique constructional typing forced by admissible same-domain reuse of measurement records on the fixed scope.

Non-optional use. Admissible constructional use is not optional once the explanatory burden is imposed on a fixed domain; any theory assigned that burden must satisfy the constraints derived here.

Meta-closure. Any purported same-domain formulation that avoids assignment structure must either fail to support admissible reuse or collapse extensionally to the assignment representation established here.

Proof. The target constructional burden is not merely to list physically realized post-measurement histories. It is to support admissible diachronic reuse of measurement records on a fixed domain. That burden requires, for each construction token $d \in \mathcal{D}$, a determination of which physically realized branch-record pair is to function as the standing realization of d for same-domain use.

If no such determination is available for some d , then the target burden is not met: the construction fails to stabilize the record-bearing continuation needed for evidential comparison, retained-record reuse, prospective-record reuse, or deliberative guidance. So every admissible realization of the burden must determine, for each d , at least one eligible

$$\rho(d) \in \text{Cand}(d) \subseteq \mathcal{B} \times \mathcal{R}.$$

This gives a map

$$\rho : \mathcal{D} \rightarrow \mathcal{B} \times \mathcal{R}.$$

It remains to show that this assignment structure is forced rather than optional. Suppose there were a same-domain realization of the target burden not equivalent to such an assignment structure. Then one of four things must occur.

First, some construction token d would lack any fixed eligible branch-record realization. But then the target burden would fail for that token, since no standing-bearing reuse of d would be fixed on the same scope.

Second, the realization could distinguish admissible uses without passage through eligible branch-record realization by introducing a further selector, classifier, or admissibility-relevant parameter. But the fixed-domain closure package excludes a second same-scope admissibility classifier, excludes admissibility-relevant parameterization, and excludes any second same-scope invariant except as bookkeeping or scope change. More specifically, if the proposed assignment-free realization appeals to branch weight, environmental salience, detector preference, centered index, or decision policy, it must either supply a declared singleton role-fixing asymmetry or become forbidden second classifier authority under KMR10 and KMR13. So that route is unavailable on the fixed scope.

Third, the purported realization could preserve same-domain admissibility while changing what counts as the relevant reuse burden. But then it would no longer realize the target constructional burden fixed in Definition 2.16; it would be a different constructional task.

Fourth, the purported realization could alter the domain itself by changing the admissibility boundary, the standing conditions, the lawful transformation class, or the target gating. That is a domain change rather than an alternative same-domain realization.

These cases are exhaustive. Therefore every admissible same-domain realization of the target burden induces, up to lawful same-scope equivalence, a canonical record-fixation assignment structure of the stated form. The assignment space used below is thus forced by the fixed-domain constructional burden rather than introduced by convenience. $\square$

Corollary 3.7 (No assignment-free same-domain realization). *There is no admissible same-domain realization of the target constructional burden that is both standing-bearing and assignment-free. Any purported assignment-free realization either fails to fix standing-bearing record continuation, introduces forbidden same-scope structure, or changes scope.*

Proof. Immediate from Theorem 3.6. $\square$

Theorem 3.8 (Branch-set evidence is target retyping unless it preserves singular record-role use). *Let G be a global branch-set, measure-valued, weight-indexed, or decoherent-histories evidential object introduced for the same measurement episode. If G is used instead of a singular standing branch-record realization for a construction token d , then one of two terminal roles is forced: either G supplies an admissible projection to a singleton standing realization for d , in which case the record-fixation assignment framework applies; or G is a different evidential target rather than a same-domain realization of the original singular record-role target. A global branch-state evidence object may be coherent for that different explanatory practice; it fails only as a same-mode solution to the original singular record-role target.*

Proof. If G preserves the original retained-record, prospective-record, deliberative, and comparative role of d , then it must determine which record-bearing role stands for d under same-domain reuse. That determination is precisely the assignment structure forced by Theorem 3.6. If G does not determine such a role, it may still be a coherent global or measure-valued evidential object, but it no longer realizes the original singular record-role target. It is therefore target retyping, not a same-domain repair. $\square$

3.1 Standing-bearing profiles of admissible assignments

For $\rho \in \mathbf{ARef}$ and $d \in \mathcal{D}$, let $\text{Prop}_\rho(d) \in \mathcal{E}$ denote the induced comparison-state for constructed hypotheses induced by taking $\rho(d)$ as the standing realization of d whenever d is used evidentially. More generally, let

$$\text{Val}_\rho(d)$$

be the standing-bearing construction value of d under ρ : for ordinary admissible diachronic construction tokens it is the induced truth value, and for evidential construction tokens it is the induced comparative ordering role for constructed hypotheses. The target-construction profile of ρ is

$$\text{Prof}(\rho) := (\text{Val}_\rho(d))_{d \in \mathcal{D}}.$$

Definition 3.9 (Evidential-order-role equivalence). Two admissible record-fixation assignments $\rho, \sigma \in \mathbf{ARef}$ are *evidential-orderly equivalent*, written $\rho \approx_{\text{role}} \sigma$, if for every evidential construction

token $d \in \mathcal{D}$,

$$\text{CR}(\text{Prop}_\rho(d)) = \text{CR}(\text{Prop}_\sigma(d)).$$

Equivalently, ρ and σ are evidential-orderly equivalent exactly when every theory-level evidential construction licensed under one assignment has the same comparative ordering role for constructed hypotheses under the other.

Lemma 3.10 (Evidential-order-role equivalence is an equivalence relation). *The relation $\approx_{\text{role}}$ on ARef is reflexive, symmetric, and transitive.*

Proof. Reflexivity holds because for every admissible assignment ρ and every evidential construction token d , we have $\text{CR}(\text{Prop}_\rho(d)) = \text{CR}(\text{Prop}_\rho(d))$. Symmetry is immediate because equality of comparative ordering role for constructed hypotheses is symmetric. Transitivity holds because if $\text{CR}(\text{Prop}_\rho(d)) = \text{CR}(\text{Prop}_\sigma(d))$ and $\text{CR}(\text{Prop}_\sigma(d)) = \text{CR}(\text{Prop}_\tau(d))$ for every relevant d , then $\text{CR}(\text{Prop}_\rho(d)) = \text{CR}(\text{Prop}_\tau(d))$ for every such d . $\square$

Definition 3.11 (Measurement-record standing equivalence). Two admissible record-fixation assignments $\rho, \sigma \in \text{ARef}$ are *standing-equivalent for the target practice*, written

$$\rho \approx_{\text{MR}} \sigma,$$

iff

$$\text{Prof}(\rho) = \text{Prof}(\sigma).$$

Thus $\rho \approx_{\text{MR}} \sigma$ exactly when the two assignments agree on every standing-bearing value of every construction token in the target explanatory practice.

Definition 3.12 (Skin-equivalent versus standing-relevant divergence). A difference between admissible record-fixation assignments is *representational skin* iff the assignments lie in the same $\approx_{\text{MR}}$ -class. It is *standing-relevant divergence* iff they lie in distinct $\approx_{\text{MR}}$ -classes.

3.2 Presentations of measurement-record content

Definition 3.13 (Measurement-record presentation). A *measurement-record presentation* is a triple (X, r, Val_X) where:

1. X is a set intended to serve as the standing-bearing content space for admissible record-fixation assignments on the fixed scope;
2. $r : \text{ARef} \rightarrow X$ is a surjection; and
3. for every construction token $d \in \mathcal{D}$ there exists a map

$$\text{Val}_X^d : X \rightarrow \mathcal{V}_d$$

for some codomain $\mathcal{V}_d$ such that

$$\text{Val}_\rho(d) = \text{Val}_X^d(r(\rho)) \quad (\rho \in \text{ARef}).$$

The presentation is *sound* if $\rho \approx_{\text{MR}} \sigma$ implies $r(\rho) = r(\sigma)$, and *complete* if $r(\rho) = r(\sigma)$ implies $\rho \approx_{\text{MR}} \sigma$.

Lemma 3.14 (Completeness of measurement-record presentations). *Every measurement-record presentation is complete.*

Proof. Assume $r(\rho) = r(\sigma)$. Then for every $d \in \mathcal{D}$,

$$\text{Val}_\rho(d) = \text{Val}_X^d(r(\rho)) = \text{Val}_X^d(r(\sigma)) = \text{Val}_\sigma(d).$$

Hence $\text{Prof}(\rho) = \text{Prof}(\sigma)$ and therefore $\rho \approx_{\text{MR}} \sigma$. $\square$

Lemma 3.15 (Soundness forced by stable record-fixation and fixed-domain uniqueness). *Within the fixed-domain closure package, every measurement-record presentation is sound.*

Proof. Assume $\rho \approx_{\text{MR}} \sigma$ but $r(\rho) \neq r(\sigma)$. Then the presentation distinguishes two putative same-scope standing-bearing contents that agree on every value in the target explanatory practice. By Definition 2.19 and Proposition 2.43, any same-scope admissibility-bearing distinction must survive either into standing classification or into licensed continuation. But $\rho \approx_{\text{MR}} \sigma$ means that the assignments agree on the full target-practice profile: there is no difference in standing-bearing use for retained-record reuse, prospective-record reuse, deliberation, or comparison. Treating $r(\rho)$ and $r(\sigma)$ as distinct therefore introduces an unwitnessed same-scope distinction. Within the fixed-domain closure package, such a distinction is either bookkeeping only or a scope change. Since X is presented as a standing-bearing same-scope content space, only bookkeeping remains. Hence $r(\rho) = r(\sigma)$ is forced. $\square$

Lemma 3.16 (Quotient universal property). *Let $q : \text{ARef} \rightarrow \text{ARef} / \approx_{\text{MR}}$ be the quotient map. If $F : \text{ARef} \rightarrow Y$ is constant on $\approx_{\text{MR}}$ -classes, then there exists a unique map $\tilde{F} : \text{ARef} / \approx_{\text{MR}} \rightarrow Y$ such that*

$$F = \tilde{F} \circ q.$$

Proof. Define $\tilde{F}([\rho]) := F(\rho)$. This is well-defined because F is constant on $\approx_{\text{MR}}$ -classes. Uniqueness follows because q is surjective. $\square$

By Theorem 3.6, the space ARef is not an optional presentation layer but the forced same-domain realization space for admissible measurement-record continuation on the fixed scope.

Theorem 3.17 (Measurement-record quotient theorem). *Let (X, r, Val_X) be any measurement-record presentation on the fixed-domain closure package. Then r is automatically sound and complete and induces a unique bijection*

$$\phi : \text{ARef} / \approx_{\text{MR}} \longrightarrow X, \quad \phi([\rho]) = r(\rho).$$

In particular, the quotient $\text{ARef} / \approx_{\text{MR}}$ is the unique standing-bearing content space for admissible measurement-record assignments on the fixed scope, up to unique isomorphism.

Proof. Completeness is Lemma 3.14. Soundness is Lemma 3.15. Define $\phi([\rho]) := r(\rho)$. Soundness implies that ϕ is well-defined. Completeness implies that ϕ is injective. Surjectivity holds because r is surjective. Uniqueness is exactly the quotient universal property of Lemma 3.16: r is constant on $\approx_{\text{MR}}$ -classes by soundness and therefore factors through q uniquely. $\square$

Definition 3.18 (Exact standing-content equivalence relation). An equivalence relation $\approx$ on $\mathbf{ARef}$ is *exact standing-content preserving on the fixed scope* iff all of the following hold:

1. every standing-bearing value on the fixed scope factors through the quotient map

$$q_{\approx} : \mathbf{ARef} \rightarrow \mathbf{ARef}/\approx;$$

equivalently, $(\mathbf{ARef}/\approx, q_{\approx}, \mathbf{Val}_{\approx})$ is a same-scope measurement-record presentation in the sense of Definition 3.13; and

2. the quotient is still claimed to carry same-scope standing-bearing content rather than mere bookkeeping; and
3. the quotient is claimed to carry *exactly and only* same-scope standing-bearing measurement-record content, not enriched metadata, proof history, physical implementation detail, candidate bookkeeping, or presentation surplus.

Theorem 3.19 (Exact standing-content equivalence-collapse theorem). *Let $\approx$ be any exact standing-content preserving equivalence relation on $\mathbf{ARef}$. Then for all $\rho, \sigma \in \mathbf{ARef}$,*

$$\rho \approx \sigma \iff \rho \approx_{\mathbf{MR}} \sigma.$$

Equivalently, any same-scope equivalence relation that claims to carry exactly and only standing-bearing measurement-record content coincides with standing-equivalence. The quotient $\mathbf{ARef}/\approx_{\mathbf{MR}}$ is therefore forced as the exact standing-content quotient, not discretionary by presentation convenience.

Proof. Assume first that $\rho \approx \sigma$ but $\rho \not\approx_{\mathbf{MR}} \sigma$. Then $\mathbf{Prof}(\rho) \neq \mathbf{Prof}(\sigma)$, so there exists a construction token $d \in \mathcal{D}$ with $\mathbf{Val}_{\rho}(d) \neq \mathbf{Val}_{\sigma}(d)$. Since $\mathbf{ARef}$ is already forced by Theorem 3.6, the issue is not whether a different presentation of the target burden was available, but whether a second same-scope exact standing-content equivalence can survive on that forced realization space. But by Definition 3.18, every standing-bearing value on the fixed scope factors through $q_{\approx}$. Since $q_{\approx}(\rho) = q_{\approx}(\sigma)$, the induced value of d would have to agree on ρ and σ , contradiction. So $\approx$ cannot be strictly coarser than $\approx_{\mathbf{MR}}$.

Assume next that $\rho \approx_{\mathbf{MR}} \sigma$ but $\rho \not\approx \sigma$. Then $\approx$ distinguishes two assignments that agree on every standing-bearing value in the target practice. Treating that additional distinction as same-scope content would amount to a second closure-relevant equivalence relation finer than standing-equivalence. By the fixed-domain closure package and Theorem 3.17, no such second same-scope equivalence survives: it is either bookkeeping only or a scope change. Since $\approx$ is assumed to carry exactly and only standing-bearing content, enriched surplus is unavailable. So $\approx$ cannot be strictly finer than $\approx_{\mathbf{MR}}$ as an exact standing-content quotient.

Thus a coarser relation would lose standing-bearing content, a finer relation would add non-standing surplus rather than remain exact standing content, and the only same-scope exact standing-content equivalence relation is $\approx_{\mathbf{MR}}$. $\square$

Corollary 3.20 (No coarser exact standing-content equivalence). *No exact standing-content same-scope equivalence relation on $\mathbf{ARef}$ can identify assignments that lie in distinct $\approx_{\mathbf{MR}}$ -classes.*

Proof. That would make the equivalence strictly coarser than $\approx_{\mathbf{MR}}$, contradicting Theorem 3.19. $\square$

Corollary 3.21 (No finer exact standing-content equivalence). *No exact standing-content same-scope equivalence relation on $\mathbf{ARef}$ can distinguish assignments that lie in the same $\approx_{\mathbf{MR}}$ -class.*

Proof. That would make the equivalence strictly finer than $\approx_{\mathbf{MR}}$, contradicting Theorem 3.19. $\square$

Corollary 3.22 (Alternative same-scope equivalence relations collapse). *No same-scope equivalence relation on $\mathbf{ARef}$ can carry exactly and only standing-bearing measurement-record content while remaining genuinely distinct from $\approx_{\mathbf{MR}}$. A coarser alternative loses standing-bearing content; a finer alternative adds enriched presentation, metadata, bookkeeping, or scope-shifting surplus rather than remaining an exact standing-content quotient.*

Proof. Immediate from Theorem 3.19 together with Corollaries 3.20 and 3.21. $\square$

Corollary 3.23 (Evidential-order divergence is quotient-relevant). *If there exist $\rho, \sigma \in \mathbf{ARef}$ and an evidential construction token $d \in \mathcal{D}$ such that*

$$\mathbf{CR}(\mathbf{Prop}_\rho(d)) \neq \mathbf{CR}(\mathbf{Prop}_\sigma(d)),$$

then $[\rho] \neq [\sigma]$ in $\mathbf{ARef} / \approx_{\mathbf{MR}}$. In particular, evidential-order divergence is not representational skin.

Proof. If $[\rho] = [\sigma]$, then by definition $\rho \approx_{\mathbf{MR}} \sigma$, so $\mathbf{Prof}(\rho) = \mathbf{Prof}(\sigma)$. That in particular forces equality of all evidential-order-role values on evidential construction tokens. The displayed inequality therefore implies $[\rho] \neq [\sigma]$. $\square$

3.3 Fixation as the bridge invariant

Definition 3.24 (Lawful identity-preserving transformation of an evidential construction). A *lawful identity-preserving transformation* of an evidential construction token $d \in \mathcal{D}$ is a same-scope re-expression of d that preserves its candidate set $\mathbf{Cand}(d)$, its target practice role, and the activated fixed-domain closure package. In particular, lawful identity-preserving transformation introduces no new selector, no new admissibility classifier, and no scope reset.

Definition 3.25 (Fixation commutation). Fixation *commutes with admissible transformation* for an evidential construction token d iff every lawful redescription of d preserves its measurement-record quotient class. Equivalently, lawful redescription cannot move the evidential use of d between distinct standing-bearing admissible assignment classes.

Lemma 3.26 (Failed fixation commutation is quotient movement). *If fixation fails to commute with admissible transformation for an evidential construction token d , then there exist lawful identity-preserving transformations of d whose admissible uses lie in distinct $\approx_{\mathbf{MR}}$ -classes.*

Proof. This is just the negation of Definition 3.25: failure of commutation means that lawful identity-preserving transformation moves the evidential use of d between distinct standing-bearing admissible assignment classes. $\square$

Theorem 3.27 (Fixation-commutation necessity for fixed-domain evidential ordering). *If fixation fails to commute with admissible transformation for an evidential construction token d , then fixed-domain evidential ordering through d is undefined on the fixed scope. Boundary-trace fixation acts on admissible role realization inside the already-forced quotient structure; it does not introduce a second quotient, a second admissibility relation, or a second standing-bearing equivalence class.*

Proof. By Lemma 3.26, failure of fixation commutation moves the admissible use of d into distinct standing-bearing quotient classes. Because the assignment structure itself is forced by Theorem 3.6, failure of fixation commutation cannot be dismissed as an artifact of representational presentation. It is failure internal to the unique same-domain constructional realization space. By Theorem 3.19, that movement cannot be absorbed by an alternative same-scope equivalence relation; it is genuine standing-bearing divergence. A theory-level evidential-order verdict, however, is supposed to be attributable to the theory on the fixed scope rather than to a further standing-bearing fixation among admissible quotient classes. So once lawful redescription moves d between distinct quotient classes, no construction-relative evidential-order verdict is fixed. The right description is therefore undefinedness of fixed-domain evidential ordering on the fixed scope, not merely a weaker or relativized verdict. $\square$

Corollary 3.28 (Fixation commutation is the bridge invariant). *On the fixed scope, quotient uniqueness and equivalence collapse become relevant to measurement only through fixation commutation: if lawful identity-preserving transformation preserves the measurement-record quotient class of an evidential construction token, then same-scope standing-bearing divergence is silent for that use; if it does not, fixed-domain evidential ordering is undefined by Theorem 3.27.*

Proof. The quotient theorem fixes the unique same-scope content space, and Theorem 3.19 closes alternative same-scope equivalences. The only remaining question for theory-level evidential use is whether lawful identity-preserving transformation preserves the relevant quotient class. When it does, same-scope standing-bearing divergence is silent for that use; when it does not, Theorem 3.27 applies. $\square$

Remark 3.29 (Status of admissibility constraints). The admissibility condition is intentionally modest but not empty. Clause (5) of Definition 3.4 closes the easiest shortcut: enlarging or pruning the assignment space by smuggling in a second same-scope admissibility classifier or a boundary selector. Theorem 3.17 then shows that once the assignment space is typed that way, only quotient-level differences count as standing-bearing, and Theorem 3.19 shows that this quotient is not a discretionary presentation among many same-scope quotients but the unique admissibility-preserving one. Theorem 3.27 then makes fixation commutation the bridge from quotient structure to the evidential use of measurement records.

4 Finite quantum occupation of the measurement-record kernel

The preceding sections state the AASC dependency at the level of determinate objecthood and then specialize it to measurement-record construction. This section occupies that structure with the finite quantum domain mechanized in Lean. The point is not to append a toy after the main argument. The finite construction is a concrete physical specialization: its outcomes, records, amplitudes, density tables, maps, and predicates are determinate only under fixed reference, admissible typing, standing-bearing equality or disequality, and irreversible act boundaries.

4.1 Finite carriers and record regimes

Let

$$\Omega = \{\uparrow, \downarrow\}, \quad \mathcal{R}_2 = \{r_\uparrow, r_\downarrow, r_\ast\}, \quad \Gamma = \{\text{fine}, \text{coarse}\}.$$

Here $r_\uparrow$ and $r_\downarrow$ are fine branch records and r_* is the merged non-selective record. Define

$$R_{\text{fine}}(\uparrow) = r_\uparrow, \quad R_{\text{fine}}(\downarrow) = r_\downarrow,$$

while

$$R_{\text{coarse}}(\uparrow) = R_{\text{coarse}}(\downarrow) = r_*.$$

In the Lean formalization these are the finite outcome, record, and pointer-regime carriers `FiniteMeasurementOutcome`, `FiniteMeasurementRecord`, and `FinitePointerRegime`, with the record maps `finiteRecordOfOutcome` and `finitePointerRecord`.

Proposition 4.1 (Kernel occupation of the finite carrier). *The finite carrier is physically determinate only as a kernel-governed object. Its outcome labels require a fixed outcome carrier; its record labels require a fixed record carrier; its fine/coarse contrast requires admissible typing; and its equality and disequality theorems require standing-bearing identity within the declared scope.*

Proof. If $\uparrow$, $\downarrow$, $r_\uparrow$, $r_\downarrow$, and r_* are treated merely as uninterpreted symbols, no physical measurement claim follows. Once they are asserted as determinate outcomes and records of one measurement construction, they must be reference-fixed, admissibly transformed, available for standing-bearing comparison, and insulated from same-act repair. These are exactly the objecthood conditions of the AASC kernel, specialized to the finite measurement-record domain. $\square$

4.2 Denominator-two branch and record weights

The finite formalization stores integer numerators with common denominator two. Thus

$$\tilde{w}(\uparrow) = \tilde{w}(\downarrow) = 1, \quad \tilde{w}(\uparrow) + \tilde{w}(\downarrow) = 2,$$

so the normalized weights are $1/2$ and $1/2$. The record weights are

	$r_\uparrow$	$r_\downarrow$	r_*
fine	1	1	0
coarse	0	0	2.

The denominator is not a cosmetic detail: it keeps the finite arithmetic exact while representing the balanced two-branch situation.

Theorem 4.2 (Fine separation and coarse merge). *On the finite carrier,*

$$R_{\text{fine}}(\uparrow) \neq R_{\text{fine}}(\downarrow),$$

but

$$R_{\text{coarse}}(\uparrow) = R_{\text{coarse}}(\downarrow) = r_*.$$

Both the fine and coarse regimes preserve total denominator-two weight.

Proof. The fine map sends the two outcomes to two distinct constructors $r_\uparrow$ and $r_\downarrow$. The coarse map sends both outcomes to r_* . The displayed weight table sums to 2 in both regimes. Hence the

fine surface distinguishes branches, the coarse surface merges them, and neither regime changes the total finite weight. $\square$

Corollary 4.3 (Non-selective coarse data do not determine a fine record). *The coarse non-selective record r_* is not a hidden choice of $r_\uparrow$ or $r_\downarrow$.*

Proof. The constructors $r_\uparrow$, $r_\downarrow$, and r_* are distinct in the declared carrier. Since the coarse map returns r_* for both outcomes, it contains no term equal to either fine record. $\square$

4.3 Balanced squared-amplitude witness

Introduce the normalized two-component vector

$$\psi = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The Lean witness stores the integer numerator vector

$$\tilde{\psi} = \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

with

$$\|\tilde{\psi}\|^2 = 1^2 + 1^2 = 2.$$

The squared-amplitude numerators are

$$\tilde{w}_{\tilde{\psi}}(\uparrow) = 1, \quad \tilde{w}_{\tilde{\psi}}(\downarrow) = 1.$$

Theorem 4.4 (Balanced-weight non-selection). *The balanced squared-amplitude witness is normalized but does not supply singleton branch selection:*

$$\neg \left(((\tilde{w}_{\tilde{\psi}}(\uparrow), \tilde{w}_{\tilde{\psi}}(\downarrow)) = (2, 0)) \vee ((\tilde{w}_{\tilde{\psi}}(\uparrow), \tilde{w}_{\tilde{\psi}}(\downarrow)) = (0, 2)) \right).$$

Proof. Both branch numerators are 1. Neither branch has numerator 2 while the other has numerator 0. Therefore the balanced weight surface differentiates and weights the two branches but does not occupy the selector role. $\square$

4.4 Pointer coupling and finite weakening

Let pointer states be $p_0, p_\uparrow, p_\downarrow, p_*$, with coupling

$$\Pi(\uparrow) = p_\uparrow, \quad \Pi(\downarrow) = p_\downarrow.$$

The fine pointer record reproduces the fine outcome record, while the coarse pointer record returns r_* for either outcome. Thus the pointer layer preserves the distinction between branch differentiation and non-selective merging.

The finite model also instantiates the manuscript’s weakening taxonomy. The declared weakening attempts are

preserveFineTrace, bookkeepingOnly, targetShiftToCoarseTrace,
secondClassifier, sameActRepair.

The first two preserve no new fine-selection authority; the last three are exactly the cases that do not preserve the fine trace. Hence the finite model realizes the general anti-weakening result: a same-domain attempt either preserves the same governance work, becomes bookkeeping, changes target, imports a second classifier, or attempts same-act repair.

4.5 Boundary-trace constructor

The finite trace-channel packet records a fine-trace regime together with declaration-before-use, stability under record-preserving maps, outcome-independent declaration, and non-generativity. If those readiness conditions hold, the finite constructor yields a boundary-trace object satisfying

Declared $\wedge$ Stable $\wedge$ OutcomeIndependent $\wedge$ NonGenerative $\wedge$ SingletonPositive.

This is a conditional constructor, not an inference from bare environmental correlation. It states exactly what a finite channel must contain to occupy the boundary-trace role.

5 The Everettian diagnosis

This section introduces a minimal Everettian measurement package and asks whether bare unitary branching supplies the fixation bridge isolated in Section 3. It proves that Everettian physics is non-selective in the sense fixed in Section 2, separates branch-local determinacy from diachronic record-fixation, and shows that decoherence, centered descriptions, successor-typing, and decision theory do not by themselves fix the standing realization of pre-measurement construction.

5.1 The Everettian base package

Only a minimal Everettian measurement package is required for the present argument. For the present purpose an Everettian measurement episode can be modeled by a pre-measurement state, an entangling measurement interaction, and a decoherence regime that renders the resulting outcome-sectors approximately dynamically autonomous for the duration of the measurement record.

Definition 5.1 (Everettian base package for a measurement episode). A measurement episode satisfies the *Everettian base package* when all of the following hold:

1. **Unitary evolution.** The total post-interaction state is obtained by a unitary measurement interaction U_M applied to the pre-measurement state.
2. **Outcome-sector multiplicity.** There exists a decomposition

$$\Psi_M = \sum_{i \in I} \alpha_i \psi_i, \quad |I| \geq 2, \quad \alpha_i \neq 0 \text{ for each } i,$$

where each ψ_i is associated with a distinct stable record/outcome package $(\text{Cont}_i, \text{Rec}_i, o_i)$.

3. **Decoherence.** The sectors ψ_i are pairwise approximately non-interfering for the purposes of the measurement record, so that each supports branch-local record stability.
4. **No selector in the bare physics.** No additional collapse postulate, hidden-variable actualization rule, or other branch-privileging physical structure is added to the bare unitary description.

The definition deliberately builds in only what an Everettian appeals to when claiming that unitary quantum theory together with decoherence already explains measurement: global unitary evolution, multiple nonzero outcome-sectors, and record stability without a collapse or actualization rule [3, 4, 10].

Approximation matters here. Decoherence, pointer robustness, environmental redundancy, and confidence measures may be graded as physical or diagnostic structure. The bivalence claimed in this manuscript applies only at the admissibility-relevant quotient for standing-bearing record use: either those structures fix a standing record role for the target construction, or they do not. The argument does not deny graded physical stability. It denies that graded stability is already singleton evidential-role fixation.

Lemma 5.2 (Branch-local determinacy). *Suppose a measurement episode satisfies Definition 5.1. Then each decohered outcome-sector contains a determinate branch-local record and a determinate branch-local observer-state.*

Proof. By clause (2) of Definition 5.1, each sector ψ_i is associated with a stable record/outcome package $(\text{Cont}_i, \text{Rec}_i, o_i)$. By clause (3), decoherence suppresses interference between the sectors for the purposes of the measurement record, so each such package is dynamically stable in the branch-local sense relevant to reporting, retained-record reuse, and within-branch experience. Hence each decohered sector contains a determinate branch-local record and branch-local observer-state. $\square$

Lemma 5.3 (Unitary multiplicity lemma). *Suppose a measurement episode satisfies Definition 5.1. Then the post-measurement state contains at least two nonzero physically realized outcome-sectors.*

Proof. By clause (2) of Definition 5.1, the post-measurement state has the form

$$\Psi_M = \sum_{i \in I} \alpha_i \psi_i$$

with $|I| \geq 2$ and $\alpha_i \neq 0$ for each $i \in I$. A nonzero coefficient means that the corresponding sector contributes nontrivially to the physical state-description of the episode. Since there are at least two such coefficients, the post-measurement state contains at least two nonzero outcome-sectors. By the intended Everettian reading, those sectors are physically realized parts of the same global state rather than merely hypothetical alternatives. Hence the episode contains at least two nonzero physically realized outcome-sectors. $\square$

Lemma 5.4 (Entanglement correlates without privileging). *Let a measurement episode satisfy Definition 5.1. Then the entangling measurement interaction correlates system, apparatus, and observer records across the distinct outcome-sectors, but does not by itself privilege one such sector as the unique diachronic continuation relevant to pre-measurement measurement construction.*

Proof. In the Everettian package, the measurement interaction produces a correlated state of the schematic form

$$\sum_{i \in I} \alpha_i |s_i\rangle |A_i\rangle |O_i\rangle,$$

where distinct system-outcome states are correlated with distinct apparatus and observer-record states. That establishes the correlation fact: each sector carries its own determinate branch-local triple and hence its own record/outcome package.

What must be shown is the absence of privileging. The entangling state specifies, for each i , what the record is in that sector. It does not specify that one i is the unique standing realization of pre-measurement same-domain subject-stage construction. To privilege one sector, the physical theory would need an additional rule of the form: exactly one correlated sector is the continuation relevant to the pre-measurement subject. But clause (4) of Definition 5.1 excludes any such added selector in the bare physics. Therefore entanglement supplies correlated multiplicity, not a unique diachronic standing realization. $\square$

Lemma 5.5 (Decoherence yields stability, not singularity). *If a measurement episode satisfies Definition 5.1, then decoherence can explain the stability and effective autonomy of each branch-local record without thereby fixing which branch is the standing realization of pre-measurement diachronic construction.*

Proof. By clause (3) of Definition 5.1, decoherence renders the sectors approximately non-interfering for the purposes of the measurement record. This yields the standard Everettian achievement: each sector supports a stable branch-local record and observer-state.

However, branch-local stability is a within-sector property. It tells us that if a sector is taken as the relevant one, its record is robust enough for local retained-record reuse and reporting. It does not add any physical fact of the form “sector i rather than sector j is the unique continuation relevant to pre-measurement construction”. Such a fact would be a selector. Since clause (4) excludes selectors in the bare physics, decoherence yields stability but not singularity of standing realization. $\square$

Lemma 5.6 (No supervenient diachronic selector). *Let T be physically non-selective with respect to a measurement process, and let O_i and O_j be two post-measurement observer-successors realized by T . Suppose that T supplies no physical law, state variable, dynamically privileged trajectory, or physically relevant asymmetry that distinguishes O_i from O_j with respect to diachronic continuation. Then no unique diachronic continuation relation privileging O_i over O_j supervenes on the physical facts delivered by T alone.*

Proof. To say that a unique diachronic continuation relation supervenes on the physical facts delivered by T is to say that those facts suffice to determine, without further addition, that one successor is the privileged continuation rather than the other. But by hypothesis, the physical description delivered by T contains no law, state variable, privileged trajectory, or other physically relevant asymmetry distinguishing O_i from O_j with respect to diachronic continuation. Any relation that nevertheless privileges O_i over O_j would therefore introduce an asymmetry not fixed by the physical facts supplied by T . Hence no such unique privileging relation supervenes on the physical facts delivered by T alone. $\square$

Corollary 5.7 (Everettian non-selection). *In a branching Everettian measurement model satisfying physical non-selection, the physical structure alone determines neither an explicit selector nor a supervenient unique diachronic standing realization for pre-measurement same-domain subject-stage construction.*

Proof. By Lemma 5.3, the Everettian base package contains at least two nonzero physically realized outcome-sectors, each with its own branch-local record and continuant. By Definition 2.44, the package contains no physical law, state variable, or dynamically privileged trajectory distinguishing one such successor as physically privileged. The absence of an explicit selector is therefore immediate. Applying Lemma 5.6 to any two such realized successors yields the absence of a supervenient unique diachronic standing realization. $\square$

Theorem 5.8 (Everettian non-selection theorem). *Every measurement episode satisfying the Everettian base package is physically non-selective in the sense of Definition 2.44.*

Proof. To be physically non-selective, a measurement episode must realize multiple stable successor histories while supplying no physical law, state variable, or dynamically privileged trajectory that singles out one such history as physically distinguished. The first condition follows from Lemma 5.3; branch-local determinacy of the realized histories follows from Lemma 5.2; and the absence of a physically distinguishing selector is built into clause (4) of Definition 5.1 and then made explicit by Corollary 5.7. Hence every measurement episode satisfying the Everettian base package is physically non-selective. $\square$

Corollary 5.9 (Bare Everett lacks the local record-kernel closeout). *The bare Everettian base package supplies unitary evolution, entanglement, branch-local records, and decoherence stability. It does not by itself supply singleton record-role fixation, an admissible boundary trace, admissibility-preserving quotient collapse, or a lawful same-domain selector. Hence it gives physical realization without evidential-role fixation.*

Proof. Theorem 5.8 gives physical non-selection: multiple stable successor histories are physically realized without a physically privileged one. KMR3 separates physical record stability from evidential-anchor standing, KMR10 blocks branch weights, environmental salience, and centered indices from acting as boundary selectors, and KMR13 requires record-role exhaustion rather than hidden fifth status. Therefore the bare package lacks the local record-kernel closeout needed for the target explanatory practice. $\square$

5.2 Branch-relative determinacy versus diachronic record-fixation

Once the Everettian base package is in place, one may grant a great deal. Each realized branch can contain a determinate local observer, a determinate local record, and a determinate local outcome. What still does not follow is that the pre-measurement subject's diachronic construction has a uniquely fixed standing realization.

Definition 5.10 (Successor-qualified history). A physically realized post-measurement history $b \in \mathcal{B}$ is *successor-qualified* for the purposes of a given piece of diachronic construction if it contains both

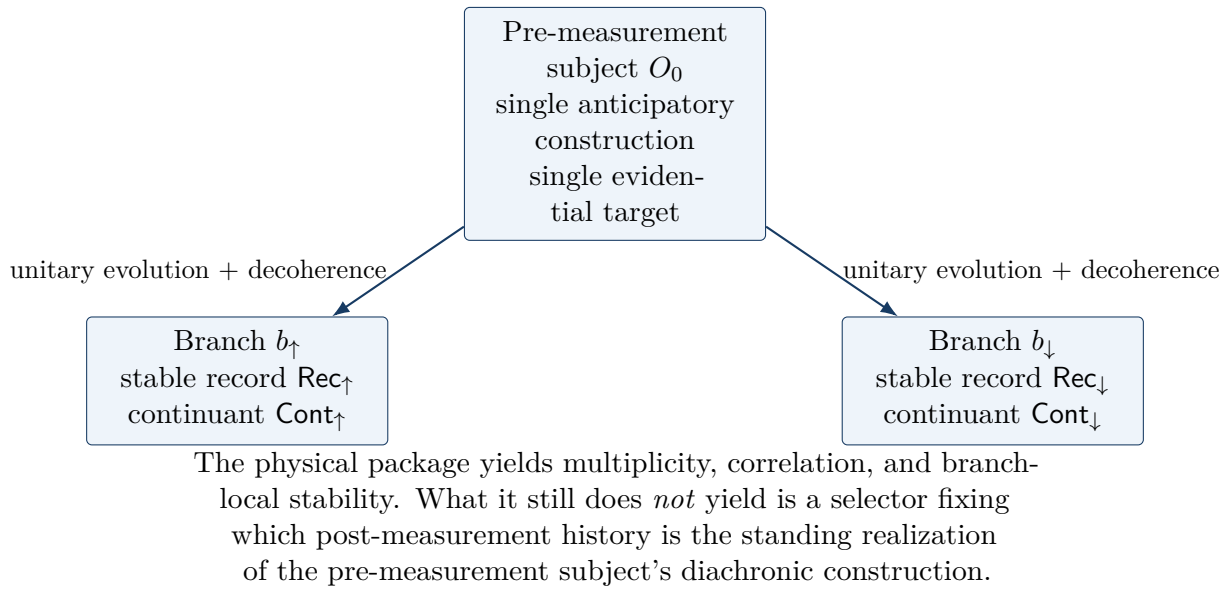


Figure 1: Minimal branching diagram for the theorem ladder. The Everettian base package yields multiple stable post-measurement records, but it does not physically select which history is the diachronic standing realization of the pre-measurement construction.

1. a determinate measurement record $\text{Rec}(b)$ appropriate to that construction, and
2. a post-measurement continuant $\text{Cont}(b)$ satisfying the continuity conditions presupposed by the construction between the pre-measurement subject and the future subject.

Lemma 5.11 (Multiplicity of successor-qualified histories yields multiplicity of admissible assignments). *Let $\mathcal{M}$ be a physically non-selective measurement episode and let d be a record-sensitive construction token in its target construction. If there exist distinct successor-qualified histories $b_i, b_j \in \mathcal{B}$ for d , then $\text{ARef}(\mathcal{M})$ contains at least two distinct admissible record-fixation assignments relevant to d .*

Proof. Because b_i is successor-qualified for the construction involving d , it satisfies the conditions that make the pair $(b_i, \text{Rec}(b_i))$ eligible to serve as a standing realization for d . So there is an admissible assignment ρ_i sending d to $(b_i, \text{Rec}(b_i))$. The same eligibility reasoning applies to b_j , yielding an admissible assignment ρ_j sending d to $(b_j, \text{Rec}(b_j))$. Because $b_i \neq b_j$, the assignments are distinct. Therefore there are at least two admissible assignments relevant to d . $\square$

Lemma 5.12 (Branch-relative determinacy does not entail diachronic determinacy). *The existence of determinate branch-relative post-measurement experiences does not determine which such branch-relative experience is the standing realization of pre-measurement diachronic same-domain subject-stage construction.*

Proof. Consider a branching model in which two post-measurement successors O_1 and O_2 each possess determinate branch-local experiences and stable branch-relative records. Now consider two admissible assignments: one taking O_1 to be the standing realization of the pre-measurement subject's diachronic construction, the other taking O_2 to be that standing realization. The branch-local determinacy of each successor remains unchanged across the two assignments. What changes

is only the diachronic successor-fixation relation. Hence branch-local determinacy does not entail diachronic constructional determinacy. $\square$

Corollary 5.13 (Determinacy within branches is compatible with global constructional underdetermination). *A theory may deliver determinate experiences and records inside every realized branch while still failing to determine which branch-history is the standing realization of pre-measurement retained-record reuse, prospective-record reuse, and evidential construction.*

Proof. This is an immediate restatement of Theorem 5.12. The theorem establishes that local determinacy in each branch and diachronic determinacy across branches are distinct conditions. The first may hold while the second fails. $\square$

Proposition 5.14 (Successor multiplicity marks a structural distinction). *The difference between branch-local determinacy and diachronic constructional determinacy marks a structural difference between two explanatory questions:*

1. *what records and observer-stages exist in each realized history; and*
2. *which such history is the standing realization of pre-measurement same-domain subject-stage and evidential construction.*

Proof. The first question is answered by the branch-local facts contained in each history: what outcome is recorded there, what observer-stage exists there, and what local retained-record reuse state exists there. The second question is answered only if the theory determines which realized history is relevant to the pre-measurement subject's diachronic construction tokens. A theory may answer the first question for every branch and still leave the second unanswered. That possibility is exactly what Theorem 5.12 shows. Therefore the distinction is structural. $\square$

Proposition 5.15 (Global branch-set fixation does not remove role-fixation). *Treating the entire branching structure as the standing physical object does not by itself fix which branch-local record functions as the evidential anchor for same-domain downstream reuse. Global branch aggregation therefore preserves the record-role fixation problem unless a singleton admissible asymmetry is supplied.*

Proof. The global branch-set description may include every realized branch and every branch-local record. That aggregation answers the physical-realization question: which records exist in the total branching structure. It does not answer the role-fixation question: which record is the standing realization of the construction token used for retained-record reuse, prospective-record reuse, deliberation, or evidential comparison. To move from the global set to one evidential role-anchor requires an internal singleton invariant, a declared provenance/indexical singleton anchor, a singleton stable boundary trace, or an explicit restriction collapsing admissible assignments to one standing-equivalence class. A branch-set or measure-valued evidential object may instead be adopted as a retyped target practice, but that is not a same-domain solution to singular record-role fixation for the original construction token. Retyping may be coherent; it is not non-degenerate realization of the original record-role target. Without such fixation, collapse, or retyping, the global branch-set construction leaves the same candidate multiplicity in place. $\square$

5.3 Decoherence and remaining indeterminacy

All discussion of evidential and experiential record-fixation in this section is conditional on the target explanatory burden fixed in Section 2. The issue is not whether decoherence is useful or real. The issue is whether decoherence, by itself, fixes diachronic record-fixation.

Definition 5.16 (Record stability under decoherence). A measurement episode $\mathcal{M}$ satisfies *decoherence-based record stability* if for every physically realized history $b \in \mathcal{B}$, the record $\text{Rec}(b)$ is dynamically stable enough to support branch-local retained-record reuse and observational reliability, and the corresponding branch remains approximately non-interfering with the others for the purposes of the episode under discussion.

Proposition 5.17 (Decoherence stabilizes records without selecting standing realizations). *Let $\mathcal{M}$ be a physically non-selective measurement episode satisfying decoherence-based record stability. Then decoherence stabilizes the records contained in each $b \in \mathcal{B}$ but does not by itself privilege one b as the unique standing realization of pre-measurement diachronic construction.*

Proof. By decoherence-based record stability, each history $b \in \mathcal{B}$ contains a dynamically stable record $\text{Rec}(b)$ that may support branch-local retained-record reuse and branch-local observational construction. This is the positive achievement of decoherence.

To show that decoherence does not by itself privilege a unique diachronic standing realization, suppose for contradiction that it does. Then decoherence would provide a selector: it would reduce the admissible record-fixation possibilities relevant to the construction to exactly one privileged assignment. But Decoherence-based record stability, as defined here, supplies only local stability and effective non-interference of the realized histories. It says that each realized branch carries a stable record; it does not say that one stable branch is physically privileged over the others as *the* continuation relevant to pre-measurement construction. Adding such a privilege would amount to adding a selector not already contained in the bare statement of decoherence-based stability. Therefore decoherence stabilizes records without by itself fixing standing realizations. $\square$

Theorem 5.18 (Decoherence non-selection theorem). *If a theory's post-measurement contribution to a measurement episode consists only of the physical realization of multiple branch-histories together with decoherence-based record stability, then the theory does not by that contribution alone fix diachronic record-fixation for measurement records.*

Proof. Assume the theory's post-measurement contribution consists only of the physical realization of multiple histories together with decoherence-based record stability. By Proposition 5.17, that contribution stabilizes the records in each realized history but does not itself provide a selector. If the episode is physically non-selective, then at least two admissible record-fixation assignments remain available. Therefore the theory's post-measurement contribution, as specified, does not fix diachronic record-fixation for the measurement records. $\square$

Proposition 5.19 (Pointer-basis stabilization is not singleton evidential-role fixation). *Pointer-basis stabilization, einselection, and environmental decoherence may fix a stable family of record-bearing sectors for a measurement episode [4, 5]. They do not, without an additional admissible same-domain asymmetry, fix which sector-record is the standing realization of a pre-measurement construction token used for retained-record reuse, prospective-record reuse, deliberation, or evidential comparison.*

Proof. Pointer-basis and einselection arguments explain why some record-bearing sectors are dynamically stable and robustly reportable within their branch. That establishes a stable candidate family, not a singleton role bearer. To turn one member of that family into the standing evidential anchor for the target construction, the theory must supply an internal singleton invariant, a declared provenance/indexical singleton, or a declared singleton boundary trace. If it supplies none of these, KMR10 and KMR13 classify the appeal to environmental robustness as support/bookkeeping or as a hidden selector. Hence pointer-basis stabilization is not yet singleton evidential-role fixation. $\square$

5.4 Centered descriptions, successor-typing, and decision theory

The most common way of resisting the previous result is to say that Everettian theory has resources for downstream handling under multiplicity: centered-world representation, successor-typing uncertainty, and decision-theoretic branch weighting [7–9]. The present claim is not that such resources are incoherent. The claim is that they presuppose a prior typing of successors.

Definition 5.20 (Centered continuation description). A *centered continuation description* for a measurement episode $\mathcal{M}$ is a pair (b, c_b) where $b \in \mathcal{B}$ is a physically realized post-measurement history and c_b is a center-marker intended to treat the post-measurement continuant in b as a candidate continuation of the pre-measurement subject for same-domain subject-stage or evidential construction.

Definition 5.21 (Centered enrichment). A *centered enrichment* of a physically non-selective measurement episode $\mathcal{M}$ is the passage from the uncentered history family $\mathcal{B}$ to a family of centered continuation descriptions

$$\mathcal{C}_M = \{(b, c_b) : b \in \mathcal{B}\}.$$

The enrichment is *selector-free* if it adds no rule that privileges exactly one center (b, c_b) as the standing realization of pre-measurement diachronic construction.

Lemma 5.22 (Centered enrichment preserves multiplicity). *Let $\mathcal{M}$ be a physically non-selective measurement episode and let $\mathcal{C}_M$ be a selector-free centered enrichment of $\mathcal{M}$. Then if $|\mathcal{B}| \geq 2$, the centered enrichment contains at least two distinct candidate centers.*

Proof. Because $\mathcal{M}$ is physically non-selective, $|\mathcal{B}| \geq 2$. So there exist distinct histories $b_i \neq b_j$ in $\mathcal{B}$. By definition of centered enrichment, both give rise to centered continuation descriptions (b_i, c_{b_i}) and (b_j, c_{b_j}) . Since the underlying histories are distinct, these centered descriptions are distinct as well. Therefore the centered enrichment contains at least two distinct candidate centers. $\square$

Proposition 5.23 (Centered enrichment does not by itself select a standing realization). *A selector-free centered enrichment of a physically non-selective measurement episode does not by itself determine which centered continuation description is the standing realization of pre-measurement diachronic measurement construction.*

Proof. By Lemma 5.22, selector-free centered enrichment preserves multiplicity of candidate centers. To determine which centered continuation description is the standing realization of pre-measurement construction, one would need a rule that singles out one center from the enriched set $\mathcal{C}_M$. But selector-free centered enrichment, by definition, adds no such rule. So the centered enrichment redescribes the multiplicity of candidate continuants; it does not eliminate that multiplicity or privilege one member of it as the unique standing realization. $\square$

Definition 5.24 (Typed successor set). A set Σ_M of post-measurement continuants is a *typed successor set* for a piece of diachronic measurement construction iff each member of Σ_M is already fixed as an admissible candidate continuation of the pre-measurement subject for the admissibility conditions of that construction.

Lemma 5.25 (Self-locating uncertainty requires a typed successor set). *A successor-typing uncertainty rule can guide same-domain subject-stage credence only if it ranges over a typed successor set.*

Proof. A successor-typing uncertainty rule distributes credence across candidate future centers or successors. For that distribution to be first-personally meaningful, the rule's domain must already consist of the candidates relevant to the subject whose uncertainty is in question. If the domain were not already typed in that way, the rule would distribute credence across a merely physical collection of post-measurement continuants without having fixed that these are the continuants relevant to the present subject's anticipatory construction. In that case the rule would fail to determine the truth conditions of same-domain subject-stage construction tokens such as "I will observe o ". So successor-typing uncertainty requires a typed successor set. $\square$

Definition 5.26 (Decision-theoretic branching rule). A *decision-theoretic branching rule* for a measurement episode $\mathcal{M}$ is any rule that assigns utilities, preferences, or credences across a set of candidate post-measurement continuants or records and is then used to guide pre-measurement deliberation.

Lemma 5.27 (Decision rules require typed successor events). *A decision-theoretic branching rule can guide action only if the successor events or records to which it assigns utilities or credences are already typed as the relevant successors for the construction being used by the agent.*

Proof. A decision-theoretic rule guides action only if its outputs bear on what the present agent should expect, prefer, or do. To have that bearing, the successor events or records over which the rule ranges must already be the successor events or records relevant to the present agent. If those items are not already typed as the relevant successors for the construction in which the decision is being made, then the rule's outputs are not yet action-guiding in the intended same-domain subject-stage sense. The rule would at best distribute weights over an untyped collection of physical continuants without thereby determining which of them are the future experiences or records to which the present agent's construction tokens refer. Therefore the decision rule requires successor events or records that are already constructionally typed for the construction in question. $\square$

Proposition 5.28 (Weights over branches are not selectors). *Assigning amplitudes, credences, utilities, or caring measures to realized branches does not by itself fix which history is the standing realization of pre-measurement diachronic construction.*

Proof. A selector determines which realized history is the relevant standing realization. A weight assignment instead orders or measures the candidate histories while leaving the domain of candidates in place. Unless one already knows what the weighted items are the candidates *for*, the weights do not fix which history the pre-measurement construction is about. Branch amplitudes, weights, or caring measures can order already-typed successors, but they cannot themselves generate singleton admissible record-role fixation. They therefore presuppose rather than provide the needed constructional typing. $\square$

Proposition 5.29 (Centered and weighted resources preserve asymmetry multiplicity). *Centered enrichment, self-locating uncertainty, successor typing, caring measures, branch weights, and anthropic filters preserve admissible multiplicity unless they introduce at least one of the following: a new admissibility-relevant selector, a privileged ontic continuation, a declared singleton provenance/indexical anchor, or singleton boundary-trace fixation on the candidate record set. Therefore they do not by themselves collapse admissible record-fixation classes.*

Proof. Each listed resource either re-expresses the candidate family, assigns weights over that family, distributes credence across already typed successors, or filters the family by a policy. If it leaves all candidates present and supplies no singleton invariant, multiplicity remains. If it privileges one candidate, then the privilege must enter through one of the normal-form sources later isolated in Theorem 8.6: internal invariant, declared provenance/indexical marker, or declared singleton boundary trace. Otherwise the privilege is an extra-domain selector, scalarized parameter, post-selection repair, or scope reset. Hence these resources preserve multiplicity unless they leave their advertised selector-free role. $\square$

Theorem 5.30 (Everettian downstream theorem). *Centered descriptions, successor-typing uncertainty, and decision-theoretic branch-weighting are downstream of the Everettian non-selection problem. They presuppose successor typing and therefore cannot by themselves supply the diachronic record-fixation whose absence constitutes the problem identified in this manuscript.*

Proof. By Proposition 5.23 and Proposition 5.29, centered enrichment without a selector preserves rather than eliminates multiplicity of candidate continuants and does not collapse the admissible asymmetry classes. By Lemma 5.25, any successor-typing treatment can guide same-domain subject-stage credence only if its domain is already a typed successor set. By Lemma 5.27, a decision-theoretic branching rule can guide action only if its successor events are already typed as the relevant successors for the present agent's construction. But Theorem 5.8 and Theorem 5.12 show that the bare Everettian physical package leaves multiple realized histories in play without fixing which one is the relevant diachronic standing realization. Proposition 5.29 gives the sharpened normal-form statement: centered, self-locating, weighted, caring-measure, and anthropic resources preserve admissible multiplicity unless they introduce a new selector, privileged ontic continuation, declared singleton provenance/indexical anchor, or singleton boundary trace. Therefore centered descriptions, successor-typing, and decision theory begin only after the missing constructional work has been done. They are downstream of the non-selection problem rather than solutions to it. $\square$

5.5 Native Everettian evidence and no fifth record role

Definition 5.31 (Native Everettian evidence package). $\text{EverettNativeEvidence}_{\text{MR}}(d)$ is the maximal same-carrier Everettian evidence package for a construction token d : the global unitary state, decohered branch structure, branch-local records, branch weights, centered or self-locating descriptions, successor-typing resources, and decision-theoretic policy are all retained, and the package is claimed to supply native evidential practice rather than an additional AASC-governed record selector.

Definition 5.32 (Everettian native-evidence role classes). For a construction token d , the relevant Everettian native-evidence roles are:

1. *support-only material*: branch-local records, decoherence structure, weights, or the global state are used as physical or probabilistic support without claiming to fix the standing evidential record role of d ;
2. *branch-relative target practice*: evidential use is explicitly indexed to a branch or successor-stage, so the target is branch-relative rather than construction-relative for the original pre-measurement token;
3. *global branch-set target practice*: the evidence object is the whole branching state, a decoherent history set, or a measure-valued branch family rather than one standing record-bearing realization of d ;
4. *admissible singleton fixation*: the package supplies an internal singleton invariant, declared provenance/indexical singleton anchor, or declared singleton boundary trace for d ;
5. *hidden selector import*: branch weight, salience, centeredness, decision policy, detector preference, or environmental appeal is used to privilege one record role without admissible declaration;
6. *assignment-relative use*: the package preserves the original target but leaves more than one admissible record-fixation class live.

Theorem 5.33 (Native Everettian evidence collapse). *For the target construction token d , any $\text{EverettNativeEvidence}_{\text{MR}}(d)$ package falls into at least one of the six roles of Definition 5.32. If the package is endpoint-standing for the original singular record-role target, its terminal role is admissible singleton fixation, hidden selector import, or assignment-relative use. There is no fifth same-domain Everettian evidence role that preserves the original singular record-role target while avoiding record-fixation.*

Proof. Consider how the package uses its Everettian resources. First, if the global state, decoherence structure, branch weights, or branch-local records are retained only as physical, probabilistic, or explanatory support, then they do not fix the standing evidential record role of d and fall under support-only material. Second, if the evidential claim is explicitly made from within a branch or successor-stage, the construction is branch-relative; it may be coherent for that different explanatory practice, but it is not the original construction-relative role of the pre-measurement token. Third, if the evidence object is the whole branch family, a measure on that family, or a decoherent history set, then Theorem 3.8 applies: either the branch-set object projects to a singleton standing realization for d , or it is a global branch-set target practice rather than the original singular record-role target. Global branch-state evidence may therefore be the correct evidence object for a different practice; it is a same-mode solution here only when it preserves the standing realization of the original construction token. If global branch-state evidence is used later to confer standing on an earlier unfixed record-use act, then it is not a third route: it is either a declared later act or update scope, or the same-act repair excluded by Remark 2.23.

Fourth, if the package supplies a singleton standing realization for d , the fixing source must be an internal invariant, a declared provenance/indexical singleton anchor, or a declared singleton boundary trace; that is admissible singleton fixation and re-enters the record-fixation framework. Fifth, if the privilege is supplied instead by branch weight, environmental salience, centered index, decision policy, detector preference, caring measure, or another undeclared asymmetry, then the

package imports a hidden selector excluded by the AMetric and same-scope closure discipline [20]. Sixth, if the package preserves the original singular record-role target while supplying no singleton fixation and no target retyping, then at least two admissible record-fixation classes remain live. The evidential verdict is therefore assignment-relative by Theorem 7.7. These cases exhaust whether the package preserves the original target and whether it fixes the standing record role. The classes may overlap as descriptive resources, but their terminal standing roles do not: support and target-retyping roles do not solve the original burden; singleton fixation re-enters the theorem ladder; hidden selector import is inadmissible; and assignment-relative use is the residual failure mode. $\square$

Section 5 has established that the standard Everettian physical package yields multiple stable branch-local records and no unique diachronic standing realization, and that successor-typing and decision-theoretic replies presuppose rather than remove that problem. The next section turns to how measurement records function as evidential anchors and what happens to fixed-domain evidential ordering when admissible record-fixation assignments diverge.

5.6 Finite coherence-removal calculation

The finite construction used later makes the Everettian non-selection point arithmetically explicit. With numerator table

$$\tilde{\rho}_\psi = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

the non-selective projective channel gives

$$\mathcal{N}(\tilde{\rho}_\psi) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Total denominator-two trace is preserved while the off-diagonal numerator is removed. This calculation is physical: it is the finite table representation of coherence removal. But it is not selection, because the non-selective output is neither the up selective branch nor the down selective branch and its record surface is the merged record r_* .

6 Finite measurement operations and physical record surfaces

The finite carrier now receives the standard physical surfaces needed for the measurement-record problem: a density table, an instrument, an environment record map, branch maps, effects, and projectors. Each surface preserves the same distinction: selective branch surfaces differentiate records; non-selective surfaces merge records without selecting a fine one.

6.1 Density table and coherence removal

Use the integer numerator table

$$\tilde{\rho}_\psi = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

corresponding to the normalized density matrix

$$\rho_\psi = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Define selective branch tables

$$\tilde{\rho}_\uparrow = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \tilde{\rho}_\downarrow = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

and the non-selective sum

$$\tilde{\rho}_{\text{ns}} = \tilde{\rho}_\uparrow + \tilde{\rho}_\downarrow = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

For a numerator table $\tilde{\rho}$, set

$$\tilde{T}(\tilde{\rho}) = \tilde{\rho}_{\uparrow\uparrow} + \tilde{\rho}_{\downarrow\downarrow}, \quad \tilde{C}(\tilde{\rho}) = \tilde{\rho}_{\uparrow\downarrow} + \tilde{\rho}_{\downarrow\uparrow}.$$

These are numerator functionals; normalized values are obtained only after division by the common denominator where that denominator has been fixed. Then

$$\tilde{T}(\tilde{\rho}_\psi) = 2, \quad \tilde{C}(\tilde{\rho}_\psi) = 2,$$

while

$$\tilde{T}(\tilde{\rho}_{\text{ns}}) = 2, \quad \tilde{C}(\tilde{\rho}_{\text{ns}}) = 0.$$

Thus finite coherence removal preserves total denominator-two trace while erasing the off-diagonal numerator. It does not turn the non-selective table into either selective branch.

6.2 Instrument decomposition

Define the selective instrument branches on the balanced witness by

$$\mathcal{I}_\uparrow(\tilde{\rho}_\psi) = \tilde{\rho}_\uparrow, \quad \mathcal{I}_\downarrow(\tilde{\rho}_\psi) = \tilde{\rho}_\downarrow,$$

and

$$\mathcal{I}_{\text{ns}}(\tilde{\rho}_\psi) = \mathcal{I}_\uparrow(\tilde{\rho}_\psi) + \mathcal{I}_\downarrow(\tilde{\rho}_\psi).$$

The instrument records are

$$R_{\mathcal{I}}(\uparrow) = r_\uparrow, \quad R_{\mathcal{I}}(\downarrow) = r_\downarrow, \quad R_{\mathcal{I}}^{\text{ns}} = r_*.$$

Theorem 6.1 (Finite instrument decomposition). *The non-selective instrument state is the sum of the two selective branches; each branch trace equals the corresponding balanced squared-amplitude numerator; the selective records are distinct; and the non-selective instrument record is r_* , not $r_\uparrow$ or $r_\downarrow$.*

Proof. The branch tables have traces 1 and 1, matching the squared-amplitude numerators. Their sum is the non-selective table. The record map sends the two selective branches to distinct fine

records but defines the non-selective record as r_* . Since r_* is distinct from both fine records, the non-selective instrument does not choose a fine record. $\square$

6.3 Environment records

Let the environment labels be $e_0, e_\uparrow, e_\downarrow, e_*$, with branch-correlating map

$$p_\uparrow \mapsto e_\uparrow, \quad p_\downarrow \mapsto e_\downarrow.$$

At fine resolution,

$$R_E(e_\uparrow) = r_\uparrow, \quad R_E(e_\downarrow) = r_\downarrow,$$

while at coarse resolution both are sent to r_* . Stable branch-correlated environmental records therefore preserve branch differentiation at fine resolution and merge at the non-selective surface. This is a finite environment-record theorem, not a full physical decoherence theory.

6.4 Projective branch maps

For

$$\rho = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

define finite projective branch maps

$$\mathcal{K}_\uparrow(\rho) = \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathcal{K}_\downarrow(\rho) = \begin{pmatrix} 0 & 0 \\ 0 & d \end{pmatrix},$$

and the non-selective channel

$$\mathcal{N}(\rho) = \mathcal{K}_\uparrow(\rho) + \mathcal{K}_\downarrow(\rho) = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}.$$

Then

$$\tilde{T}(\mathcal{K}_o(\tilde{\rho})) = \tilde{w}_{\tilde{\rho}}(o),$$

and, on the balanced witness,

$$\tilde{T}(\mathcal{N}(\tilde{\rho}_\psi)) = \tilde{T}(\tilde{\rho}_\psi), \quad \tilde{C}(\mathcal{N}(\tilde{\rho}_\psi)) = 0.$$

The channel preserves trace on the balanced witness, removes off-diagonal coherence, recovers the instrument branches, and still does not select a fine record at the non-selective surface.

Lemma 6.2 (Finite quantum correspondence). *The finite carrier is a role-faithful two-branch quantum specialization. The numerator vector $\tilde{\psi} = (1, 1)$ represents the balanced state $\psi = (1, 1)/\sqrt{2}$; the numerator density table $\tilde{\rho}_\psi = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ represents $\rho_\psi = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$; the maps $\mathcal{K}_\uparrow$ and $\mathcal{K}_\downarrow$ implement diagonal projection in the declared basis; branch-map traces agree with density weights; effect weights*

agree with branch-map traces; and zero-product repeatability aligns the projector and branch-map surfaces.

Proof. Each clause is a displayed finite calculation on the declared two-branch carrier. The normalization convention supplies the common denominator two, the branch maps retain exactly the relevant diagonal numerator, the trace functional reads that retained numerator, and the projector product and branch-map composition both annihilate the opposite branch. The lemma is finite and role-faithful; it is not a reconstruction of general quantum measurement theory. $\square$

6.5 Finite effects and repeatability

The finite effect carrier is

$$\mathcal{E}_2 = \{0, E_\uparrow, E_\downarrow, I\}.$$

The weight functional is

$$\tilde{e}_\uparrow(\tilde{\rho}) = \tilde{\rho}_{\uparrow\uparrow}, \quad \tilde{e}_\downarrow(\tilde{\rho}) = \tilde{\rho}_{\downarrow\downarrow}, \quad \tilde{e}_I(\tilde{\rho}) = \tilde{T}(\tilde{\rho}), \quad \tilde{e}_0(\tilde{\rho}) = 0.$$

Complementary effect completeness is

$$E_\uparrow \oplus E_\downarrow = I.$$

Here $\oplus$ denotes the declared complementary-effect join on this finite carrier. It is not asserted as a general effect-algebra operation; the formalization totalizes the finite function for definitional convenience, while the manuscript uses only the declared complementary completeness case.

The projector product satisfies

$$P_o^2 = P_o, \quad P_\uparrow P_\downarrow = P_\downarrow P_\uparrow = 0.$$

At the map level,

$$\mathcal{K}_o \circ \mathcal{K}_o = \mathcal{K}_o, \quad \mathcal{K}_\uparrow \circ \mathcal{K}_\downarrow = \mathcal{K}_\downarrow \circ \mathcal{K}_\uparrow = 0, \quad \mathcal{N} \circ \mathcal{N} = \mathcal{N}.$$

Repeatability preserves the branch already supplied to a selective operation. It does not convert the non-selective operation into a selector. The public v0.9.1 projector layer aligns effect-level zero-product orthogonality with branch-map annihilation: selective repeatability preserves a supplied branch, while the non-selective operation remains a merged record surface rather than a selector.

7 Measurement records as evidential anchors and fixed-domain evidential ordering

Section 3 isolated the quotient structure of admissible record-fixation assignments and identified fixation commutation as the bridge invariant. The present section uses that bridge to turn the quotient into a measurement theorem about evidence. Measurement records are evidential anchors; once that role is explicit, divergence among admissible quotient classes is no longer a representational curiosity but a question about whether the theory fixes its own evidential output.

Central hinge. The quotient theorem identifies which divergences among admissible record-fixation assignments are standing-bearing, and Section 3 proves that fixation commutation is the bridge invariant for theory-level evidential use. The present section therefore asks a narrower question: when quotient-relevant divergence survives into comparative ordering role for constructed hypotheses, does the theory fix its own evidential output?

For readability, this section often abbreviates “fixed-domain ordering of constructed hypotheses under admissible record reuse” as *fixed-domain evidential ordering*. The full phrase remains in definition and theorem headings where the exact constructional role is load-bearing.

7.1 Measurement records as evidential anchors

Definition 7.1 (Evidential anchor). A measurement record $r \in \mathcal{R}$ is an *evidential anchor* for a diachronic measurement construction token $d \in \mathcal{D}$ just in case the theory-level use of d depends on taking r as the standing realization of d and then using the induced comparison-state for constructed hypotheses $\text{Prop}_\rho(d)$ to compare rival hypotheses in $\mathcal{H}$.

Definition 7.2 (Theory-level evidential ordering). A *fixed-domain evidential-ordering verdict* is a comparative evidential verdict attributable to the theory on the fixed scope rather than to an unresolved fixation among admissible record-fixation assignments. Equivalently, such a verdict is well-defined only if the theory’s admissible use does not leave the comparative ordering role for constructed hypotheses of its measurement records unsettled across admissible standing-equivalence classes. This is the abbreviated form of fixed-domain ordering of constructed hypotheses under admissible record reuse.

Definition 7.3 (Comparative hypothesis ordering induced by a record). Let $d \in \mathcal{D}$ be an evidential construction token and let $\rho \in \text{ARef}$ be an admissible record-fixation assignment. The *comparative hypothesis ordering induced by d under ρ* is the function

$$\text{Ord}_\rho^d(H_i, H_j) := \text{CR}(\text{Prop}_\rho(d))(H_i, H_j) \quad (H_i, H_j \in \mathcal{H}).$$

A fixed-domain evidential ordering built from d is construction-relative only if this ordering is invariant on the admissible quotient classes relevant to d .

Proposition 7.4 (Evidential use of records is assignment-sensitive). *Let $d \in \mathcal{D}$ be a diachronic measurement construction token used as an evidential anchor. Then every admissible record-fixation assignment $\rho \in \text{ARef}$ determines a corresponding comparative hypothesis ordering Ord_ρ^d on $\mathcal{H}$. Therefore any unresolved variation among admissible quotient classes is already variation in evidential use unless all such orderings coincide.*

Proof. By definition of admissible record-fixation assignment, $\rho(d)$ is a physically realized and admissible branch-record pair for d . By definition of the induced comparison-state for constructed hypotheses, $\rho(d)$ determines $\text{Prop}_\rho(d) \in \mathcal{E}$. By definition of comparative ordering role for constructed hypotheses, $\text{CR}(\text{Prop}_\rho(d))$ is a function on ordered pairs of hypotheses. So Ord_ρ^d is well-defined for every admissible assignment. If different admissible quotient classes induce different orderings, then the record is doing different comparative evidential work under those classes. Hence unresolved

variation among admissible quotient classes is already variation in evidential use unless all orderings coincide. $\square$

7.2 The representational hinge

Lemma 7.5 (Two-case exhaustion for evidential record-fixation). *Let $d \in \mathcal{D}$ be used evidentially. Then one of the following mutually exclusive alternatives holds:*

1. *all admissible record-fixation assignments lie in a single $\approx_{\text{MR}}$ -class on the evidential use of d ; or*
2. *there exist admissible record-fixation assignments whose evidential values for d lie in distinct $\approx_{\text{MR}}$ -classes.*

No third same-scope case exists.

Proof. Either the admissible assignments agree on the standing-bearing evidential value of d , or they do not. If they do, then they lie in one $\approx_{\text{MR}}$ -class on that use. If they do not, then there exist at least two distinct $\approx_{\text{MR}}$ -classes. By Definition 2.19 and Proposition 2.43, no third same-scope standing-bearing locus remains. $\square$

Theorem 7.6 (No auxiliary parameterization of fixed-domain evidential ordering). *Let T be a theory and let ARef be the class of its admissible record-fixation assignments for the fixed measurement setting. If comparative evidential verdicts for T vary across admissible $\approx_{\text{MR}}$ -classes, then those verdicts are not attributable to T as such. They are attributable only to the pair consisting of T together with the fixed quotient class. Consequently, such verdicts do not constitute fixed-domain evidential ordering in the sense relevant to the target explanatory practice.*

Proof. By Definition 7.2, a fixed-domain evidential-ordering verdict must be attributable to the theory on the fixed scope rather than to a further unresolved fixation among admissible record-fixation assignments. Suppose instead that distinct admissible $\approx_{\text{MR}}$ -classes yield different comparative orderings for some evidential construction token d . Then the comparative support relation between hypotheses is not fixed by T alone. It varies with the fixed admissible quotient class. In that case the verdict is attributable only to $(T, [\rho])$ for some fixed class $[\rho] \in \text{ARef} / \approx_{\text{MR}}$, not to T as such. Such verdicts are therefore assignment-parameterized rather than theory-level. $\square$

Theorem 7.7 (Theory-level evidential-stabilization criterion). *For an evidential construction token $d \in \mathcal{D}$, fixed-domain evidential ordering is well-defined if and only if the map*

$$[\rho] \longmapsto \text{Ord}_\rho^d$$

from $\text{ARef} / \approx_{\text{MR}}$ to comparative hypothesis orderings is constant. Equivalently, if there exist admissible assignments whose comparative ordering role for constructed hypotheses differ, then no theory-level evidential-ordering verdict is fixed by the theory as such.

Proof. If the displayed map is constant, then all admissible assignments induce the same comparative ordering for d , so the verdict is attributable to the theory on the fixed scope rather than to a fixed admissible quotient class. Conversely, if the map is not constant, then there exist admissible assignments inducing different comparative orderings. By Corollary 3.23, that divergence is quotient-relevant and therefore standing-bearing rather than representational skin. By Theorem 7.6, the

resulting verdict is attributable only to the pair consisting of the theory together with a fixed admissible quotient class. Hence no theory-level evidential-ordering verdict is fixed by the theory as such. $\square$

Corollary 7.8 (Evidential-order consequence). *If there exist admissible record-fixation assignments that differ in comparative ordering role for constructed hypotheses, then evidential ordering is assignment-relative rather than construction-relative on the fixed scope.*

Proof. Immediate from Theorem 7.7. $\square$

7.3 Fixation as the bridge invariant

Definitions 3.24 and 3.25, together with Theorem 3.27 and Corollary 3.28, were established already in Section 3. They are placed there because fixation is the bridge invariant between quotient structure and all later applications. The present section uses that bridge to state the evidential-stabilization criterion and the stabilization consequence for physically non-selective theories.

Theorem 7.9 (Record-fixation stabilization necessity). *If a physically non-selective theory is used to underwrite fixed-domain evidential ordering and admissible diachronic construction whose measurement records function as evidential anchors, then the theory or its admissible interpretive completion must restrict admissible record-fixation assignments determinately enough that fixation commutes with admissible transformation on the target construction.*

Proof. By Lemma 7.5, admissible assignments either collapse into a single standing-bearing class on the target construction or they do not. If they do not, then by Theorem 7.7 and Theorem 3.27, fixed-domain evidential ordering through measurement records is undefined on the fixed scope. But the standing conditional of Section 2 assumes that the theory is being used to underwrite theory-level evidential ordering and admissible diachronic construction in which measurement records function as evidential anchors. Therefore the role-divergent case is incompatible with the theory's intended explanatory use. The only remaining option is that the theory or its admissible interpretive completion restricts admissible record-fixation assignments determinately enough for fixation to commute with admissible redescription on the target construction. $\square$

Proposition 7.10 (Selector and representational-infrastructure normal form). *After support-only, bookkeeping, repair, and scope-change readings are removed, any successful stabilization strategy for admissible measurement-record-fixation terminates in one of two normal forms:*

1. selector form, in which singular record-fixation is fixed by privileging one history or one successor-relation as privileged; or
2. representational-infrastructure form, in which admissible assignments remain multiple but are restricted so that they collapse into one standing-equivalence class on the target construction.

No third same-scope stabilization form remains.

Proof. If a successful stabilization strategy fixes a unique standing realization outright, it has selector form. If it does not fix a unique standing realization outright but still preserves fixed-domain evidential ordering, then by Theorem 7.9 it must restrict admissible assignments strongly enough that fixation commutes and the assignments collapse into one standing-equivalence class on the

target construction. That is representational-infrastructure form. By Proposition 2.43, no third same-scope locus of admissibility-bearing work remains. $\square$

Proposition 7.11 (Scope of the main theorem). *Theorems 7.7 and 7.9 concern fixed-domain evidential ordering on the fixed scope. They do not deny that branch-internal construction may be perfectly determinate once a branch is taken as fixed.*

Proof. The hypotheses of Definitions 7.2 and 3.25 explicitly concern theory-level evidential use on the activated fixed-domain closure package. Nothing in those definitions denies local branch-internal determinacy once a particular branch-history is held fixed. The theorems only state that such local determinacy does not by itself suffice for fixed-domain evidential ordering on the fixed scope. $\square$

7.4 Compatibility does not discharge fixation

The finite stack supplies a concrete case in which all declared branch and weight surfaces agree while the standing fine-record role remains unfixed at the non-selective level. This is exactly the separation needed by the confirmation argument: stable evidential support and standing record-role occupancy are not the same structural achievement.

Proposition 7.12 (Compatibility does not discharge fixation). *Suppose the finite outcome, pointer, instrument, channel, effect, environment, trace, and weight surfaces commute on their declared branch content, and suppose every declared non-selective surface yields only the merged record r_* . Then cross-surface compatibility does not determine which fine record occupies the standing evidential role.*

Proof. Compatibility identifies the branch presentations with one another and identifies the non-selective presentations with r_* . It does not identify r_* with either $r_\uparrow$ or $r_\downarrow$. Therefore the compatibility theorem preserves the missing-selector boundary rather than eliminating it. $\square$

8 Boundary-trace fixation as the record-role hardening layer

The quotient and commutation machinery above shows what happens once admissible record-fixation assignments are in play. This section adds the remaining fixation layer: before a physically realized record can function as a theory-level evidential anchor, the construction must fix the role by which that record is used. Branch-local stability is not yet role-anchor fixation.

Boundary-trace fixation is not a constructive mechanism. Boundary-trace fixation is not introduced as a new physical posit, explanatory dynamics, collapse law, hidden selector, or ontological branch-selection mechanism. It is the constraint-theoretic classification that remains available after internal invariants, provenance/indexical markers, support-only roles, bookkeeping, hidden selectors, repair routes, and target shifts have been exhausted on the fixed measurement-record scope. Its status is therefore that of a forced residual admissible normal form rather than a constructed physical solution.

Definition 8.1 (Record-bearing role). A *record-bearing role* for a construction token $d \in \mathcal{D}$ is the role in which a physically realized record is used as the standing realization of d for retained-record reuse, prospective-record reuse, deliberation, or evidential comparison.

Definition 8.2 (Declared boundary trace). For a candidate set $\text{Cand}(d)$, a *declared boundary trace* is an instance of the boundary-trace admissibility relation, represented by the fixed-domain predicate

$$\text{Tr}_{Ch}^d : \text{Cand}(d) \rightarrow \{0, 1\}$$

whose value records whether a candidate branch-record pair is coupled to the declared detector, channel, apparatus, calibration line, environmental interface, or boundary condition relevant to the fixed measurement domain. Its trace source must be part of the declared measurement-domain architecture: apparatus channel, detector coupling, calibration lineage, predeclared record-location relation fixed by the measurement-domain architecture, environmental interface, or an explicitly declared boundary condition. The trace is admissibility-relevant only when it satisfies the boundary-trace admissibility relation of Definition 3.2: it is declared before role use, stable under channel-preserving identity transformations, outcome-independent, non-generative, and sourced in that declared architecture rather than by later outcome success or post-measurement salience.

Definition 8.3 (Singleton boundary-trace fixation). A declared boundary trace fixes the record-bearing role for d iff

$$\{x \in \text{Cand}(d) : \text{Tr}_{Ch}^d(x) = 1\} = \{x^*\}.$$

If the positive trace set is empty or non-singleton, the trace-fixation operator is undefined. Undefinedness is fail-closed, not partial reference, graded standing, or an invitation to add a tie-breaker.

Example 8.4 (Singleton trace witness). Let

$$\text{Cand}(d) = \{(b_U, r_U), (b_D, r_D)\}, \quad \text{Tr}_{Ch}^d : \text{Cand}(d) \rightarrow \{0, 1\}$$

be a declared channel trace. If

$$\text{Tr}_{Ch}^d(b_U, r_U) = 1, \quad \text{Tr}_{Ch}^d(b_D, r_D) = 0,$$

then the positive trace set is the singleton $\{(b_U, r_U)\}$ and the record-bearing role is fixed to that candidate. If both values are 0, the trace supplies no role-bearer. If both values are 1, the trace is non-singleton and supplies no singular role-bearer. If the trace value depends on outcome desirability, future evidential success, branch weight alone, observer preference, a decision policy, later calibration, or retrospective report success, then it is not a declared outcome-independent boundary trace; it is selector import, scalarized parameterization, or same-act repair. Singleton trace fixation fixes the evidential role of the record-bearing continuation. It does not actualize one branch, delete the other branch-local records, or add a collapse law.

Boundary trace is not an oracle. A boundary trace cannot inspect the environment, detector, future data, observer preference, branch weight, or later calibration result to choose a favorable record. Permitted trace status is declared before role use, stable under channel-preserving redescription, outcome-independent, bivalent, and non-generative. Anything else is support, bookkeeping, target change, repair, or hidden selector authority.

Theorem 8.5 (No arbitrary singleton trace). A *singleton-valued predicate on $\text{Cand}(d)$* fixes a record-bearing role only if its trace source is independently declared as part of the measurement domain

before role use and satisfies the stability, bivalence, outcome-independence, and non-generativity conditions of Definition 3.2. Otherwise the singleton predicate is a hidden selector, bookkeeping label, or scope change rather than admissible boundary-trace fixation.

Proof. For any nonempty candidate set, one can define an arbitrary predicate that is positive on exactly one candidate. Such a predicate has no admissibility force merely by being singleton-valued. To fix a record-bearing role on the same domain, the predicate must be sourced in a declared measurement architecture and preserved under lawful channel-preserving redescription. If it is chosen after the outcome, tuned to future evidential success, based on branch salience or preference, or otherwise not part of the declared domain, then it supplies exactly the selector authority excluded by the local kernel and same-scope closure discipline. Singleton cardinality is therefore necessary for singular fixation but not sufficient for admissible fixation. $\square$

Theorem 8.6 (Record-role asymmetry normal form). *Let $d \in \mathcal{D}$ be a diachronic measurement construction token with $|\text{Cand}(d)| \geq 2$. Any putative same-domain source of singular record-role fixation for d has a terminal classification in one of the following normal-form classes:*

1. *internal constructional invariant;*
2. *declared provenance, indexical, or centered successor marker;*
3. *declared singleton boundary trace;*
4. *inadmissible extra-domain preference, including pragmatic salience, semantic intention, scalar coupling strength, posterior success, decision weight, delayed calibration, branch convenience, or undeclared environmental appeal.*

The first three are the only admissible same-domain asymmetry sources. The fourth is not an admissible source: if declared as new structure it is a scope change, and if left undeclared it is selector smuggling. When a proposal carries more than one descriptive feature, its terminal classification is determined by the feature that actually fixes the standing record role.

Proof. A same-domain record-role selector must fix one candidate in $\text{Cand}(d)$ while preserving the fixed record practice, admissibility boundary, and lawful transformations. If the decisive feature is internal to the measurement construction, it is class (1). If it is a declared origin, index, centered relation, or production-history marker, it is class (2). If it is neither internal nor provenance/indexical but is still declared in the measurement domain, then the only remaining same-domain location is the boundary between the candidate record and the declared apparatus/channel/environmental interface; because standing is bivalent, the relevant object is the bivalent trace of that boundary relation, giving class (3). All other proposals either reduce to one of these classes after explicit declaration or decide by a preference not invariantly fixed in the domain. That is class (4). $\square$

Theorem 8.7 (Asymmetry exhaustion for admissible measurement-record fixation). *Let $d \in \mathcal{D}$ be a measurement construction token whose candidate record-bearing realizations are not already singleton-fixed. If internal constructional invariants and declared provenance/indexical markers do not produce a singleton record-role anchor for d , then any remaining admissible same-domain singular fixation of d must occur through singleton boundary-trace fixation. If no declared, stable, bivalent, outcome-independent boundary trace has exactly one positive candidate in $\text{Cand}(d)$, then the record-bearing role for d is undefined on the same scope.*

Proof. By Theorem 8.6, every decisive same-domain asymmetry source is internal, provenance/indexical, boundary-trace, or inadmissible extra-domain preference. The first two classes are unavailable by hypothesis. The fourth class is not admissible on the fixed scope. If left undeclared, it is selector smuggling; if declared, it changes the domain or the target construction. Therefore the only admissible remaining class is a declared boundary trace. Because standing is bivalent and the role is singular, that trace can fix d only in the singleton-positive case of Definition 8.3. Empty, non-singleton, unstable, scalarized, outcome-conditioned, or delayed traces leave the role unfixed rather than weakly fixed. $\square$

Corollary 8.8 (Residual boundary-trace forcing). *After internal and provenance/indexical exhaustion, boundary-trace fixation is not an optional presentation of record-role anchoring. It is the residual admissible determinacy route on the fixed scope.*

Proof. This is the contrapositive form of Theorem 8.7: a same-domain record-role fixation that is neither internal nor provenance/indexical and is not boundary-trace mediated must fall into the inadmissible extra-domain class. $\square$

8.1 Boundary-trace fixation as physical record-role anchoring

Boundary-trace fixation does not add a collapse law, actualize an ontic world, privilege a hidden trajectory, or produce a physical outcome. It fixes the admissible physical record role by which a physically realized record is allowed to function as the standing record for a construction token on a declared domain. The distinction is essential: physical role anchoring concerns the determinate standing of an already available candidate record under the kernel roles; it need not be identified with a particular collapse dynamics.

Proposition 8.9 (Boundary traces are abstract role-fixing invariants). *A boundary trace is not a detector, apparatus, classical readout, environmental mechanism, or laboratory technology. It is a fixed admissibility-preserving asymmetry relation on the candidate record set, represented as a bivalent predicate and preserved by the identity transformations of the declared measurement domain. Concrete detectors, calibration lines, apparatus channels, or environmental interfaces matter only insofar as they instantiate that abstract trace relation.*

Proof. The formal object in Definition 8.2 is the predicate $\text{Tr}_{Ch}^d : \text{Cand}(d) \rightarrow \{0, 1\}$, not the physical device or environment considered in open-ended form. A device or environment may contain indefinitely many facts, most of which do no admissibility work. Only the stable, declared, outcome-independent boundary-status relation on $\text{Cand}(d)$ enters the fixed-domain role assignment. Therefore the theorem is not technologically contingent; it is a constraint on admissible asymmetry. $\square$

Proposition 8.10 (Evidential anchoring requires role-anchor fixation). *A physically realized branch-local record is not yet a theory-level evidential anchor merely because it is stable within its branch. It becomes an evidential anchor for d only when the record-bearing role for d is fixed by an admissible same-domain asymmetry in the sense of Theorem 8.6.*

Proof. Branch-local stability establishes that a record can support local reuse once that branch is already taken as the relevant continuation. The evidential-anchor role requires more: it requires that the record be the standing realization of d for the fixed downstream evidential practice. The

fixing source must be an internal invariant, a declared provenance/indexical marker, or a declared singleton boundary trace; otherwise it is selector smuggling or scope change. $\square$

Corollary 8.11 (Bare Everett lacks boundary-trace fixation). *The bare Everettian base package does not by itself supply singleton boundary-trace fixation for the pre-measurement record-bearing role. It supplies branch-local records, entanglement correlations, and decoherence stability, but not a declared, stable, outcome-independent, singleton channel predicate that fixes one branch-record pair as the standing realization of the pre-measurement construction.*

Proof. Theorem 5.8 shows that the bare package realizes multiple stable successor histories while supplying no physical selector. A singleton boundary trace would be a declared same-domain asymmetry fixing exactly one candidate branch-record pair by channel status. The bare unitary, entangling, decohering package contains no such declared channel predicate: decoherence supplies branch-local stability for each candidate, not singleton positive trace status for one candidate. $\square$

Theorem 8.12 (Everettian asymmetry exhaustion theorem). *For the bare Everettian base package applied to the target measurement construction, no admissible same-domain singular record-role fixation is supplied by any of the three admissible asymmetry classes: internal constructional invariant, declared provenance/indexical marker, or declared singleton boundary trace. Hence bare Everettian structure does not by itself fix the record-bearing role for pre-measurement evidential use on the fixed scope.*

Proof. Internal fixation fails because Theorem 5.8 and Corollary 5.7 show that the bare physical package realizes multiple stable successor histories while supplying no law, state variable, dynamically privileged trajectory, or physically relevant singleton asymmetry. Provenance/indexical fixation fails because centered descriptions and successor typing preserve candidate multiplicity unless an additional rule privileges one center or successor as the standing realization; by Theorem 5.30, those resources presuppose rather than supply the missing typing. Boundary-trace fixation fails by Corollary 8.11: the bare package contains no declared singleton channel predicate on the candidate branch-record set. By Theorem 8.7, once those three classes fail, the only remaining moves are inadmissible selector import, scope change, or constructional deflation. $\square$

Theorem 8.13 (Boundary-trace inheritance theorem). *If retained-record reuse, prospective-record reuse, deliberation, and fixed-domain evidential ordering inherit a common constructional lineage through a measurement record, then they also inherit the fixation status of the role-anchor for that lineage. An unfixed or non-singleton boundary trace at the record-bearing role cannot be repaired separately by any one downstream practice.*

Proof. By Theorem 11.13, the downstream practices factor through the same measurement-record quotient and inherit quotient-relevant lineage divergence together. A boundary trace, when present, fixes the record-bearing role that supplies that lineage. If the trace is absent, non-singleton, unstable, scalarized, or delayed, then the role-anchor is not fixed on the same scope. Any downstream attempt to fix a record anyway introduces a new selector, outcome-conditioned repair, or scope change. $\square$

8.2 Finite trace-channel constructor

The finite formalization exposes the boundary-trace role as a constructor. A finite channel packet C contains a fine-trace regime together with declaration before use, stability under record-preserving

maps, outcome-independent declaration, and non-generativity. From those fields one obtains a boundary-trace object with singleton-positive status:

$$\text{FiniteTraceReady}(C) \Rightarrow \text{BoundaryTraceFixation}(C).$$

This theorem does not infer singleton trace from bare decoherence or environment correlation. It states the readiness conditions that a finite physical channel must already satisfy to occupy the boundary-trace role.

Corollary 8.14 (Finite boundary occupation). *When a finite trace channel satisfies the record-fixation readiness conditions, it is an explicit finite occupation of the manuscript's general boundary-trace normal form.*

9 Cross-surface compatibility and the finite measurement boundary

The finite construction is not a sequence of unrelated encodings. It proves that the outcome, pointer, instrument, channel, effect, environment, trace, and Born-weight surfaces route the same finite branches and the same non-selective merged record. This is the mathematical center of the integrated Lean development.

9.1 Fine-record compatibility

Let the canonical fine-record map be

$$R^\circ(\uparrow) = r_\uparrow, \quad R^\circ(\downarrow) = r_\downarrow.$$

For every $o \in \Omega$, the outcome-indexed selective record maps satisfy

$$R^\circ(o) = R_{\text{pointer}}(o) = R_{\text{instrument}}(o) = R_{\text{channel}}(o) = R_{\text{effect}}(o).$$

The environment fine-record surface is basis-indexed in the formalization, so it is routed through the pointer/environment compatibility theorem rather than inserted into this outcome-indexed display. In the present notation this says $R_E(b_o) = R_{\text{pointer}}(o)$ for the basis state b_o corresponding to outcome o . This states equality of physical record presentations, not mere adjacency of modules.

Theorem 9.1 (Fine-record compatibility). *All declared selective fine-record surfaces agree branch-by-branch on the canonical finite record.*

Proof. Each selective map sends $\uparrow$ to $r_\uparrow$ and $\downarrow$ to $r_\downarrow$. The proof is a two-case calculation over Ω . $\square$

9.2 Coarse-record compatibility

The canonical coarse record is r_* . The declared non-selective or coarse surfaces agree as

$$R_{\text{pointer}}^{\text{coarse}} = R_{\text{instrument}}^{\text{ns}} = R_{\text{channel}}^{\text{ns}} = R_{\text{effect}}^{\text{ns}} = R_{\text{environment}}^{\text{coarse}} = r_*.$$

Theorem 9.2 (Coarse-record compatibility). *Every declared non-selective or coarse surface in the finite stack agrees on the merged record r_* .*

Proof. The non-selective records are not admitted merely by being named. They sit inside the finite role package: branch weights normalize, selective branches retain branch records, non-selective channels preserve total numerator weight while erasing fine off-diagonal coherence, and the corresponding instrument, channel, effect, and environment surfaces all route the same merged record. This theorem records the common coarse value only; the claim that no such surface functions as a fine-record selector is reserved for Theorem 9.4, after the selector disjunction has been defined. $\square$

9.3 Weight compatibility

For an encoded density table ρ and outcome o ,

$$\tilde{T}(\mathcal{K}_o(\tilde{\rho})) = \tilde{e}_o(\tilde{\rho}) = \tilde{w}_{\tilde{\rho}}(o).$$

For the balanced amplitude/density pair,

$$\tilde{T}(\mathcal{K}_o(\tilde{\rho}_\psi)) = \tilde{e}_o(\tilde{\rho}_\psi) = \tilde{w}_{\tilde{\rho}_\psi}(o) = \tilde{w}_{\tilde{\psi}}(o) = 1,$$

with normalized value $1/2$ after division by the common denominator two. The branch-map trace, effect weight, density weight, and squared-amplitude weight agree in numerator form. Agreement of weights is not selection of one record.

9.4 Branch separation

The fine records remain separated:

$$r_{\uparrow} \neq r_{\downarrow}.$$

Combining this with Theorem 9.1 yields branch separation on every selective fine-record surface. Compatibility therefore does not collapse branch difference; it shows that each surface tracks the same branch difference.

9.5 Finite measurement-role adequacy

Theorem 9.3 (Finite measurement-role adequacy). *The finite stack is a constrained measurement-role package rather than an arbitrary naming convention. It jointly satisfies denominator-two normalization, amplitude/record weight agreement, density-trace and amplitude agreement on the balanced witness, selective instrument decomposition, non-selective branch-sum formation, trace preservation with coherence removal, branch-map trace agreement with density weight, effect-weight agreement with branch-map trace, zero-product finite projector orthogonality, branch-map annihilation, selective branch-record separation, and non-selective merged-record agreement.*

Proof. Each condition is proved by a finite calculation or finite case split on the declared carrier. The amplitude and density layers supply the normalized numerator arithmetic; the instrument and branch-map layers supply selective/non-selective decomposition; the effect layer supplies the weight/trace equality; the projector layer supplies idempotence and zero cross-products; and the record surfaces supply fine separation and coarse merger. The no-selector theorem below therefore follows from a role-faithful finite measurement package, not from attaching arbitrary labels to records. $\square$

9.6 The finite selector disjunction

Define the candidate non-selective selectors:

$$\text{Sel}_{\text{finite}} := \text{Sel}_{\text{state}} \vee \text{Sel}_{\text{instrument}} \vee \text{Sel}_{\text{environment}} \\ \vee \text{Sel}_{\text{channel}} \vee \text{Sel}_{\text{effect}} \vee \text{Sel}_{\text{repeatability}}.$$

Here $\text{Sel}_{\text{state}}$ means that the non-selective state determines a fine record; $\text{Sel}_{\text{instrument}}$ means that the non-selective instrument record is $r_{\uparrow}$ or $r_{\downarrow}$; and the remaining predicates say the analogous thing for the environment, channel, effect, and repeatability surfaces.

Theorem 9.4 (No hidden selector on the declared non-selective surfaces). *On the declared finite measurement stack,*

$$\neg \text{Sel}_{\text{finite}}.$$

Proof. Assume $\text{Sel}_{\text{finite}}$. Split the disjunction into its six cases. The state case contradicts the fine/coarse record theorem. The instrument case contradicts Theorem 6.1. The environment case contradicts the finite environment non-selection result. The channel case contradicts the channel non-selection result. The effect case contradicts the effect non-selection result. The repeatability case contradicts the repeatability non-selection result. Thus every disjunct is impossible. $\square$

9.7 Closed stack and boundary certificate

The finite closed stack consists of record dynamics, squared-amplitude weights, density/instrument behavior, environment records, branch maps, effects, projectors/repeatability, and cross-surface compatibility. Its global invariants are:

1. no declared non-selective surface supplies a fine-record selector;
2. branch traces, effect weights, density weights, and squared-amplitude weights agree;
3. selective fine records and non-selective coarse records agree across the declared surfaces.

Theorem 9.5 (Finite measurement-boundary certificate). *On the declared two-branch kernel-governed quantum domain, no enumerated non-selective surface supplies a fine-record selector; all selective fine-record surfaces agree branch-by-branch; all non-selective coarse-record surfaces agree on r_* ; and $r_{\uparrow} \neq r_{\downarrow}$.*

Proof. The first clause is Theorem 9.4. The second is Theorem 9.1. The third is Theorem 9.2. The fourth is the branch-separation theorem of the finite record carrier. $\square$

Theorem 9.6 (Finite compatible-nonselection theorem). *On the declared two-branch kernel-governed quantum measurement carrier, branch weights are normalized; squared-amplitude weights, density weights, branch-map traces, and effect weights agree in numerator form; selective fine-record surfaces agree branch-by-branch; non-selective surfaces agree on the merged record; fine records remain separated; and the state, instrument, environment, channel, effect, and repeatability surfaces fail to supply a fine-record selector. Consequently,*

$$\text{weighting} + \text{coherence removal} + \text{branch correlation} \\ + \text{effect completeness} + \text{zero-product orthogonality} + \text{repeatability} \not\Rightarrow \text{fine-record selection}.$$

Proof. The normalization and weight-compatibility clauses come from the denominator-two amplitude, density, branch-map, and effect calculations. The record clauses are Theorems 9.1 and 9.2. Branch separation is the fine-record separation theorem. The selector clause is Theorem 9.4. The displayed implication is the conjunction of those facts. $\square$

Integrated AASC measurement theorem. Let $\mathfrak{M}$ instantiate a non-degenerate fixed-domain measurement-record target. Then the AASC kernel is active by necessity; the local measurement-record packet types admissible record construction; the finite quantum specialization exhibits compatible branch, weight, density, instrument, channel, effect, environment, and repeatability surfaces; none of the declared non-selective surfaces supplies a fine-record selector; and therefore normalized weighting, coherence removal, record stability, effect completeness, zero-product orthogonality, and repeatability do not discharge the fine-record role-fixation burden. Any complete same-domain account must occupy an admissible fixation locus.

10 A toy model of evidential-order divergence

Abstract theorems are necessary, but here a concrete toy model helps show exactly what the problem looks like in practice. The goal is not to model a realistic laboratory in every detail. The goal is to exhibit, in the smallest possible setting, how a single branching measurement episode can generate opposed fixed-domain orderings of constructed hypotheses under admissible record reuse when admissible record-fixation assignments are left unresolved. In boundary-trace terms, the same model also witnesses absence of singleton trace-fixation: unless one candidate record is trace-positive under a declared channel predicate, the evidential role remains unfixed on the same scope.

10.1 The model

Definition 10.1 (Toy branching evidential ordering model). *A toy branching evidential ordering model* is a measurement scenario $\mathcal{M}_{\text{toy}}$ with the following data:

1. two rival hypotheses $\mathcal{H} = \{H_U, H_D\}$;
2. two physically realized post-measurement histories $\mathcal{B} = \{b_U, b_D\}$;
3. two corresponding branch records $r_U := \text{Rec}(b_U)$ and $r_D := \text{Rec}(b_D)$;
4. a single diachronic construction token $d \in \mathcal{D}$ whose intended English gloss is “this is the measurement record I am using as evidence”;
5. two admissible record-fixation assignments $\rho_U, \rho_D \in \text{ARef}(\mathcal{M}_{\text{toy}})$ with

$$\rho_U(d) = (b_U, r_U), \quad \rho_D(d) = (b_D, r_D);$$

6. induced comparison-state for constructed hypotheses $E_U := \text{Prop}_{\rho_U}(d)$ and $E_D := \text{Prop}_{\rho_D}(d)$ with comparative ordering role for constructed hypotheses satisfying

$$\text{CR}(E_U)(H_U, H_D) = 1, \quad \text{CR}(E_D)(H_U, H_D) = -1.$$

Intuitively, E_U is a record that favors H_U over H_D , while E_D favors H_D over H_U . The physical non-selectivity of the scenario is represented by the fact that both branch-record pairs are realized and both are admissible assignments for the same diachronic construction token d .

Proposition 10.2 (Hypothesis-order instability witness). *Let H_U, H_D be rival hypotheses, let d be the fixed measurement construction token in a toy branching evidential ordering model, and let ρ_U, ρ_D be admissible record-fixation assignments mapping that same pre-measurement construction to realized records r_U, r_D . If under ρ_U the induced evidential proposition favors H_U over H_D , while under ρ_D the induced comparison-state for constructed hypotheses favors H_D over H_U , then comparative evidential ordering is not fixed by the theory alone.*

Proof. By Definition 10.1, ρ_U and ρ_D are admissible record-fixation assignments for the same diachronic construction token d , and

$$\text{CR}(\text{Prop}_{\rho_U}(d))(H_U, H_D) = 1, \quad \text{CR}(\text{Prop}_{\rho_D}(d))(H_U, H_D) = -1.$$

So under ρ_U the record referred to by d favors H_U over H_D , while under ρ_D the record referred to by d favors H_D over H_U . The comparative ordering of the hypotheses therefore depends on which admissible assignment is used. Hence comparative evidential ordering is not fixed by the theory alone. $\square$

Theorem 10.3 (Hypothesis-order reversal in the toy model). *In every toy branching evidential ordering model, the comparative ordering of H_U and H_D reverses across admissible record-fixation assignments. Consequently, the fixed-domain evidential ordering built from d is assignment-relative rather than construction-relative.*

Proof. By Definition 10.1,

$$\text{Ord}_{\rho_U}^d(H_U, H_D) = \text{CR}(E_U)(H_U, H_D) = 1$$

and

$$\text{Ord}_{\rho_D}^d(H_U, H_D) = \text{CR}(E_D)(H_U, H_D) = -1.$$

So under ρ_U , the relevant record favors H_U over H_D , whereas under ρ_D , the relevant record favors H_D over H_U . The comparative ordering is therefore reversed across admissible assignments.

Suppose the fixed-domain evidential ordering built from d were construction-relative. Then, by Definition 7.2 and Theorem 7.7, the induced comparative ordering would have to be invariant across admissible assignments. But it is not. Therefore the construction token is assignment-relative rather than construction-relative. $\square$

Corollary 10.4 (The same physical branching episode can support opposed evidential-order verdicts). *In the toy model, the same physical branching episode supports both of the following verdicts under different admissible assignments:*

$$\begin{array}{ccc} \text{“the record referred to by } d \text{ confirms } H_U \text{ over } H_D \text{”} & \text{and} & \text{“the record referred to by } d \\ & & \text{confirms } H_D \text{ over } H_U \text{”}. \end{array}$$

Proof. The first verdict holds under ρ_U because $\text{Ord}_{\rho_U}^d(H_U, H_D) = 1$. The second verdict holds under ρ_D because $\text{Ord}_{\rho_D}^d(H_U, H_D) = -1$, which is equivalent to saying that the record favors H_D

Assignment	Record referred to by d	Boundary-trace status	Induced comparison state	Comparative role on (H_U, H_D)
ρ_U	(b_U, r_U)	unfixed unless singleton positive	E_U	favours H_U over H_D
ρ_D	(b_D, r_D)	unfixed unless singleton positive	E_D	favours H_D over H_U

Table 3: The toy model makes the danger concrete: the same branching measurement episode supports opposed hypothesis orderings when admissible record-fixation assignments remain unresolved; without a singleton boundary trace, the record role fails closed.

over H_U . By Theorem 10.3, both verdicts arise from the same branching episode under different admissible assignments. $\square$

Theorem 10.5 (Toy-model fixation failure). *In the toy model, if no declared, stable, outcome-independent boundary trace is singleton-positive on $\{(b_U, r_U), (b_D, r_D)\}$, then admissible evidential fixation for d is undefined on the fixed scope. The model therefore witnesses not only evidential-order reversal under divergent assignments, but fail-closed absence of record-role fixation.*

Proof. The candidate set for d contains the two branch-record pairs (b_U, r_U) and (b_D, r_D) . If both are trace-negative, the trace supplies no positive role-bearer. If both are trace-positive, the trace supplies no unique role-bearer. If no trace is declared, the alleged environmental or apparatus relation is not typed on the fixed domain. In each case Definition 8.3 leaves the trace-fixation operator undefined. By Theorem 8.7, after internal and provenance/indexical non-selection, no admissible same-domain source remains to fix d . $\square$

Remark 10.6. Table 3 is not a replacement for the theorem ladder; it is a visibility aid. The point of the toy model is to make plain that the problem is not some exotic representational subtlety. If admissible assignments are allowed to diverge in comparative ordering role for constructed hypotheses, the same physical episode can be used to support opposed verdicts about which hypothesis the evidence favors.

11 Diachronic reuse, anticipation, and deliberative guidance

The point of the measurement-record quotient is not confined to evidential comparison. The same records support retained-record reuse constructions, anticipatory construction tokens, and deliberative guidance. If those practices are treated as admissible on the fixed scope, then they must all inherit a common constructional lineage. This section makes that dependence explicit.

11.1 Retained-record reuse and prospective-record reuse

Definition 11.1 (Retained-record reuse construction). *A retained-record reuse construction is a diachronic construction token whose admissibility conditions depend on taking a present subject-stage to stand in the relevant same-scope lineage relation to a past record-bearing history.*

Definition 11.2 (Prospective-record reuse construction). An *prospective-record reuse construction* is a diachronic construction token whose admissibility conditions depend on taking a present subject-stage to stand in the relevant same-scope lineage relation to one or more future record-bearing histories.

Definition 11.3 (Constructional lineage). For an admissible record-fixation assignment ρ , the *constructional lineage* induced by ρ is the same-scope chain of subject-stage, successor-typing, and record-fixation relations through which the construction tokens in $\mathcal{D}$ receive their standing-bearing values under ρ .

Definition 11.4 (Diachronic construction token). A *diachronic construction token* is any construction token in the target explanatory practice whose standing-bearing value depends on locating a present subject-stage, a past or future measurement record, or both within the same constructional lineage.

Theorem 11.5 (Diachronic closure). *Any admissible diachronic construction factors through the same standing-preserving equivalence relation on $\mathbf{ARef}$, namely the measurement-record quotient $\mathbf{ARef}/\approx_{\text{MR}}$. Equivalently, there is no admissible diachronic operator whose standing-bearing content is independent of that quotient.*

Proof. By Definition 11.4, a diachronic construction token belongs to the target explanatory practice only insofar as it receives a standing-bearing value from the same record-sensitive profile map Prof used to define $\approx_{\text{MR}}$. By Theorem 3.17, every standing-bearing content space for admissible measurement-record-fixation assignments factors through the unique quotient $\mathbf{ARef}/\approx_{\text{MR}}$. So if a diachronic construction token failed to factor through that quotient, two assignments in the same $\approx_{\text{MR}}$ -class would have to yield different standing-bearing values for the construction token, contradicting the definition of $\approx_{\text{MR}}$ and Theorem 3.19. Hence any admissible diachronic construction token factors through the same standing-preserving equivalence relation. $\square$

Lemma 11.6 (Retained-record reuse admissibility conditions depend on record-fixation). *If two admissible assignments differ on which past record is the standing realization of a retained-record reuse construction, then they need not assign the same admissibility value to that construction token.*

Proof. A retained-record reuse construction is true only if the present subject-stage is related, under the relevant diachronic lineage, to the past record the construction token is about. Changing which record is taken as the standing realization can therefore change whether the construction token is true. Hence retained-record reuse admissibility conditions depend on record record-fixation. $\square$

Lemma 11.7 (Prospective-record reuse admissibility conditions depend on successor typing). *If two admissible assignments differ on which future record-bearing history is the standing realization relevant to a prospective-record reuse construction, then they need not assign the same admissibility value to that construction token.*

Proof. An prospective-record reuse construction attributes a future outcome to the present subject-stage. If admissible assignments differ on which future record-bearing history is the relevant continuation, then they may differ on whether the attributed outcome occurs in that history. So prospective-record reuse admissibility conditions depend on successor typing and record-fixation. $\square$

11.2 Deliberative guidance and deliberation

Definition 11.8 (Agential deliberation construction). *An agential deliberation construction token is a construction token whose standing-bearing use guides present action by appeal to future measurement-record outcomes treated as belonging to the same diachronic practice-package.*

Lemma 11.9 (Agential guidance requires typed future records). *If a deliberative construction token is to guide present action by appeal to future measurement outcomes, then the future records appealed to must be typed as belonging to the same constructional lineage as the present subject-stage.*

Proof. Deliberation links a present act to future consequences for the same diachronic practice. If the future records appealed to are not typed as belonging to the relevant lineage, then the deliberative construction token has no determinate target for guidance. So agential guidance requires typed future records. $\square$

Theorem 11.10 (Decision-theoretic dependence for literal same-domain constructional prospective-record reuse). *Any literal same-domain constructional decision-theoretic use of branching quantum theory presupposes prior successor typing and record-fixation. It does not create them.*

Proof. A decision-theoretic rule can distribute weights or preferences only over already identified future alternatives. In the present setting those alternatives are future record-bearing histories. If successor relations and the relevant records are not already typed, then the decision rule lacks a determinate space of future options to which its weights or preferences apply. So decision-theoretic use presupposes rather than creates record-fixation. $\square$

Proposition 11.11 (Branch-aggregative retyping). *A framework may avoid literal same-domain constructional prospective-record reuse by retyping deliberation into branch-aggregative or impersonal terms. Such a framework does not remove the need for constructional typing; it changes the construction whose admissibility conditions are at issue.*

Proof. If a framework replaces literal same-domain constructional prospective-record reuse by impersonal aggregation over branches, then it no longer uses the same diachronic same-domain construction tokens as the target explanatory practice. The move therefore avoids the strongest lineage requirement only by changing the construction at issue. That is retyping, not elimination of the prior dependence. $\square$

11.3 A unified lineage theorem

Theorem 11.12 (Identity-preserving lineage theorem). *On the fixed-domain closure package, retained-record reuse, prospective-record reuse, deliberation, and fixed-domain evidential ordering all presuppose a fixed identity-preserving constructional lineage through the measurement records used in those practices.*

Proof. By Lemma 11.6, retained-record reuse constructions depend on which past records are the relevant standing realizations. By Lemma 11.7, prospective-record reuse constructions depend on successor typing and future record-fixation. By Lemma 11.9, agential guidance depends on typed future records. By Proposition 7.4 and Theorem 7.7, fixed-domain evidential ordering depends on the standing-bearing evidential use of the same records. These four practices therefore do not rest

on four independent structures. They rest on one identity-preserving constructional lineage through the relevant measurement records. $\square$

Theorem 11.13 (Diachronic inheritance theorem). *On the fixed scope, retained-record reuse, prospective-record reuse, deliberation, and fixed-domain evidential ordering all factor through the same measurement-record quotient. Consequently, any quotient-relevant divergence in lineage assignment is inherited by all four practices together rather than by one of them in isolation.*

Proof. By Theorem 11.5, every admissible diachronic construction token factors through the unique quotient $\text{ARef}/\approx_{\text{MR}}$; this is a special case of the general measurement-record quotient theorem. Retained-record reuse and prospective-record reuse constructions depend on which record-bearing histories are taken as their standing realizations (Lemmas 11.6 and 11.7); agential deliberation depends on the same typed future records (Lemma 11.9); and fixed-domain evidential ordering depends on the standing-bearing evidential use of those records (Theorem 7.7). Therefore all four practices factor through the same measurement-record quotient. If divergence is quotient-relevant, then it is inherited by each practice through that shared factorization. $\square$

Theorem 11.14 (Diachronic practice dependence theorem). *If admissible record-fixation assignments remain divergent on the fixed scope in a way that changes the constructional lineage of the relevant measurement records, then retained-record reuse, prospective-record reuse, deliberation, and fixed-domain evidential ordering become assignment-relative together. No one of the four can be stabilized in isolation while the shared quotient-level lineage remains divergent.*

Proof. By Theorem 11.12, the four practices share a common constructional lineage. If admissible assignments diverge in a way that changes that lineage, then by Lemmas 11.6 and 11.7 they can assign different truth conditions to retained-record reuse and prospective-record reuse constructions; by Lemma 11.9 they can change the future records relevant to deliberation; and by Corollary 7.8 they can change fixed-domain evidential ordering. Since the dependence is shared, the instability is shared. The four practices therefore become assignment-relative together. $\square$

Corollary 11.15 (Structural priority over probabilistic treatment). *Questions about branch-weighting, credence, or rational choice are downstream of the lineage issue. They presuppose, rather than settle, the fixed constructional lineage required by the target explanatory practice.*

Proof. Immediate from Theorems 11.10 and 11.12. A probabilistic or decision-theoretic treatment can operate only once the relevant future records and successor relations have already been typed into a fixed lineage. $\square$

12 Persistence-bearing bridge for measurement records

The preceding section shows that retained-record reuse, prospective-record reuse, deliberation, and fixed-domain evidential comparison share one constructional lineage through measurement records. This bridge section states the physics-facing consequence of that lineage result. It does not import the full forced-existence arc as an additional premise. It uses that arc only to identify the kind of structure a measurement record must have when it functions as a standing-bearing object in a physically realized downstream practice: a continuation locus whose role is fixed as determinate

enough for persistence-bearing reuse. The continuation-locus language is used as non-load-bearing orientation; the local measurement theorems in this manuscript supply the operative premises.

Physics-facing bridge. The manuscript does not derive concrete ontology from quantum measurement. It makes the narrower claim that, once measurement records are used in a physically realized downstream practice, their standing-bearing use cannot be treated as inscription, semantic bookkeeping, or branch-local description alone. The record must function as a persistence-bearing continuation locus for the practice in which it is retained, anticipated, deliberately used, and compared as evidence.

Definition 12.1 (Measurement-record continuation locus). For a diachronic measurement construction token $d \in \mathcal{D}$, a *measurement-record continuation locus* is the role-structured candidate record-bearing locus through which the standing-bearing value of d is preserved across same-domain retained-record reuse, prospective-record reuse, deliberation, and evidential comparison. Equivalently, it is the continuation-locus reading of the fixed record-bearing role whose admissible realizations are represented by $\text{Cand}(d)$.

Definition 12.2 (Persistence-bearing evidential anchor). A measurement record is a *persistence-bearing evidential anchor* for d iff its role as the standing record for d is fixed in a way that preserves standing-bearing consequence across lawful same-domain continuation. The phrase does not mean that the manuscript has derived a concrete object, observer, spacetime region, or material substrate. It means that the record's evidential use participates in a persistence-bearing lineage rather than in inert notation, semantic reference, trace succession, or representational skin.

Proposition 12.3 (Measurement-record roles are continuation loci). *If a measurement record is used as a standing-bearing evidential anchor in the target explanatory practice, then the corresponding record-bearing role is also a measurement-record continuation locus.*

Proof. By Definition 2.16, the target practice uses measurement records in retained-record reuse, prospective-record reuse, deliberation, and evidential comparison. By Theorem 11.12, those uses presuppose one identity-preserving constructional lineage through the relevant records. The role by which a record bears evidential standing is therefore also the role by which same-domain continuation through that record is preserved. That is the measurement-record continuation locus of Definition 12.1. $\square$

Theorem 12.4 (Persistence-bearing evidential fixation bridge). *On the fixed-domain closure package, admissible record-fixation for the target measurement practice is simultaneously:*

1. *evidential-role fixation: fixing which record may function as the standing evidential anchor for d ;*
2. *continuation-locus fixation: fixing the role-locus through which the same record-bearing construction persists across downstream reuse; and*
3. *persistence-bearing lineage fixation: fixing the lineage through which retained-record reuse, prospective-record reuse, deliberation, and evidential comparison remain construction-relative.*

Consequently, in a physically non-selective regime, unresolved admissible multiplicity underfixes not only a comparative evidential verdict but also the persistence-bearing measurement-record lineage on the same domain.

Proof. By Proposition 12.3, a standing evidential record-role in the target practice is a continuation locus. By Theorem 11.13, retained-record reuse, prospective-record reuse, deliberation, and fixed-domain evidential comparison all factor through the same measurement-record quotient. By Theorem 11.14, quotient-relevant divergence in that lineage makes those practices assignment-relative together. Thus record-fixation fixes the lineage through which the record-bearing practice retains standing-bearing consequence across lawful downstream reuse. The larger continuation and persistence vocabulary supplies a structural reading: a fixed role-anchor for a continuation locus, when retained under irreversible standing-bearing continuation, is a persistence-bearing regime rather than trace-only or representational skin. The present theorem uses only the local measurement instance of that architecture and does not import any additional theorem as a premise. $\square$

Corollary 12.5 (Physical realization alone does not supply persistence-bearing lineage). *Multiple stable branch-local records may be physically realized. That fact alone does not fix which record-bearing continuation locus is the standing persistence-bearing lineage for the target practice. A singleton admissible fixation asymmetry, or an explicit change of target construction, is still required.*

Proof. Physical realization supplies branch-local records. By Theorem 12.4, the target practice requires the role-locus through which those records bear standing across downstream continuation. If the physically non-selective regime leaves several candidates available and no singleton admissible asymmetry fixes one role-locus, then the lineage remains underfixed by Theorem 8.7. Treating the whole branch family, an aggregate branch measure, or a semantic description as the standing object changes the target construction unless it also fixes the record-bearing role used by retained-record reuse, prospective-record reuse, deliberation, and evidential comparison. $\square$

Proposition 12.6 (Not-real-physics objection and target-domain status). *Within the target explanatory practice, the measurement-record problem is not bookkeeping about descriptions alone. Once the records are required to stand in a physically realized downstream practice, admissible record-fixation constrains persistence-bearing continuation loci. A response that treats the records as abstract-only, semantic-only, trace-only, or skin-only changes the target to a deflationary or bookkeeping practice rather than satisfying the original one.*

Proof. The target practice fixes measurement records as objects of retained reuse, prospective reuse, deliberative guidance, and evidential comparison. By Theorem 12.4, those roles require persistence-bearing lineage fixation. Abstract-only or semantic-only treatment can describe candidate records, but it does not preserve the standing-bearing continuation role unless it also supplies the fixation structure required by the measurement-record quotient and asymmetry-exhaustion results. If it refuses that role, it is constructional deflation; if it supplies the role, it re-enters the admissible fixation architecture. $\square$

13 Interpretation taxonomy: where is singular record-fixation fixed?

The previous sections have now done enough work that the interpretation discussion can be stated as a same-scope exhaustion theorem. The relevant question is not simply which interpretation one prefers. It is this: *where, if anywhere, is the work of fixing singular record-fixation done as*

determinate enough for measurement records to function as theory-level evidence on the fixed scope?

The taxonomy classifies the role of singular record-fixation, not the metaphysical self-description of an interpretation. Relational, perspectival, consistent-histories, pragmatist, or decoherent-histories programs must therefore enter by the fixation work they perform: dynamic selection, ontic or perspectival privileging, restriction to one constructional class, or refusal of the target construction.

13.1 Interpretation packages and the loci of record-fixation work

Definition 13.1 (Interpretation package). An *interpretation package* for a measurement scenario $\mathcal{M}$ is a triple

$$\mathfrak{I} = (T, \Lambda, \mathcal{D}^*),$$

where:

1. $T = (\Omega, \Delta)$ is the physical theory, with ontology or state-description Ω and dynamics Δ ;
2. Λ is the class of further resources appealed to in fixing the standing realizations of measurement construction, including added ontology, representational restrictions, or construction-level policies; and
3. $\mathcal{D}^*$ is the class of diachronic measurement constructions the package is intended to underwrite.

Definition 13.2 (Discourse-committing and construction-deflationary packages). An interpretation package $\mathfrak{I}$ is *construction-committing* if it takes the construction tokens in $\mathcal{D}^*$ to be admissible and evidential-stabilization-guiding through the measurement records of the scenario. It is *construction-deflationary* if it does not take the theory to underwrite those construction tokens in the required admissible diachronic sense.

Definition 13.3 (Dynamic selection). A construction-committing package realizes *dynamic selection* if singular record-fixation is fixed by the physical evolution law itself: for the relevant measurement episode, the dynamics Δ alone suppresses or removes outcome multiplicity so that exactly one post-measurement history remains eligible as the standing realization relevant to $\mathcal{D}^*$.

Definition 13.4 (Ontic privileging). A construction-committing package realizes *ontic privileging* if singular record-fixation is fixed by additional ontological structure not determined by the bare dynamics alone. Equivalently, the physical description still requires an extra state variable, trajectory, configuration, or comparable ontic fact to designate which history is the actual standing realization.

Definition 13.5 (Restriction-to-one-constructional-class). A construction-committing package realizes *restriction-to-one-constructional-class* if the physical theory remains non-selective, but admissible record-fixation assignments are restricted so that the surviving assignments form a single $\approx_{\text{MR}}$ -class on the target construction.

13.2 From stabilization to structural exhaustion

Lemma 13.6 (Discourse commitment forces stabilization). *If an interpretation package $\mathfrak{I}$ is construction-committing, then it must stabilize admissible record-fixation assignments so that they avoid the role-divergent case of Lemma 7.5.*

Proof. A construction-committing package takes fixed-domain evidential ordering, retained-record reuse, prospective-record reuse, deliberation, or some combination of them to be admissible and guided by measurement records. By Theorem 7.9, any such use requires admissible record-fixation assignments to collapse so that fixation commutes with admissible transformation on the target construction. So construction commitment forces stabilization. $\square$

Definition 13.7 (Structural locus of singular-record-fixation fixing). A same-scope resource that purports to fix singular record-fixation does its standing-bearing work at one of the following terminal loci:

1. **boundary locus:** by acting as an admissibility-relevant selector or parameter at the pre-admissible boundary;
2. **quotient locus:** by refining the quotient-level standing-bearing content itself through a new invariant or privileged continuation relation; or
3. **representation locus:** by restricting how already-admissible representatives or uses are allowed to count without changing the quotient-level physical package.

Theorem 13.8 (Structural locus trichotomy). *Any admissibility-preserving same-scope selector has its terminal operation at one of the loci in Definition 13.7. No fourth same-scope locus exists.*

Proof. By Proposition 2.43, any same-scope admissibility-bearing difference must appear either in standing classification or in licensed continuation. The activated fixed-domain closure package closes any second same-scope invariant except as bookkeeping or scope change, and the role-decomposition results imported with that closure package leave no fourth admissibility-bearing role beside anchor-level, quotient/tensor-level, and representational/skin-level work. A selector that acts before quotient-level content is fixed therefore acts at the boundary; a selector that changes standing-bearing content acts on the quotient; and a selector that leaves quotient content untouched but changes how representatives may count acts via representation. If it does none of these, it is bookkeeping only and not a selector. So no fourth same-scope locus remains. $\square$

Corollary 13.9 (Boundary selectors are forbidden). *If a same-scope selector operates at the boundary locus, it introduces admissibility-relevant parameterization or selection and is therefore forbidden on the fixed scope.*

Proof. By Theorem 13.8, a boundary-locus selector acts before quotient-level content is fixed. But Theorem 2.42 and Lemma 2.41 exclude same-scope admissibility-relevant parameterization or selection at the boundary. So a boundary selector is forbidden. $\square$

Corollary 13.10 (Quotient refinements force new invariant structure). *If a same-scope selector operates at the quotient locus, then it refines standing-bearing content by a new invariant or privileged continuation relation. On the fixed scope that is not an additional neutral interpretation layer; it is a scope change relative to the non-selective base package.*

Proof. A quotient-locus selector changes which standing-bearing content counts as the relevant continuation. By Theorem 3.19, no alternative same-scope admissibility-preserving quotient survives beside $\approx_{\text{MR}}$. So any quotient refinement either coincides with the existing quotient or introduces a new same-scope invariant. The latter is excluded by the activated fixed-domain closure package except as scope change. $\square$

Corollary 13.11 (Representation restrictions yield restriction-to-one-constructional-class or deflation). *If a same-scope selector operates only at the representation locus while leaving the physical package non-selective, then either it restricts admissible representatives to one quotient class (representational restriction) or it refuses the target admissible downstream construction (constructional deflation).*

Proof. A purely representational selector cannot change boundary status or quotient-level content without leaving the representation locus. If the target construction remains admissible and construction-relative, the only way such a selector can matter is by restricting admissible representatives so that their use collapses into one quotient class; that is restriction-to-one-constructional-class. If the target construction is instead refused, the package is construction-deflationary. No third representation-locus option remains. $\square$

Theorem 13.12 (Selector-exhaustion theorem). *Any admissibility-preserving selector has a terminal role in one of the following classes:*

1. *introduce boundary parameterization or a boundary selector, which is forbidden;*
2. *refine standing-bearing quotient content by a new invariant or privileged continuation relation, which is a scope change on the fixed scope;*
3. *restrict representation so that admissible assignments collapse into one quotient class, which is restriction-to-one-constructional-class; or*
4. *refuse the target admissible downstream construction, which is constructional deflation.*

No fifth same-scope selector route exists. In particular, delayed calibration, retrospective evidential stabilization, posterior branch privilege, Bayesian after-selection, future-data repair, and post hoc re-identification are not additional same-scope routes; each is either post-selection repair, scope reset, or illicit selector injection.

Proof. By Theorem 13.8, every same-scope selector operates at the boundary, on the quotient, or via representation. Boundary-locus selectors are forbidden by Corollary 13.9. Quotient-locus selectors refine standing-bearing content and therefore introduce a new invariant or privileged continuation relation; by Corollary 13.10 this is a scope change on the fixed scope. Representation-locus selectors fall under Corollary 13.11: they are either representational restriction or constructional deflation. Temporal repair variants do not evade the trichotomy. If the later material changes the channel, calibration line, posterior branch status, or evidential target, it changes the fixed domain or target construction. If it does not change the domain, it supplies no new invariant. Treating it as retroactively decisive is post-selection repair or hidden selector import. These cases are exhaustive, so no fifth same-scope selector route remains. $\square$

Corollary 13.13 (Forbidden fifth-case selector attempts). *Branch weights, caring measures, centered worlds, self-locating uncertainty, detector salience, environmental robustness, pragmatic success, delayed calibration, posterior Bayesian update, and future-data repair do not constitute a fifth same-domain solution to singular record-fixation. Each is either support/bookkeeping, a declared role-fixing trace, a scope change, a forbidden second classifier, or same-act repair.*

Candidate resource	Normal-form location	Result on the fixed scope
Centered worlds or centered descriptions	Declared indexical/provenance marker if fixed; otherwise candidate re-description	Does not fix a record role unless one center is declared as the singleton anchor.
Self-locating uncertainty	Indexical/provenance domain over typed successors	Presupposes the typed successor set; does not create role fixation.
Branch weights or amplitudes	Scalar ranking over multiple candidates	Measures or orders candidates; does not fix a singleton role-bearer without declared bivalent structure.
Caring measures or utilities	Scalarized decision parameter	Downstream guidance only; inadmissible as a same-scope selector unless retyped as new domain structure.
Anthropic selection	Extra-domain preference unless fixed as a declared invariant	Not a same-domain selector unless reduced to an internal invariant, provenance/indexical anchor, or singleton boundary trace.
Detector/environment appeal	Boundary-trace candidate	Admissible only as declared, stable, bivalent, outcome-independent singleton trace.

Table 4: Standard Everettian and quasi-Everettian resources under the record-role asymmetry normal form. The table blocks the residual loophole that centered, anthropic, decision-theoretic, or weighted resources constitute a hidden fifth source of fixation.

Proof. Each listed resource either leaves all admissible candidates in place, in which case it is support or bookkeeping; fixes a singleton by a declared stable trace, in which case it belongs to an already named route; changes the relevant domain or target; adds selector authority beside the gate; or uses later information to repair prior unfixed status. By Theorem 13.12 there is no fifth same-scope locus. $\square$

Proposition 13.14 (Post-selection routes do not fix same-scope records). *Delayed calibration, retrospective evidential stabilization, posterior branch privilege, Bayesian after-selection, and future-data repair do not fix a measurement record for the same evaluated act. They either define a fresh domain, add a new admissibility-relevant parameter, or condition fixation on later evidential success.*

Proof. Same-scope record-role fixation must be available before the record is reused as an evidential anchor. A delayed or posterior route changes the channel package, the admissibility-relevant information, or the outcome-ordering after the original role-fixation failure. Treating that later enrichment as if it fixed the earlier act is same-act repair. Treating it explicitly as new structure is a scope reset. Neither case supplies same-scope fixation for the original construction. $\square$

Theorem 13.15 (Interpretation taxonomy). *Every interpretation package $\mathfrak{I}$ for a measurement scenario has a terminal classification in one of the following structural classes:*

1. **Dynamic selection.** *Singular record-fixation is fixed by physical dynamics.*
2. **Ontic privileging.** *Singular record-fixation is fixed by added ontological structure or a privileged trajectory.*

3. **Restriction-to-one-constructional-class.** *The physical theory remains non-selective, but admissible record-fixation assignments are restricted to a single standing-equivalence class.*

4. **Discourse deflation.** *The package does not take the theory to underwrite the target class of admissible diachronic measurement construction.*

No fifth structural locus remains under the fixed-domain closure package.

Proof. Take any interpretation package $\mathfrak{J}$. If it is not construction-committing, it is construction-deflationary and we are in class (4).

Assume instead that $\mathfrak{J}$ is construction-committing. By Lemma 13.6, it must stabilize admissible record-fixation assignments so that quotient-relevant divergence is avoided. If the stabilizing work is done by changing the physical dynamics so that only one continuation remains eligible, the package realizes dynamic selection. If the stabilizing work is done by additional ontological structure not fixed by the bare dynamics, the package realizes ontic privileging. If the physical theory remains non-selective and the stabilization is achieved only by restricting admissible representatives to one quotient class, the package realizes restriction-to-one-constructional-class. By Theorem 13.12, no fifth same-scope selector route exists. Hence the listed classes are exhaustive and mutually exclusive. $\square$

13.3 The comparative interpretation matrix

Class	Selector locus	What is fixed	Representative strategies
Dynamic selection	Physical dynamics	One continuation by law	Collapse dynamics, objective reduction
Ontic privileging	Added ontology	One continuation by extra physical fact	Hidden-variable, actual-history, privileged trajectory
Restriction-to-one-constructional-class	Admissible record-fixation assignments	One standing-equivalence class of admissible assignments	Everettian record-fixation programs, typed successor restrictions
Discourse deflation	Nowhere inside the theory	The target construction is refused	Instrumentalism, QBism, other deflationary responses

Table 5: Comparative interpretation matrix. The classification question is not which interpretation is metaphysically best, but where the work of fixing singular record-fixation is done if it is done at all.

13.4 Where the familiar interpretations land

Proposition 13.16 (Collapse strategies instantiate dynamic selection). *Collapse strategies such as GRW instantiate class (1) of Theorem 13.15: they place the work of singular-record-fixation in the physical dynamics itself.*

Proof. A collapse theory changes the dynamics so that, for the relevant measurement episode, one outcome history is physically singled out rather than multiple stable histories remaining equally realized. That is exactly dynamic selection. $\square$

Proposition 13.17 (Hidden-variable strategies instantiate ontic privileging). *Hidden-variable or actual-history strategies instantiate class (2) of Theorem 13.15: they place the work of singular-record-fixation in added ontological structure rather than in the bare dynamics alone.*

Proof. Such strategies retain a multiplicity in the wavefunction-level description but add further structure — for example an actual configuration or privileged trajectory — that designates which history is the one relevant to diachronic construction. That is ontic privileging. $\square$

Proposition 13.18 (Bare Everett is not yet a completed construction-committing strategy). *The bare Everettian package is physically non-selective and therefore does not by itself realize class (1), class (2), or class (3). To underwrite the target construction, it must either be completed by representational restriction or move to constructional deflation.*

Proof. Theorem 5.8 and Corollary 5.7 show that the bare Everettian package is physically non-selective. So it does not realize dynamic selection. By construction it adds no extra ontology that privileges one branch, so it does not realize ontic privileging. It realizes restriction-to-one-constructional-class only if admissible record-fixation assignments are restricted so that they collapse into one standing-equivalence class. Bare Everett alone does not do that. Therefore it is not yet a completed construction-committing strategy. To underwrite the target construction, it must either be completed by restriction-to-one-constructional-class or retreat to constructional deflation. $\square$

Corollary 13.19 (Bare Everettian exclusion of the first three loci). *If one retains the bare Everettian package without adding new dynamics, without adding ontological selection structure, and without imposing an admissible same-scope fixation restriction on record-fixation assignments, then the package realizes none of the first three loci of Theorem 13.12. Relative to the target explanatory practice, only constructional deflation remains.*

Proof. Theorem 5.8 excludes dynamic selection in the bare Everettian package. Proposition 13.18 excludes ontic privileging and shows that the bare package does not by itself collapse admissible assignments into a single standing-equivalence class. The sharper record-role statement is Theorem 8.12: the bare package supplies no internal singleton invariant, no declared provenance/indexical singleton anchor, and no declared singleton boundary trace. Therefore, absent additional same-scope structure, the bare package realizes none of the first three loci of Theorem 13.12. If the target explanatory practice is nonetheless refused, the package moves to constructional deflation. $\square$

Proposition 13.20 (Everettian record-fixation strategies instantiate restriction-to-one-constructional-class). *Any Everettian strategy that keeps the physical theory non-selective while restricting admissible diachronic record-fixation determinately enough for records to function as evidence falls under class (3) of Theorem 13.15.*

Proof. Such a strategy leaves the underlying physical package non-selective, so it is neither dynamic selection nor ontic privileging. If it still underwrites fixed-domain evidential ordering, retained-record reuse, prospective-record reuse, and deliberative guidance, then by Lemma 13.6 it must stabilize admissible record-fixation assignments. Since the work is not being done by a physical selector, it is being done by restricting admissible assignments themselves. That is exactly restriction-to-one-constructional-class. $\square$

Proposition 13.21 (QBist and instrumentalist strategies instantiate constructional deflation). *QBist and instrumentalist strategies [14] fall under class (4) of Theorem 13.15 insofar as they do not take quantum theory to underwrite the target class of admissible diachronic measurement construction.*

Proof. On QBist and related instrumentalist views, quantum states and outcomes are not taken to describe an objective diachronic measurement history in the sense needed by the standing conditional. Instead they organize expectations, betting commitments, or instrumental predictions. The package therefore does not aim to secure singular record-fixation for the target construction through the theory itself. That is not a failed stabilization strategy; it is a refusal of the target explanatory burden. Hence these views instantiate constructional deflation. $\square$

Remark 13.22 (Relational and perspectival views). Some relational, perspectival, consistent-histories, pragmatist, or decoherent-histories views may not advertise themselves under any of the four standard labels [15]. The present taxonomy still applies because it classifies the location of record-fixation work rather than the interpretation's preferred self-description. If such a view is construction-committing and physically non-selective, it must either supply restriction-to-one-constructional-class or add a physical, ontic, perspectival, or history-selecting fixing resource. If it instead treats the relevant diachronic construction as non-fundamental or merely perspective-bound in a way that refuses theory-level admissibility, it moves toward constructional deflation. The exact placement of particular relational programs is therefore an application question, not a challenge to the taxonomy.

13.5 Finite realization of the selector taxonomy

The finite stack instantiates the selector taxonomy on concrete physical surfaces. Balanced weights provide branch weights but not singleton fixation. The density table and non-selective channel provide coherence removal and trace preservation but not fine-record selection. The instrument provides selective branches and a non-selective sum; the environment provides branch correlation; the effect surface provides outcome weights; and the repeatability surface provides idempotent branch action. None of those non-selective surfaces becomes a selector merely by commuting with the others.

Finite surface	What it supplies	What it does not supply
Balanced weights	Equal branch numerators and normalization.	Singleton branch fixation.
Density/channel	Coherence removal and trace preservation.	Fine-record selection.
Instrument	Selective branches and non-selective branch sum.	Selection by the non-selective output.
Environment	Stable branch correlation.	Unique standing record.
Effect	Outcome weight and complementary completeness.	Record-role fixation.
Projector/repeatability	Idempotent branch action and zero cross-product.	Branch origination or selection.

Thus the finite disjunction exhausted in Theorem 9.4 is not the universal selector taxonomy; it is the physically explicit same-domain occupation of the general support/selector distinction.

14 Refutation-burden audit and denial conditions

At this point the relevant objections are no longer free-floating. Each survives only by denying a labeled definition, lemma, or theorem. Table 6 states the denial burden in its strict form before the representative objections are spelled out in brief.

Same-mode denial fork. A same-mode denial must preserve the singular construction-relative record-use target and then deny a named link: record-use-to-kernel forcing, AMetric no-selector discipline, assignment-space forcing, exact standing-content quotient collapse, Everettian non-selection, boundary-trace exhaustion, fixation commutation, diachronic inheritance, or selector exhaustion. A branch-relative, global-branch, instrumental, or decision-theoretic replacement that does not preserve the original construction token is target retyping; a replacement that does preserve it re-enters the kernel and quotient theorems.

Table 6: Strict denial burden for the measurement-record proof.

Proof link	Successful denial burden	Location
Target practice fixed	Show that measurement records can function as retained, prospective, deliberative, and evidential anchors without fixed-domain record construction.	Sec. 2
Record use instantiates K_{MR}	Exhibit a non-degenerate same-domain record-use regime preserving reference, standing, admissibility, and irreversibility while lacking the local kernel roles.	Sec. 2
AMetric boundary discipline	Show that branch weights, environmental salience, centered indices, detector preference, or decision policies can decide record standing without selector authority.	Sec. 2
AASC is optional vocabulary	Break the activation theorem, deny the standing conditional, or exhibit a same-domain record-use practice preserving target determinacy, standing-bearing reuse, act-time finality, and lawful re-description while avoiding the imported closure package.	Sec. 2

Proof link	Successful denial burden	Location
Adequacy conditions allegedly smuggle in target content	Preserve the same non-degenerate measurement-record object while denying target determinacy, standing-bearing evaluability, act-time finality, or same-regime fidelity; otherwise the objection is target refusal or retyping, not same-target refutation.	Sec. 2
Assignment-space forcing is avoidable	Supply a standing-bearing assignment-free realization of measurement-record reuse.	Sec. 3
You chose the quotient	The exact standing-content collapse theorem, Theorem 3.19.	Sec. 3
Alternative exact standing-content equivalence relation remains available	Theorem 3.19 together with Corollaries 3.20 and 3.21.	Sec. 3
Branch-local determinacy is enough	Theorem 5.12.	Sec. 5
Everett still contains an implicit selector	Theorem 13.12 or Corollary 13.19.	Sec. 13
Detector/environment fixes the record	Show detector or environment fixation without declared stable bivalent singleton trace.	Sec. 8
Pointer basis or einselection fixes the record	Show that pointer-basis stabilization collapses candidate record-role multiplicity to singleton evidential-role fixation without an additional admissible asymmetry.	Sec. 5
Weakening-resistance can be bypassed	Define a strict weaker same-domain regime preserving the target practice while permitting evidentially usable unfixed record roles without second classifier, repair, or scope shift.	Sec. 2
Evidential stabilization can be relativized or weakened	Theorems 7.7 and 3.27.	Sec. 7
Deliberative guidance, retained-record reuse, and prospective-record reuse are separate issues	Theorems 11.5, 11.12, and 11.13.	Sec. 11

Proof link	Successful denial burden	Location
There may be a fifth interpretive family	Theorems 13.8 and 13.12.	Sec. 13
An alternative selector outside the taxonomy remains available	Theorems 13.8 and 13.12.	Sec. 13
No cost-free target retyping	Show that a branch-relative, global-branch, instrumental, self-locating, or decision-theoretic evidence object preserves the original singular construction token without projecting back to record-fixation assignments.	Sec. 2
QBism or instrumentalism refutes the theorem ladder	Proposition 14.11.	Sec. 14

No row in Table 6 is rhetorical. Each identifies the theorem link that must be broken, the target that must be refused, or the same-domain alternative that must be supplied.

14.1 Target and dependency objections

14.1.1 Branch-relative determinacy is enough

Proposition 14.1 (Branch-relative determinacy is downstream of the target). *Branch-relative record determinacy is not denied. It fails only when it is used as the same construction-relative evidential record object without supplying admissible fixation of that role.*

Proof. A branch-relative record can be locally stable, reportable, and scientifically useful. The target here is stronger: retained, prospective, deliberative, and evidential reuse of the same record object. If the branch-relative record preserves that target, it enters the local kernel and quotient analysis; if it does not, it is a different target. $\square$

14.1.2 Successor-relative typing already solves the problem

Proposition 14.2 (Successor typing changes the downstream object). *A successor-relative record is admissible as a successor-relative object. It is not the same construction-relative pre-measurement record unless it preserves the original record-use target.*

Proof. The successor-relative object is typed after the branch-relative successor is fixed. The target of the manuscript is the standing evidential record for the original construction token. Preserving that target returns the view to the kernel and quotient theorems; abandoning it is target retyping. $\square$

14.1.3 Decision theory replaces record fixation

Proposition 14.3 (Decision-theoretic ordering is downstream of record typing). *Decision-theoretic weights or preferences may guide action, but they do not by themselves identify which fine record is the standing construction-relative evidential object.*

Proof. A decision rule orders acts or preferences over branch-structured futures. The present target asks for admissible record-role fixation for retained, prospective, deliberative, and evidential reuse. A decision rule that preserves that target requires the local kernel; a decision rule that does not preserve it is a coherent different practice. $\square$

Theorem 14.4 (Indeterminacy of the target breaks same-target science). *If a practice claims the same non-degenerate measurement-record target while leaving the record-bearing role unfixed, then theory-level comparison and reuse become assignment-relative rather than construction-relative.*

Proof. The target requires a stable record-bearing role. If incompatible admissible assignments remain live and no quotient collapses them, later evidential ordering depends on assignment choice. That is not a stable construction-relative scientific record for the original target. $\square$

14.1.4 The target is too demanding or ad hoc

Proposition 14.5 (The target burden is explicit rather than ad hoc). *The manuscript does not smuggle in a favored solution. It fixes a target: measurement records used as retained, prospective, deliberative, and evidential anchors. The target can be refused, but it cannot be retained while dropping the kernel roles required by non-degenerate objecthood.*

Proof. The target is stated before the interpretation taxonomy and before the finite construction. Any view can decline it by adopting branch-relative, global-state, instrumental, or decision-theoretic evidence. That is a target exit, not a same-target solution. $\square$

Proposition 14.6 (The target-demand objection is a target-refusal option). *If a critic rejects retained, prospective, deliberative, and evidential record reuse as the target, the manuscript classifies the critic's practice as different rather than false.*

Proof. The theorem is conditional on the target. Refusing the target removes the theorem's application; it does not break the theorem link for theories that continue to use measurement records as standing anchors. $\square$

14.2 Physics objections

14.2.1 This is not real physics

Proposition 14.7 (The physics objection reverses the dependency). *AASC is not imposed after physics has already supplied determinate objects. It identifies the necessity conditions under which a state, outcome, carrier, record, transformation, or endpoint is a determinate physical object at all.*

Proof. The finite construction uses amplitude vectors, density tables, trace functionals, projective branch maps, effects, projector identities, environment records, and compatibility theorems. A standing objection must attack target instantiation, a kernel necessity condition, a finite definition, a matrix identity, a compatibility proof, the selector exhaustion, or the scope of the conclusion. Merely calling the result “not physics” refuses the dependency direction rather than refuting it. $\square$

14.2.2 The finite model is only two-dimensional

The finite theorem has a declared two-branch carrier. That is a scope statement, not a defect. The theorem proves simultaneous compatibility across multiple physical surfaces and shows that weighting, coherence removal, effect completeness, zero-product orthogonality, and repeatability do not by themselves produce a fine-record selector on that declared carrier. Generalization is a separate theorem target.

14.2.3 The conclusion is hard-coded

The record carrier is declared, but the theorem's force is not that arbitrary names cannot select. The force lies in the finite measurement-role adequacy theorem: amplitude, density, instrument, environment, channel, effect, projector, and repeatability structures commute on branch and weight content while non-selective surfaces jointly remain at the merged record. The no-selector theorem then eliminates a finite disjunction of concrete candidate selector surfaces inside that constrained package.

14.2.4 Pointer basis, detector, or environment fixes the record

Proposition 14.8 (Pointer-basis stabilization is candidate-family fixation, not role-fixation). *To maintain that pointer-basis stabilization or einselection fixes the measurement record in the required sense, one must deny Proposition 5.19 or show that the stable pointer family contains a declared singleton admissible role-bearer for the target construction.*

Proof. Pointer-basis stabilization explains why certain branch-local records are stable and robust. The target practice requires a further role statement: which candidate record is the standing evidential anchor for pre-measurement retained-record reuse, prospective-record reuse, deliberation, and evidential comparison. Without a singleton internal invariant, declared provenance/indexical anchor, or singleton boundary trace, the pointer family remains a stable candidate family rather than a fixed evidential role. $\square$

Proposition 14.9 (Environmental fixing requires declared singleton trace). *To maintain that the detector, apparatus, calibration line, or environment fixes the measurement record, one must show that the relevant relation is a declared, stable, bivalent, outcome-independent boundary trace with singleton positive value on the candidate record set.*

Proof. The environment as such is not a same-domain selector. It may contain many distinctions among candidate records, but only distinctions typed into the fixed measurement domain can bear admissibility. Boundary-trace fixation permits role fixation only in the singleton-positive case. $\square$

14.3 Formalization objections

14.3.1 The formalization merely packages assumptions

Some paper-facing declarations package hypotheses or route imported theorem surfaces. The finite modules are different: they define explicit carriers, maps, integer tables, trace and coherence functionals, branch operations, effects, and projector products, then prove exact identities and incompatibilities by computation, case analysis, and composition. Appendix E separates concrete

finite computation, cross-surface compatibility, selector exclusion, aggregate packaging, and audit declarations.

14.3.2 Why should the projector/effect layer be trusted?

The controlling public v0.9.1 formal artifact contains the zero-effect finite effect carrier, assigns that effect zero weight, and defines finite projector orthogonality by zero cross-products. The same layer also retains branch-map cross-annihilation and repeatability facts. The manuscript's projector/effect claims therefore cite the released v0.9.1 formal artifact rather than an unrevised preliminary source. The historical correction from the earlier release is recorded in Appendix C and in the revision memorandum, not used as a live premise in the main theorem.

14.4 Interpretive objections

14.4.1 The comparative matrix is unfair

Proposition 14.10 (The comparative matrix is classificatory rather than partisan). *The selector taxonomy does not privilege one interpretation family by rhetoric. To reject the comparative matrix, one must identify a fifth same-scope locus of singular-record-fixation not covered by Theorem 13.15.*

Proof. Theorem 13.15 classifies every package by the same structural question: where is singular record-fixation fixed, if it is fixed at all? The classification is unfair only if a fifth same-scope locus exists. The theorem denies exactly that. $\square$

14.4.2 Instrumentalism or QBism dissolves the problem

Proposition 14.11 (Discourse deflation exits the standing conditional). *Instrumentalist or QBist strategies do not refute the theorem ladder. They exit it by refusing the standing conditional under which the manuscript runs.*

Proof. QBist and related instrumentalist strategies are constructional deflation. Such strategies do not claim that the theory underwrites the target class of admissible diachronic measurement construction. Because the standing conditional is refused, the need for record-fixation stabilization inside the theory is likewise refused. That makes the strategy a coherent exit from the target problem, not a counterexample to the conditional theorem ladder. $\square$

14.4.3 The manuscript confuses representation with physics

Proposition 14.12 (The result is conditional on explanatory use). *The manuscript does not replace physical analysis with representational legislation. To maintain the opposite, one must deny that the theory is being used to underwrite admissible diachronic construction through measurement records.*

Proof. The manuscript takes physical packages as given and asks what follows if they are assigned a particular explanatory burden. If that burden is removed, the conditional result does not apply. That is not confusion between representation and physics; it is the explicit scope condition of the theorem. $\square$

15 Scope, established results, and nonclaims

This section separates the three theorem levels now present in the manuscript: the general AASC necessity result, the finite quantum specialization, and the formal audit layer.

15.1 What the manuscript has established

1. AASC kernel necessity: every non-degenerate determinate object, state, carrier, record, transformation, comparison, or endpoint requires admissibility, standing, reference, and irreversibility at the relevant scope.
2. A measurement-record specialization: a record used as the same standing physical/evidential object across retained, prospective, deliberative, comparative, or report-bearing use must satisfy the local measurement-record packet.
3. Fixed-domain assignment-space forcing and a measurement-record quotient theorem for standing-bearing constructional content.
4. Same-scope equivalence collapse: a rival quotient claiming exactly the same standing-bearing content collapses to the measurement-record quotient.
5. Everettian non-selection for the bare unitary, entangling, decohering package on the record-fixation target.
6. Fixation commutation, diachronic inheritance, persistence-bearing reuse, asymmetry exhaustion, anti-weakening, and selector exhaustion.
7. A finite quantum specialization with an explicit two-branch carrier, denominator-two weights, amplitude/density agreement, selective and non-selective instruments, environment records, projective branch maps, complementary effects, zero-product projectors, repeatability, fine/coarse record compatibility, weight compatibility, and a finite no-selector theorem.
8. A formalization crosswalk identifying concrete finite computations, finite structural theorems, cross-surface compatibility theorems, selector-exclusion theorems, aggregate spines, assumption-routing declarations, and audit declarations.

15.2 What the manuscript does not claim

The following exclusions are theorem-scope boundaries, not retreats from physical content. The result constrains determinate physical measurement structure without claiming to reconstruct every mathematical structure used in quantum theory.

1. It does not derive Hilbert-space quantum mechanics.
2. It does not derive general POVM theory.
3. It does not derive full projection-valued-measure theory.
4. It does not derive full operator algebra.
5. It does not provide a general Kraus representation theorem.

6. It does not reconstruct complete positivity.
7. It does not provide a categorical reconstruction of measurement theory.
8. It does not derive the continuum Born rule.
9. It does not derive universal physical decoherence theory.
10. It does not introduce a new collapse dynamics.
11. It does not prove that no physical collapse process occurs in nature.
12. It does not prove physical branch selection.
13. It does not prove interpretive truth.
14. It does not classify every possible physical measurement mechanism.

15.3 Positive physical scope

These exclusions do not make the result nonphysical. The theorem establishes necessary structure for determinate physical measurement objects and proves that a concrete finite quantum measurement stack satisfies a cross-surface non-selection boundary. In particular, the absence of a hidden selector on the declared finite non-selective surfaces is not equivalent to a theorem that no physical collapse process occurs in nature; it is the finite theorem that those declared surfaces do not themselves supply the fine-record selector.

15.4 Bell-scope pairing

The present manuscript establishes the record side of the quantum determinacy problem: non-degenerate measurement-record objecthood requires admissible record-role fixation. The paired Bell-scope manuscript establishes the compatibility side: Bell witness reuse requires fixed setting/outcome and detector-channel interfaces, entangled preparations are typed as joint standing classes carrying tensor-level compatibility, and no same-domain signal, metric-transport, marginal-modulation, or distant-repair typing survives [24]. Together they identify the kernel-governed determinacy structure underlying quantum record and witness use.

16 Conclusion

The main result is conditional but exact in the target-instantiation sense. The manuscript does not claim that quantum mechanics requires new dynamics or that Everettian physics is false. It proves a kernel-role determinacy theorem: AASC does not choose a physical mechanism for record production, but it identifies the admissibility conditions under which physical record occurrence can stand as non-degenerate evidential record objecthood and exhausts alternatives that fail those conditions. If a physically non-selective theory underwrites admissible diachronic construction in which measurement records function as evidential anchors, then admissible record-fixation assignments for those records must collapse into a single standing-bearing class on the fixed scope. Otherwise evidential ordering, retained-record reuse, prospective-record reuse, and deliberation become assignment-relative rather than construction-relative.

The local dependency order is part of the result. Non-degenerate determinate measurement objecthood activates the kernel of admissibility, local object-standing, reference, and irreversibility at the relevant physical scope; measurement-record evidential use then specializes that kernel into the stronger measurement-record standing required for retained use, prospective use, deliberation, and comparison. AMetric no-selector discipline then blocks hidden branch-weight, environmental, detector, centered-index, and decision-policy selectors; anti-weakening shows that relaxing the kernel either preserves the same governance work, lowers the record to support or bookkeeping, shifts scope, imports a second classifier, or permits same-act repair.

The theorem ladder reaches the measurement conclusion through five steps. First, admissible record-fixation assignments are typed in fixed-domain form and their standing-bearing content factors through a unique measurement-record quotient. Second, the bare Everettian package is physically non-selective relative to the target construction. Third, fixed-domain evidential ordering requires fixation commutation under admissible transformation; quotient-relevant divergence makes that ordering undefined on the fixed scope. Fourth, asymmetry exhaustion shows that, after internal and provenance/indexical routes fail, singleton stable boundary-trace fixation is the residual same-domain admissible route. Fifth, retained-record reuse, prospective-record reuse, deliberation, and evidential comparison share one identity-preserving constructional lineage through the relevant records. The interpretation taxonomy then locates where that lineage can be fixed, if it is fixed at all.

In the exhausted physically non-selective regime, once internal invariants and provenance/indexical asymmetries fail, singleton stable boundary-trace fixation is the residual same-domain route. It is not an optional interpretive supplement.

Physical realization and evidential-role fixation therefore remain distinct. The existence of multiple stable branch-local records supplies physical realization; it does not supply the role-anchor by which one record may function as the standing evidential anchor for same-domain reuse.

The boundary-trace analysis also clarifies why appeals to apparatus, detector channel, or environment cannot be used loosely. Those relations fix a record-bearing role only when they are part of the declared measurement domain as stable, bivalent, outcome-independent traces and only when the trace-positive set is a singleton. Otherwise they add a selector, scalar parameter, delayed calibration, or different domain.

The resulting diagnosis differs from the standard presentation of the measurement problem. On the target construction, the first failure is not yet that a physically non-selective theory does not produce one outcome. The first failure is that the theory may leave underfixed the standing realizations of the very records through which outcomes are reported, remembered, anticipated, and used as evidence. In that precise sense, the measurement problem is a problem of record-fixation to measurement records before it is a problem of outcome production.

The persistence-bearing bridge sharpens the physics-facing interpretation without changing the theorem ladder. Measurement records in the target practice are role-fixed continuation loci through which standing-bearing consequences are retained across reuse. Unresolved multiplicity therefore underfixes not only a comparative evidential verdict but also the persistence-bearing measurement-record lineage by which the downstream practice remains construction-relative. This is not a proof that Everettian physics is false or that new ontology is required. It is a conditional result about what a physically realized measurement-record practice must fix to retain standing-bearing use on the same domain.

The present result therefore does not derive contradiction from multiplicity itself. Multiple branch-local records may exist and remain physically stable. The result is sharper: physical occurrence without admissible record-role fixation does not amount to ontic non-degenerate record determinacy for the construction token under audit.

The result is not a denial of the practical success of quantum theory, decoherence, or branch-relative record talk. It is a conditional theorem about a non-degenerate determinate record target. If the target is prediction, branch-local reporting, or instrumental update, the present theorem may classify the practice as different rather than false. But when measurement records are used as construction-relative evidential anchors, target-relative degeneracy is bivalent: either the record-use regime preserves the local kernel and admits standing-bearing reuse, or it fails to instantiate the target as non-degenerate construction. A different target may be legitimate. It is not the same work.

The manuscript therefore does not ask physics to become more philosophical. It identifies the governance already required whenever physics treats measurement records as reusable evidential objects. The kernel does not require scientists to adopt AASC terminology. It identifies the governance conditions already required when measurement records are used as the same evidential objects across memory-like retention, replication, comparison, deliberation, and future inference. The kernel is not an optional interpretation placed on top of that practice. It is the minimal structure that makes the practice construction-relative rather than assignment-relative.

The measurement problem is therefore not first a demand for collapse, hidden variables, or decision-theoretic weights. It is first a demand for admissible record-fixation under the kernel conditions of fixed-domain construction. Kernel denial has unbearable cost for the target practice: it removes record reference, record standing, admissible use, irreversibility, or the no-selector boundary that makes evidential reuse construction-relative rather than assignment-relative.

The theorem is therefore conditional only in the precise sense in which any necessity theorem for a target is conditional: the target must be instantiated. Once construction-relative evidential record use is instantiated as a non-degenerate determinate object, the kernel is already active. Refusing branch-independent record fixation, adopting global branch-state evidence, or moving to decision-theoretic policy may be coherent alternative epistemic projects, but they do not preserve the same target unless they recover the kernel roles under another description.

Boundary-trace fixation is therefore not a speculative addition to quantum dynamics. It is the forced residual normal form for evidential-role fixation once the target is fixed and inadmissible same-domain alternatives are closed. This is a kernel-role determinacy result, not a dynamics-first measurement theory; its correctness turns on target instantiation, kernel forcing, imported closure links, quotient construction, and exhaustion of faithful same-domain alternatives, not on whether the proof resembles a conventional measurement-theory mechanism.

The measurement-record case shows the downstream force of the kernel roles. Reference prevents silent target drift; standing separates physical record occurrence from evidential eligibility; admissibility closes selector import, bookkeeping, carrier shift, and target retyping; irreversibility blocks later success from repairing an unfixed same act. Boundary-trace fixation is not a new mechanism added to quantum dynamics. It is the forced residual admissible normal form left by these roles after the same-domain alternatives have been exhausted.

The endpoint is therefore stronger than a taxonomy of interpretations. It is a determinacy result. A physical record occurrence that lacks admissible record-role fixation may still be a physical event,

branch-local record, memory state, detector state, or environmental trace. It does not thereby stand as the same non-degenerate evidential object. Once that standing is claimed, the kernel roles govern the object.

Physical selection is not rejected. It is a stronger admissible route when a theory supplies it. Collapse dynamics, ontic variables, detector-channel dynamics, or boundary-condition physics may each realize record-role fixation if they preserve reference, standing, admissibility, irreversibility, quotient stability, and downstream reuse without hidden selector import or same-act repair. The target of this manuscript is prior to choosing among those mechanisms: any candidate mechanism must first satisfy the record-determinacy constraint proved here.

This result pairs with the Bell-scope no-signaling compatibility theorem. The present manuscript proves that record determinacy requires fixation; the companion manuscript *Bell-Nonlocal Quantum Entanglement as Boundary-Level Compatibility: A Kernel-Role Determinacy Theorem for Bell-Witness Fixation, No-Signaling, and Tensor-Level Compatibility* proves that Bell witness determinacy is boundary-level compatibility carried by a joint standing class and that no same-domain signal, metric-transport, marginal-modulation, or distant-repair typing survives [24]. Together they identify the kernel-governed determinacy structure underlying quantum record and witness use.

AASC is not an interpretation applied after quantum mechanics has already supplied determinate physical objects. The kernel is the necessity structure under which those objects, states, records, and transformations possess determinate standing. The finite formalization occupies that structure in an explicit two-branch quantum domain. It proves that normalized weights, coherence removal, branch correlation, instrument decomposition, channel closure, effect completeness, zero-product projector orthogonality, and repeatability can all agree while every declared non-selective surface remains fixed only at the merged record. The missing work is therefore not branch production, weighting, or stabilization. It is admissible fine-record role fixation.

$\begin{aligned} \text{determinate measurement objecthood} &\Rightarrow K \\ &\Rightarrow \text{admissible measurement construction} \\ &\Rightarrow \text{compatible finite quantum surfaces} \\ &\Rightarrow \neg \text{non-selective fine-record selection} \\ &\Rightarrow \text{admissible fixation locus.} \end{aligned}$

A Appendix: formal fixed-scope quotient presentation

This appendix gives a compact formal presentation of the quotient machinery used in the main body. Its job is narrow: to state in one place the fixed scope, the admissible assignment space, the standing equivalence relation, the quotient universal property, and the reason evidential-order-role divergence is quotient-relevant rather than representational skin.

A.1 Fixed scope and admissible assignment space

Fix a physically non-selective measurement setting on the activated fixed-domain closure package of Definition 2.19. Let ARef be the set of admissible record-fixation assignments from Definition 3.4. For each $\rho \in \text{ARef}$ and for each construction token $d \in \mathcal{D}$, let $\text{Val}_\rho(d)$ denote the standing-bearing

construction value of d on the fixed scope.

Definition A.1 (Target-construction profile map). The *target-construction profile map* is

$$\text{Prof} : \text{ARef} \rightarrow \prod_{d \in \mathcal{D}} \mathcal{V}_d, \quad \text{Prof}(\rho) := (\text{Val}_\rho(d))_{d \in \mathcal{D}},$$

where $\mathcal{V}_d$ is the codomain of the standing-bearing value attached to d .

Definition A.2 (Standing equivalence on admissible assignments). For $\rho, \sigma \in \text{ARef}$, define

$$\rho \approx_{\text{MR}} \sigma \iff \text{Prof}(\rho) = \text{Prof}(\sigma).$$

The quotient of admissible assignments by this relation is written $\text{ARef} / \approx_{\text{MR}}$.

A.2 Presentations and quotient uniqueness

Definition A.3 (Standing presentation). A *standing presentation* for admissible measurement-record content is a pair (X, r) such that $r : \text{ARef} \rightarrow X$ and every standing-bearing value on the fixed scope factors through r .

Lemma A.4 (Automatic completeness). *Every standing presentation is complete: if $r(\rho) = r(\sigma)$, then $\rho \approx_{\text{MR}} \sigma$.*

Proof. If $r(\rho) = r(\sigma)$, then every standing-bearing value factors through the same point of X . Hence the values of every construction token in $\mathcal{D}$ agree under ρ and σ . So $\text{Prof}(\rho) = \text{Prof}(\sigma)$ and therefore $\rho \approx_{\text{MR}} \sigma$. $\square$

Lemma A.5 (Automatic soundness on the fixed-domain closure package). *Every standing presentation is sound on the fixed-domain closure package: if $\rho \approx_{\text{MR}} \sigma$, then $r(\rho) = r(\sigma)$.*

Proof. Suppose $\rho \approx_{\text{MR}} \sigma$ but $r(\rho) \neq r(\sigma)$. Then the presentation distinguishes two same-scope contents that agree on every standing-bearing value in the target construction. Within the fixed-domain closure package, any same-scope admissibility-bearing distinction must survive either into standing classification or into licensed continuation. Since no such difference survives by hypothesis, the distinction is not standing-bearing. Treating it as such therefore violates the fixed-domain uniqueness package. Hence $r(\rho) = r(\sigma)$. $\square$

Lemma A.6 (Quotient universal property). *Let $q : \text{ARef} \rightarrow \text{ARef} / \approx_{\text{MR}}$ be the quotient map. If $F : \text{ARef} \rightarrow Y$ is constant on $\approx_{\text{MR}}$ -classes, then there exists a unique map $\tilde{F} : \text{ARef} / \approx_{\text{MR}} \rightarrow Y$ such that $F = \tilde{F} \circ q$.*

Proof. Define $\tilde{F}([\rho]) := F(\rho)$. This is well-defined because F is constant on the $\approx_{\text{MR}}$ -classes. Uniqueness follows from surjectivity of q . $\square$

Theorem A.7 (Quotient uniqueness). *Let (X, r) be any standing presentation for admissible measurement-record content. Then there is a unique bijection*

$$\phi : \text{ARef} / \approx_{\text{MR}} \longrightarrow X, \quad \phi([\rho]) = r(\rho).$$

Hence $\mathbf{ARef} / \approx_{\text{MR}}$ is the unique same-scope content space for standing-bearing measurement-record-fixation assignments, up to unique isomorphism.

Proof. Completeness is Lemma A.4 and soundness is Lemma A.5. The proof of bijectivity is then identical to the proof of Theorem 3.17: soundness gives well-definedness, completeness gives injectivity, surjectivity comes from surjectivity of r , and uniqueness is the quotient universal property of Lemma A.6. $\square$

Theorem A.8 (Exact standing-content equivalence collapse on the fixed scope). *Let $\approx$ be any same-scope equivalence relation on $\mathbf{ARef}$ that claims to carry exactly and only standing-bearing measurement-record content, and through which every standing-bearing value factors. Then for all $\rho, \sigma \in \mathbf{ARef}$,*

$$\rho \approx \sigma \iff \rho \approx_{\text{MR}} \sigma.$$

Thus no finer or coarser exact same-scope standing-content equivalence remains for measurement-record content.

Proof. If $\approx$ were coarser than $\approx_{\text{MR}}$, it would identify assignments with different profile values, so the profile map would fail to factor through $\mathbf{ARef} / \approx$. If it were finer than $\approx_{\text{MR}}$, it would distinguish assignments that agree on every standing-bearing value. Such a finer relation is enriched presentation, metadata, bookkeeping, or scope-changing surplus, not exact standing content. By the fixed-domain closure package and same-scope operator closure, those surplus distinctions are excluded from the exact content quotient [20]. Hence the relations coincide. $\square$

A.3 Why evidential-order divergence is not representational skin

Corollary A.9 (Evidential-order-role divergence is quotient-relevant). *If two admissible assignments $\rho, \sigma \in \mathbf{ARef}$ disagree on the comparative ordering role for constructed hypotheses of any evidential construction token, then they belong to distinct classes in $\mathbf{ARef} / \approx_{\text{MR}}$.*

Proof. Comparative ordering role for constructed hypotheses is one coordinate of the target-construction profile. So a difference in that coordinate implies $\text{Prof}(\rho) \neq \text{Prof}(\sigma)$ and hence $\rho \not\approx_{\text{MR}} \sigma$. $\square$

Remark A.10. This is the formal point that blocks the easiest philosophical dismissal of the manuscript. The manuscript does not merely claim that two representational descriptions feel different. It proves that if they differ in comparative ordering role for constructed hypotheses, then the difference survives the unique same-scope exact standing-content quotient of admissible measurement-record content. That is standing-bearing divergence, not representational skin.

B Appendix: structural package, exact imports, and finite formalization release posture

This appendix states exactly which imported fixed-domain closure results are used as background and how they are used. The point is transparency, not retrospective promotion. The present manuscript is conditional on these results only at the fixed-domain roles stated below. The canonical source for the local kernel import is *Non-Degenerate Construction and the Kernel of Admissibility*:

An Exhaustion Theorem for Standing, Reference, and Irreversibility [16]. The narrow closure spine used by this manuscript also includes claim-standing/report legitimacy [17], anchor/tensor/skin role decomposition [19], same-scope operator closure by exhaustion [20], and singleton admissible-interior uniqueness [18].

B.1 Imported fixed-domain package

Table 7: Import discipline for the measurement-record paper.

Imported result	Function in the measurement manuscript
Kernel $K = \{\text{Adm}, \text{St}, \text{Ref}, \text{Irr}\}$	Forces the local fixed-domain record-use conditions: admissible entry, standing for reuse, reference to the same record/outcome/evidential target, and irreversibility of unfixed or failed record-role status.
Non-derivability and no lower generator	Blocks record standing from being generated by labels, branch weights, reports, memories, or decision policies without already performing the kernel role.
AMetric boundary	Blocks branch-weight, environmental, detector, centered-index, probability-ranking, and decision-policy selectors at the boundary.
Bivalent admissibility quotient	Blocks unfixed, weighted, deferred, branch-relative, or environmentally suggested status from becoming a third standing-bearing record status.
Standing as reuse-stable admissibility	Makes evidential record use dependent on admissible downstream reuse rather than physical presence alone.
Unique admissible interior	Blocks plural same-scope evidential interiors or rival admissible record-use spaces for the same target practice [18].
Conservation of standing	Blocks later calibration, posterior success, future data, or branch-relative report from creating same-act record standing.
Report preservation	Blocks record reports from exceeding target, carrier, fixation, quotient, and trace support.
ATS role discipline	Separates anchors, standing-bearing tensor content, and skin; branch labels and centered descriptions do not become standing-bearing by being written [19].

Imported result	Function in the measurement manuscript
Boundary-trace fixation	Supplies the residual admissible route once internal and provenance/indexical asymmetries are exhausted: a declared, stable, bivalent, outcome-independent singleton trace.
Same-scope operator closure and anti-weakening	Blocks weakened regimes, hidden fifth routes, independent same-domain operators, unfixed evidence-bearing records, hidden selectors, repair, or second classifiers [20].

The narrow load-bearing closure spine is [16–20]. Additional continuation-locus vocabulary is used only as orientation; it is not part of the load-bearing imported closure spine and does not replace the local measurement-record theorem ladder.

B.2 How those imports are used here

The imported package is used in the following controlled places.

1. In Section 2, local measurement-record adequacy forces K_{MR} , locally activates the imported fixed-domain closure spine, proves the kernel-denial cost, and proves the no-strict-weakening result for evidentially usable unfixed record roles.
2. In Section 3, the packet forces assignment-space typing and soundness of measurement-record presentations, yielding quotient uniqueness on the fixed scope.
3. In Section 7, it turns evidential-order divergence into a standing-bearing difference rather than a mere representational redescription.
4. In Section 8, AMetric no-selector discipline and boundary-trace fixation supply the residual same-domain asymmetry theorem.
5. In Section 13, it closes the no-fifth-locus theorem by ruling out boundary selectors, quotient refinements, hidden repair, and additional same-scope selector routes.
6. In Section 12 and Appendix D, it supplies the narrow continuation-locus and persistence-bearing interpretation of measurement-record lineage.

No proof in the main body uses the imported closure spine for anything beyond these fixed-domain constraints.

B.3 Lean 4 finite measurement-boundary stack and public audit reference

The Lean 4 formalization is an integrated finite specialization over declared structures and hypotheses. The controlling public formal artifact is v0.9.1, which includes the zero-effect finite effect carrier and zero-product finite projector/repeatability layer. The retained formal route includes the paper-facing theorem surface, record-construction refinement, finite record dynamics, Hilbert/Born,

density/instrument, decoherence-channel, Kraus/channel, POVM/effect, projector/repeatability, measurement-compatibility, closed-stack, measurement-boundary, and audit-manifest layers [21–23].

The release has three distinct audit levels. First, concrete finite arithmetic, basis, pointer, branch, instrument, channel, effect, projector, compatibility, and boundary facts are proved directly. Second, the named spines package those facts and their bridges; some aggregate endpoints report the standard Lean `propext` dependency arising in negation, equality, or extensionality bridge steps. Third, the audit manifest exposes the endpoint names, concrete boundary facts, inventory, mechanization-state declaration, and non-claim boundary as Lean-readable declarations. The manifest is an audit-discoverability layer, not an additional physical theorem.

Table 8: Primary Lean modules and manuscript-facing roles.

Module	Role in the released stack
<code>TheoremRegister.lean</code>	Verbatim theorem register for the paper-facing surface.
<code>PaperStatements.lean</code>	Manuscript-facing theorem declarations.
<code>RecordConstructionRefinement.lean</code>	Admissible-record-construction refinement and local kernel routing.
<code>FiniteRecordDynamics.lean</code>	Denominator-2 outcome/record weights, fine/coarse record behavior, weakening exhaustion, and boundary-trace bridge.
<code>FiniteHilbertBornWitness.lean</code>	Two-basis integer-amplitude witness, squared-amplitude Born weights, and pointer records.
<code>FiniteDensityInstrument.lean</code>	Balanced density table, selective branches, non-selective merge, and instrument records.
<code>FiniteDecoherenceChannel.lean</code>	Environment records, fine/coarse traces, idempotence, and non-selective decoherence behavior.
<code>FiniteKrausChannel.lean</code>	Projective branch maps, non-selective channel, trace/coherence facts, and channel records.
<code>FinitePOVMEffect.lean</code>	Finite projective effects, completeness, effect/Born/trace agreement, and effect records.
<code>FiniteProjectorRepeatability.lean</code>	Projector idempotence, zero-product orthogonality, repeatability, annihilation, and record preservation.
<code>FiniteMeasurementCompatibility.lean</code>	Agreement among outcome, pointer, instrument, channel, effect, projector/repeatability, decoherence, Born, and coarse-record surfaces.
<code>FiniteMeasurementClosedStack.lean</code>	One aggregate endpoint packaging the finite measurement layers.

Module	Role in the released stack
<code>FiniteMeasurementBoundary.lean</code>	No hidden non-selective fine-record selector, branch-record agreement, coarse-record agreement, and branch separation.
<code>FiniteMeasurementAuditManifest.lean</code>	Lean-readable capstone, fact inventory, endpoint list, state declaration, and non-claim boundary.

Endpoint group	Named endpoints
Earlier retained layers	<code>reasonableDeepestFiniteMeasurementRecord_spine</code> <code>reasonableFiniteHilbertBornMeasurement_spine</code> <code>reasonableFiniteDensityInstrumentMeasurement_spine</code> <code>reasonableFiniteDecoherenceMeasurement_spine</code> <code>reasonableFiniteKrausMeasurement_spine</code> .
Released v0.9.1 measurement-boundary layers	<code>reasonableFinitePOVMMeasurement_spine</code> <code>reasonableFiniteProjectorMeasurement_spine</code> <code>reasonableFiniteMeasurementCompatibility_spine</code> <code>reasonableFiniteMeasurementClosedStack</code> <code>reasonableFiniteMeasurementBoundary_certificate</code> <code>reasonableFiniteMeasurementAuditManifest</code> .

The measurement-boundary certificate proves, for the enumerated finite surfaces:

- `no_hidden_finite_nonselective_selector`;
- `fine_branch_records_agree`;
- `coarse_nonselective_records_agree`;
- `branch_separation_agreement`.

The audit manifest packages those facts with the capstone endpoint and retained endpoint inventory. The strongest truthful summary is: a finite measurement-boundary and audit-manifest stack over declared carriers and hypotheses with zero-product finite projector orthogonality.

Audit item	Recorded public route
Repository	<code>somamaley-ux/measurement-problem-reference</code> .
Release posture	Public v0.9.1 release; zero-effect and zero-product finite projector layer are part of the controlling formal artifact.
Archive	DOI 10.5281/zenodo.19542014.
Standalone build	<code>lake build</code> .
Axiom audit	<code>lake env lean</code> <code>Checks/Axiom/</code> <code>MeasurementProblemReferenceAxiomCheck.lean</code> .
Source scan	Search the Lean trees for <code>sorry</code> , <code>admit</code> , and <code>unsafe</code> ; the released audit records none.
Axiom posture	Concrete finite facts must be reported declaration-by-declaration in the v0.9.1 audit; bundled spines and aggregate endpoints may inherit the standard Lean <code>propext</code> dependency through bridge and extensionality steps. This is not a custom axiom or a proof gap.

Formalization boundary. The formalized finite stack is finite, conditional, and manuscript-facing. It does not reconstruct general POVM theory, full projection-valued-measure theory, full operator algebra, categorical measurement theory, Hilbert-space quantum mechanics, physical decoherence theory, the continuum Born rule, physical selection, interpretive truth, or a physical no-collapse theorem. Its value is that it makes the record-fixation route, finite measurement surfaces, selector-boundary facts, and audit boundary explicit in Lean.

B.4 Local measurement-record specialization and integrated finite occupation

The imported closure spine does not by itself solve the measurement problem. What the present manuscript adds is a fixed-domain quotient-and-uniqueness analysis of measurement-record construction in physically non-selective theories. Its distinctive results are:

1. local measurement-record kernel forcing and denial-cost analysis;
2. no-strict-same-domain-weakening for evidentially usable unfixed record roles;
3. the explicit forced assignment space for diachronic measurement-record construction;
4. the measurement-record quotient theorem;
5. the Everettian non-selection theorem in this measurement-specific setting;
6. the asymmetry-exhaustion upgrade showing that bare Everett supplies no internal, provenance/indexical, or boundary-trace source of singular record-role fixation;
7. the fixation-commutation criterion for fixed-domain ordering of constructed hypotheses under admissible record reuse;
8. the common-lineage theorem for comparison, retained-record reuse, prospective-record reuse, and deliberation; and
9. the persistence-bearing bridge showing that, on the target physically realized practice, record-fixation is also continuation-locus fixation and persistence-bearing lineage fixation.

So the relation between the present manuscript and the imported closure spine is asymmetric. The imported results fix and justify the closure package; the present manuscript verifies its measurement-record activation conditions and applies it to admissible measurement-record construction.

B.5 Immutable release provenance

Item	Value
Repository	somamaley-ux/measurement-problem-reference
Controlling tag	v0.9.1
Controlling commit	332037dffe02c420084d4975023d436dfd40290f
Release URL	https://github.com/somamaley-ux/measurement-problem-reference/releases/tag/v0.9.1
Release source archive	https://github.com/somamaley-ux/measurement-problem-reference/archive/refs/tags/v0.9.1.zip
Zenodo archive	https://doi.org/10.5281/zenodo.19542014
Scope	Finite measurement-boundary release with zero-effect finite effect carrier, zero-product projector/repeatability layer, compatibility, closed-stack, boundary certificate, and audit manifest.

C Appendix: audit maps, denial matrix, and reading path

This appendix collects three audit aids: a dependency map, a denial-condition matrix, and a compact reading path. It adds no premises to the theorem ladder.

C.1 Dependency map for the theorem ladder

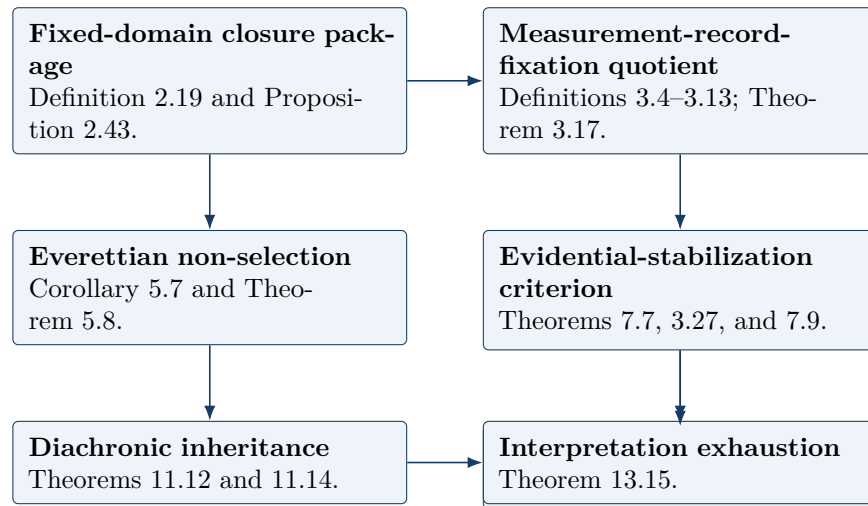


Figure 2: Dependency map for the hardened theorem ladder. The quotient theorem and the Everettian non-selection theorem are the two main inputs to the evidential ordering and lineage theorems; the interpretation taxonomy is downstream of both.

C.2 Denial-condition matrix

Table 9: Denial-condition matrix for standard objections.

Objection	To maintain it, one must deny	Where to look
AASC is optional vocabulary	Break the activation theorem, deny the standing conditional, or exhibit a same-domain record-use practice preserving target determinacy, standing-bearing reuse, act-time finality, and lawful redescription while avoiding the imported closure package	Sec. 2
Different admissible assignments are just representational packagings	The equivalence-collapse theorem or Corollary 3.23	Sec. 3; App. A
Branch-local determinacy is enough	Lemma 5.12	Sec. 5
Decision theory solves the problem	Theorem 11.10	Sec. 11
Evidential stabilization can be relativized without loss	Theorems 7.7 and 3.27	Sec. 7
Retained-record reuse, prospective-record reuse, and deliberative guidance are separate issues	Theorems 11.12 and 11.13	Sec. 11
There may be a fifth interpretive family	Theorem 13.12	Sec. 13
This is not real physics	Theorem 12.4 or Definition 2.16	Sec. 12; App. D
Instrumentalism or QBism refutes the theorem ladder	Proposition 14.11	Sec. 14

C.3 Compact reading path

A short but faithful route through the manuscript is:

1. Section 2: fixed-domain closure package, target construction, and standing conditional.
2. Section 3: Definitions 3.4–3.13, Theorem 3.17, and Corollary 3.23.
3. Section 5: Corollary 5.7 and Theorem 5.8.
4. Section 7: Theorems 7.7, 3.27, and 7.9.

5. Section 10: Theorem 10.3.
6. Section 11: Theorems 11.12 and 11.14.
7. Section 12: Theorem 12.4.
8. Section 13: Theorem 13.15.
9. Section 14: only the objection subsections still thought live after the first seven steps.

C.4 Lean release and audit path

The public finite measurement-boundary stack is audited through the standalone repository commands

```
lake build
lake env lean Checks/Axiom/MeasurementProblemReferenceAxiomCheck.lean
rg -n "\\b(sorry|admit|unsafe)\\b" -g "*.lean" MaleyLean Checks
```

The first command builds the standalone Lean package. The second prints the axiom dependencies of the paper-facing, refinement, finite-measurement, boundary, and audit-manifest endpoints. The third is the sharp source scan used to exclude unfinished or unsafe declarations. The support file `FORMAL_AUDIT_EXCERPTS_v0.9.1.md` records the public release audit excerpts and declaration-level audit interpretation used by this package; `SOURCE_SCAN_OUTPUT_v0.9.1.md` records the public source-scan status and reproduction command.

Audit stratum	Status and interpretation
Concrete finite facts	Direct arithmetic, basis, pointer, instrument, channel, effect, projector, compatibility, and selector-boundary theorems; see Appendix E for declaration-level audit status.
Bundled finite spines	Aggregate routes connecting the concrete facts; some inherit the standard Lean propext dependency through negation, equality, or extensionality bridges.
Audit manifest	Lean-readable declaration of the capstone, concrete facts, endpoint inventory, mechanization-state string, and non-claim boundary; it adds no new physics.
General manuscript theorem	Human-facing fixed-domain quotient, record-fixation, and selector-exhaustion proof. It is not claimed to be fully derived by the finite Lean stack.

The phrase “standard Lean **propext** dependency” records the ordinary proposition-extensionality dependency reported by bundled equality or extensionality proofs. It is neither a custom project axiom nor evidence of an admitted proof.

C.5 Role of the appendix

This appendix is procedural. The dependency map, denial-condition matrix, and reading path expose the architecture of the argument without adding premises to it. Table 6 in Section 14 is the main-text compression of the same refutation-burden logic.

C.6 v0.9.1 release record

The public v0.9.1 Lean release at commit 332037dffe02c420084d4975023d436dfd40290f is the controlling formal artifact for the finite effect/projector layer. Its immutable source archive is <https://github.com/somamaley-ux/measurement-problem-reference/archive/refs/tags/v0.9.1.zip>. It adds `zeroEffect`, assigns it zero weight, makes projector cross-products return `zeroEffect`, defines finite orthogonality against zero, and retains the downstream compatibility, closed-stack, boundary-certificate, and audit-manifest route.

The reproducibility commands for the public Lean repository are:

```
lake build
lake env lean Checks/Axiom/MeasurementProblemReferenceAxiomCheck.lean
rg -n "\b(sorry|admit|unsafe)\b" -g "*.lean" MaleyLean Checks
```

The manuscript package itself does not vendor the full Lean source tree; the public repository and v0.9.1 release are the reproducibility target. The packaged audit excerpt files preserve the decisive public-release facts used in the manuscript: the zero-effect carrier, zero-product projector orthogonality, branch-map annihilation, selector-boundary endpoints, no `sorry/admit/unsafe` scan status, and the standard `propext` qualification for aggregate bundles.

D Appendix: continuation-locus and realized-persistence bridge

This appendix records how the present measurement-record manuscript relates to the continuation-locus and realized-persistence vocabulary. The connection is intentionally narrow. The local measurement theorem does not require a further existence arc as an additional premise. The vocabulary supplies a structural interpretation of what the local theorem already forces: a standing-bearing measurement record is a role-fixed continuation locus participating in downstream persistence-bearing use.

D.1 Import discipline

The bridge uses the continuation-locus vocabulary in three restricted ways.

1. It explains why record-fixation is not semantic bookkeeping when the target practice is physically realized retained-record reuse, prospective-record reuse, deliberation, and evidential comparison.
2. It supplies continuation-locus vocabulary for the record-bearing roles already required in Sections 11 and 12.
3. It locates the same fixation form within a larger progression from non-vacuous evaluability to continuation locus, boundary-trace role-anchor fixation, realized persistence, standing localization, and concrete-domain participation.

It does not add a collapse postulate, hidden variable, observer ontology, spacetime ontology, biological embodiment, or empirical law.

D.2 Source chain

Table 10: Continuation-locus vocabulary as a non-load-bearing bridge vocabulary for measurement-record fixation.

Source result	Structural contribution	Use in this manuscript
Null-regime closure	Total vacuity cannot function as an admissibility-bearing endpoint inside fixed-domain evaluability.	Background motivation only; not a local premise.
Constraint-bundle forcing	Non-degenerate evaluability requires non-vacuous invariant role-constraint architecture.	Explains why record reuse cannot be empty bookkeeping.
Continuation-locus induction	Non-degenerate continuation-bearing governance induces continuation-locus equivalence classes.	Supplies the continuation-locus reading of measurement-record roles.
Boundary-trace fixation of continuation loci	Under internal non-selection, a declared stable singleton boundary trace fixes a role-anchor equivalence class for a continuation locus; otherwise fixation fails closed.	Aligns with boundary-trace record-role fixation in Section 8.
Persistence-regime realization	Irreversible standing-bearing continuation cannot remain trace-only, inert, abstract-only, repairable, or skin-level; it induces a persistence-regime equivalence class.	Gives the persistence-bearing reading of retained record lineage.
Standing localization	Realized standing-bearing persistence with lawful repair semantics induces bounded standing localization.	Clarifies why standing-bearing reuse is not global, diffuse, or freely reindexable.

Source result	Structural contribution	Use in this manuscript
Concrete-domain closure	Realized repair-localized bounded persistence is non-dischargeable into abstraction, formal inscription, semantic reference, trace succession, or skin-only presentation.	Answers the “not real physics” objection at the target-practice level: physical record reuse cannot be discharged into mere representation while retaining the same burden.

D.3 What the bridge adds and does not add

The bridge adds a physics-facing interpretation of the local theorem. In the main body, a measurement record is a physically realized structure used as an evidential anchor. The lineage theorem shows that retained-record reuse, prospective-record reuse, deliberation, and evidential comparison depend on one identity-preserving record lineage. The persistence bridge reads that lineage as a continuation-locus structure: when it is retained under standing-bearing downstream use, it has the form of persistence-bearing lineage rather than inert notation.

The bridge does not turn the measurement manuscript into a general existence theorem. It does not derive concrete objects, observers, biological systems, or spacetime regions from quantum measurement. It does not prove that Everettian physics is false. It says only that if the target practice is a physically realized measurement-record practice, then the record-fixation problem is not exhausted by representation-level bookkeeping. The record-bearing role must be fixed as a determinate continuation locus for standing-bearing reuse.

D.4 Relation to the target-practice objection

The objection that the manuscript is “not real physics” is best answered by distinguishing two claims. The manuscript does not present a new physical dynamics. It presents a structural constraint on the admissible use of physically realized measurement records. Once a theory is used to underwrite retained memory, prospective anticipation, deliberative guidance, and evidential comparison through records, those records are no longer descriptive marks alone. They are persistence-bearing continuation loci in the target practice. If the theory leaves those loci unfixed, the standing-bearing measurement-record lineage of the realized practice is underfixed.

E Appendix: mathematical and Lean crosswalk

This appendix records the correspondence between the manuscript mathematics and the public Lean 4 formalization. It separates concrete finite computation from cross-surface compatibility, selector exclusion, aggregate packaging, assumption routing, and audit declarations. The public

v0.9.1 release at commit 332037dffe02c420084d4975023d436dfd40290f is the controlling finite formal artifact for the zero-effect and zero-product projector layer.

E.1 Object crosswalk

Manuscript object	Lean declaration	Role
Ω	<code>FiniteMeasurementOutcome</code>	Two-outcome carrier.
$\mathcal{R}_2$	<code>FiniteMeasurementRecord</code>	Fine records and merged non-selective record.
Γ	<code>FinitePointerRegime</code>	Fine/coarse record regime.
$R_{\text{fine}}, R_{\text{coarse}}$	<code>finitePointerRecord</code>	Selective and non-selective record maps.
$\tilde{\psi}$	<code>FiniteKet2, balancedKet2</code>	Integer-amplitude numerator carrier.
$\tilde{\rho}$	<code>FiniteDensity2</code>	Density numerator table.
$\mathcal{I}_o$	<code>projectiveInstrumentBranch2</code>	Instrument branch table.
$\mathcal{K}_o$	<code>projectiveBranchMap2</code>	Projective branch map.
E_o	<code>FiniteProjectiveEffect2</code>	Finite effect surface with <code>zeroEffect</code> in v0.9.1.
R°	<code>compatibleFineRecordForOutcome</code>	Canonical fine-record map.

E.2 Theorem, proof-class, and audit ledger

Manuscript theorem	Lean declaration or declaration family	Proof class and audit status
Fine separation	<code>finite_fine_trace_distinguishes_records</code>	Finite case analysis / decidable disequality; axiom-free concrete fact.
Coarse merge	<code>finite_coarse_trace_merges_outcomes</code>	Definitional equality; axiom-free concrete fact.
Denominator-two normalization	<code>finiteOutcomeWeight2_normalized;</code> <code>finiteFineRecordWeight2_normalized;</code> <code>finiteCoarseRecordWeight2_normalized</code>	Arithmetic reduction; axiom-free concrete facts.
Balanced-weight non-selection	<code>balancedKet_born_not_singleton_selection2</code>	Denominator-two arithmetic contradiction; axiom-free concrete fact.
Density trace and coherence	<code>balancedPureDensity_trace2;</code> <code>balancedPureDensity_coherence2;</code> <code>nonSelectiveDensity_coherence_zero</code>	Matrix-entry calculation; axiom-free concrete facts.
Instrument decomposition	<code>projectiveInstrument_nonselective_eq_density;</code> <code>projectiveInstrument_branch_trace_matches_born</code>	Definitional equality and trace calculation; axiom-free concrete facts.
Environment non-selection	<code>coarseEnvironment_chooses_no_fine_record</code>	Case split / contradiction; axiom-free concrete fact.
Branch-map trace compatibility	<code>branchMap_trace_matches_density_born</code>	Matrix-entry calculation; axiom-free concrete fact.
Effect/trace/Born agreement	<code>projectiveEffect_weight_matches_density_born;</code> <code>projectiveEffect_weight_matches_branch_trace</code>	Equality calculation; axiom-free concrete facts.
Zero-product projector orthogonality	<code>projectiveEffects_orthogonal2</code>	v0.9.1 finite zero-effect calculation; axiom-free concrete fact.
Branch-map annihilation	<code>branchMap_cross_annihilates</code>	Matrix-entry calculation; axiom-free concrete fact.
Fine-record compatibility	<code>pointer_instrument_channel_effect_records_compatible</code>	Cross-surface compatibility theorem; axiom-free concrete compatibility fact.
Coarse-record compatibility	<code>coarse_records_compatible;</code> <code>decohered_coarse_matches_compatible_record</code>	Cross-surface compatibility theorem; axiom-free concrete compatibility facts.
Finite measurement-role adequacy	Spine family: record dynamics, Hilbert/Born, density/instrument, Kraus/channel, effect, projector/repeatability, and compatibility.	Derived manuscript theorem assembled from formalized finite spines; concrete components are axiom-free, while bundled spines may report standard <code>propext</code> .

Manuscript theorem	Lean declaration or declaration family	Proof class and audit status
Finite quantum correspondence	Amplitude, density, branch-map, effect, projector, and annihilation declarations listed above.	Finite role-faithfulness lemma assembling concrete equalities; supporting declarations are axiom-free concrete facts.
No hidden finite selector	<code>no_hidden_finite_nonselective_selector</code>	Selector-disjunction elimination; concrete boundary fact reported axiom-free in the released audit class.
Finite compatible-nonselection	Boundary certificate plus the compatibility and no-selector declarations.	Manuscript synthesis of checked compatibility and boundary theorems; components concrete, while the aggregate certificate may inherit standard <code>propext</code> .
Closed stack	<code>reasonableFiniteMeasurementClosedStack</code>	Aggregate conjunction package; may inherit standard <code>propext</code> .
Boundary certificate	<code>reasonableFiniteMeasurementBoundary_certificate</code>	Aggregate finite boundary theorem; concrete boundary facts are separately audited.
Audit manifest	<code>reasonableFiniteMeasurementAuditManifest</code>	Lean-readable audit declaration and endpoint inventory; no new physics; may inherit standard <code>propext</code> .

E.3 Axiom and audit interpretation

The released audit distinguishes concrete finite facts from aggregate endpoints. The concrete arithmetic, basis, pointer, instrument, channel, effect, projector, compatibility, and selector-boundary facts listed above are direct finite calculations, case splits, or definitional equalities and are recorded declaration-by-declaration in `FORMAL_AUDIT_EXCERPTS_v0.9.1.md`. In this manuscript ledger, direct concrete facts such as `finite_fine_trace_distinguishes_records`, `balancedKet_born_not_singleton_selection2`, `projectiveEffect_weight_matches_branch_trace`, `projectiveEffects_orthogonal2`, `branchMap_cross_annihilates`, and `no_hidden_finite_nonselective_selector` are separated from aggregate endpoints such as `reasonableFiniteMeasurementClosedStack`, `reasonableFiniteMeasurementBoundary_certificate`, and `reasonableFiniteMeasurementAuditManifest`. Bundled spines, the closed stack, the boundary certificate as an aggregate endpoint, and the audit manifest may inherit the standard Lean `propext` dependency through negation, equality, extensionality, or bundle-bridge steps. A standard Lean `propext` dependency is neither a custom project axiom nor an admitted proof. The audit manifest exposes the capstone endpoint, concrete selector-boundary facts, aggregate endpoint list, mechanization-state declaration, and non-claim boundary as Lean-readable declarations.

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